Quantitative Volatility Corpus

Volume II: Multi-Name Volatility, Correlation, and Numerical Methods

Quant Corpus Consolidation Committee

Contents

Chapter 1

Basket Volatility and the Correlation–Volatility Relation

A long basket (or index) straddle is *long correlation*: the basket volatility rises when the constituents move together, so the position is worth more when they correlate. The opposite side of that risk is the dispersion trade — long single-name volatility, short basket volatility — which is short correlation. This chapter builds the objects with which that risk is measured and re-marked: the basket volatility identity, the basket vol spread ξ, the CVC spread (the price of the dispersion package), the exact relation $\rho = \rho(\sigma_1, \sigma_2, \xi)$, and the correlation sensitivity Cega. It also extends the no-arbitrage ladder to two names — the rungs $\xi \geq 0$ (equivalently $\rho \leq 1$) and $\rho \geq -1$ — and fixes the two re-marking conventions, sticky-ρ and sticky-CVC, whose difference generates the shadow Greeks of the routed-sensitivities chapter. A single worked example ($\sigma = 20\%$, $\rho = 0.50$, $\xi \approx 2.7$ vol points (2.68 exact), Cega $= +0.2$ per correlation point — a $+1$ P&L under the desk's gap bump) is opened here and carried through the next two chapters so that every number reconciles.

1.1 The basket volatility identity

Definition 1.1 (Two-name equal-weight basket)**.** Let S_1, S_2 be two underlyings entering a basket with equal weights $w_1 = w_2 = \frac{1}{2}$; the basket level is $S_B = \frac{1}{2}(S_1 + S_2)$. Let $\sigma_1, \sigma_2 > 0$ be their implied volatilities (per annum, in absolute terms; a *vol point* means 0.01) and let $\rho \in [-1, 1]$ be the implied correlation of their log-returns. The basket implied volatility is

$$\sigma_B \;=\; \sqrt{\tfrac{1}{4}\sigma_1^2 + \tfrac{1}{4}\sigma_2^2 + \tfrac{1}{2}\rho\,\sigma_1\sigma_2}, \tag{1.1}$$

the standard deviation of $\frac{1}{2}(\mathrm{d}\log S_1 + \mathrm{d}\log S_2)$ when each leg is lognormal with the stated vols and correlation. σ_B is defined in log (geometric) coordinates while the level S_B is arithmetic; Lemma **??** states and error-bounds the bridge between the two at the price interface.

Assumption 1.2 (Constituent vols are ATM vols)**.** The vols entering (**??**) are the single-name at-the-money volatilities of the two surfaces,

$$\sigma_i \;\equiv\; \sigma_{\mathrm{ATM},i}(\ell_{F,i}, T), \qquad i = 1, 2, \tag{1.2}$$

evaluated at the current forward, where $\ell_{F,i} := \ln(F_i/F_{\mathrm{ref},i})$ is the per-name forward log-level — the per-name instance of ℓ_F and F_{ref}, ratified with Volume II scope by the note to constraint C1 of the notation contract — and T is the common expiry. Under the standing zero-rate, zero-dividend assumption the forwards collapse to spot, $F_i = S_i$ (flagged here and at each later use). Every derivative taken through σ_i in this volume inherits the single-name surface dynamics of Volume I through (**??**).

Remark 1.3 (Why ATM, exactly). The decomposition $\sigma(m, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$ with the normalisation $\sigma_{\text{smile}}(0, T) = 0$ gives $\sigma(0, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T)$ exactly: at the money ($m = \ln(K/F) = 0$, strike at the forward) the implied vol *is* the ATM component, with no smile contamination. The CVC spread of Section **??** is built from at-the-money options, so it is a relation among the $\sigma_{\text{ATM},i}$ and σ_B alone — which is what makes Assumption **??** the consistent reading of (**??**) rather than a mere convention.

Conventions used throughout the chapter.

- **Vol points.** Vols and the spread ξ are quoted in vol points; 1 vol point $= 0.01$. A per-unit-σ derivative is divided by 100 to express it per vol point; this is the source of every "/100".
- **Vega normaliser.** For a book of value V, the basket vega per vol point is $\mathcal{V}_B = \frac{1}{100}\frac{\partial V}{\partial \sigma_B}$.
- **Correlation bump (the "/10").** Correlation is bumped not by a fixed absolute amount but by a fixed fraction $\omega = 0.1$ of its remaining distance to 1: $\rho^* = \rho + \omega(1-\rho) = \rho + (1-\rho)/10$. This is a convex move toward 1 and can never leave $[-1, 1]$. The bump $(1-\rho)/10$ is a fraction of the remaining gap, not a 0.10 absolute shift, and it is one-sided: always toward $\rho = 1$. It defines the gap-bump P&L ΔV_ω of Section **??**; the ratified Cega itself is quoted per correlation point.

1.2 The spread ξ and the no-arbitrage ladder

Definition 1.4 (Maximal basket vol and the spread ξ)**.** At $\rho = 1$ the two legs are perfectly correlated and (**??**) collapses to the arithmetic mean,

$$\sigma_B^{\rho=1} \;=\; \sqrt{\tfrac{1}{4}(\sigma_1+\sigma_2)^2} \;=\; \tfrac{1}{2}(\sigma_1+\sigma_2) \;=:\; \bar{\sigma}, \tag{1.3}$$

the largest value σ_B can take at fixed single-name vols. Define the *basket vol spread*

$$\boxed{\xi \;:=\; \tfrac{1}{2}(\sigma_1+\sigma_2) - \sigma_B \;=\; \bar{\sigma} - \sigma_B \;\geq\; 0.} \tag{1.4}$$

ξ is the gap between the perfectly-correlated basket vol and the actual basket vol. It is zero iff $\rho = 1$ and grows as correlation falls; it is the diversification benefit of the basket, measured in vol points, and it is the natural sign-definite "how far from perfectly correlated" coordinate for the dynamics studied in this volume.

The two-name book carries a second natural volatility: the Margrabe volatility Σ_M of the *difference* of the logs, the pricing vol of the exchange option studied two chapters on. The two are complementary.

Lemma 1.5 (Sum–difference variance split)**.** *With* $\Sigma_M^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2$*,*

$$\sigma_B^2 + \tfrac{1}{4}\Sigma_M^2 \;=\; \tfrac{1}{2}\left(\sigma_1^2 + \sigma_2^2\right). \tag{1.5}$$

Proof. Add (**??**) squared to a quarter of the definition of Σ_M^2; the cross terms $\pm\frac{1}{2}\rho\sigma_1\sigma_2$ cancel. □

Correlation rotates a fixed total variance budget between the sum portfolio (the basket) and the difference portfolio (the spread): ρ up moves variance from Σ_M to σ_B, and conversely. This is why the basket book and the spread book of the exchange-option chapter are opposite sides of the same correlation risk.

Proposition 1.6 (Two-name rungs of the no-arbitrage ladder). *For $\sigma_1, \sigma_2 > 0$, the correlation parameter satisfies $\rho \in [-1, 1]$ if and only if*

$$\tfrac{1}{2}|\sigma_1 - \sigma_2| \ \le\ \sigma_B \ \le\ \bar{\sigma}, \qquad \textit{equivalently} \qquad 0 \ \le\ \xi \ \le\ \min(\sigma_1, \sigma_2), \tag{1.6}$$

with the upper σ_B bound attained iff $\rho = 1$ and the lower iff $\rho = -1$.

Proof. σ_B^2 in (**??**) is affine and increasing in ρ. At $\rho = 1$ it equals $\frac{1}{4}(\sigma_1 + \sigma_2)^2$, at $\rho = -1$ it equals $\frac{1}{4}(\sigma_1 - \sigma_2)^2$; taking square roots gives the σ_B bounds. For the ξ form, the upper σ_B bound is $\xi \ge 0$ by (**??**), and at $\rho = -1$, $\xi = \bar{\sigma} - \frac{1}{2}|\sigma_1 - \sigma_2| = \min(\sigma_1, \sigma_2)$. □

Remark 1.7 (What each rung means on the desk). *Upper rung* ($\xi \ge 0$, $\rho \le 1$): a basket cannot be more volatile than the weighted average of its constituents. In price space this is a model-free payoff dominance — a basket of calls dominates the call on the basket (Proposition **??** below), and for straddles the same state-by-state dominance is the triangle inequality $\frac{1}{2}|S_1 - K_1| + \frac{1}{2}|S_2 - K_2| \ge |S_B - K_B|$ (from $|a| + |b| \ge |a + b|$) — so a market quoting $\sigma_B > \bar{\sigma}$ offers a static arbitrage: sell the basket straddle, buy the single-name straddles, and the package can only pay at expiry. That it is also put on for a *credit* needs one more step: at equal spots the ATM straddle $\mathrm{Str}(\sigma) = 2S\big[2N(\frac{1}{2}\sigma\sqrt{T}) - 1\big]$ is increasing and concave in σ ($\mathrm{Str}'(\sigma) = 2S\sqrt{T}\,\varphi(\frac{1}{2}\sigma\sqrt{T}) > 0$, $\mathrm{Str}''(\sigma) = -\frac{1}{2}ST^{3/2}\sigma\,\varphi(\frac{1}{2}\sigma\sqrt{T}) < 0$), so $\sigma_B > \bar{\sigma}$ gives $\mathrm{Str}(\sigma_B) > \mathrm{Str}(\bar{\sigma}) \ge \frac{1}{2}\mathrm{Str}(\sigma_1) + \frac{1}{2}\mathrm{Str}(\sigma_2)$: the basket straddle sold brings in more than the single-name straddles cost. The same monotonicity-plus-concavity step is what converts the price inequality $\Pi \ge 0$ into the vol inequality $\sigma_B \le \bar{\sigma}$ at equal spots. *Lower rung* ($\xi \le \min(\sigma_1, \sigma_2)$, $\rho \ge -1$): by Lemma **??** it is equivalent to $\Sigma_M \le \sigma_1 + \sigma_2$ — the spread cannot be more volatile than the sum of the legs' vols. A basket vol below $\frac{1}{2}|\sigma_1 - \sigma_2|$ implies $\rho < -1$: the implied 2×2 correlation matrix fails to be positive semi-definite, and the difference portfolio is priced as if it had more variance than any pair of random variables can deliver.

Remark 1.8 (The ladder, continued). Volume I established the single-name rungs: butterfly convexity of $C(K, T)$ in strike and calendar monotonicity of total variance — rungs (A2) and (A3) of the admissible parameter set defined in the surface chapter of Volume I, with the static-arbitrage consequences proved there — the static no-arbitrage of one surface. Proposition **??** adds the first two multi-name rungs, which constrain the *joint* quotes of two surfaces and a correlation. The n-name rung — positive semi-definiteness of the full correlation matrix Ω, checked by Schur complements — is climbed in the admissibility chapter of this volume.

1.3 The CVC spread: basket of calls versus call on the basket

Definition 1.9 (CVC spread). Fix a common expiry $T > 0$. Under the standing zero-rate, zero-dividend assumption, let $C(S, \sigma)$ be the at-the-money price — strike equal to the forward, which here collapses to spot, $F = S$ (flagged) — of a European call on an underlying of level S and implied volatility σ, and let C_B denote the price of the at-the-money call on the basket level S_B itself (strike at the basket forward, collapsing to S_B). The *CVC spread* is the price difference between a basket of single-name at-the-money calls and the at-the-money call on the basket,

$$\Pi \ :=\ w_1\, C(S_1, \sigma_1) + w_2\, C(S_2, \sigma_2) - C_B. \tag{1.7}$$

By Remark **??**, every single-name vol entering (**??**) is an ATM vol: $\sigma_i = \sigma_{\mathrm{ATM},i}(\ell_{F,i}, T)$ per Assumption **??**. For computation the basket call is carried at the *geometric (lognormal-basket) convention* $C_B = C(S_B, \sigma_B)$: the basket level is arithmetic while σ_B is the vol of the log-average, and Lemma **??** below prices the identification error at $O\Big(S(\sigma\sqrt{T})^3\Big)$ at equal spots — inside every remainder used in this chapter.

Proposition 1.10 (Model-free non-negativity). $\Pi \geq 0$, *independently of any model. At expiry, with each strike set at its leg's forward and the basket strike at the basket forward,*

$$\tfrac{1}{2}(S_1 - K_1)^+ + \tfrac{1}{2}(S_2 - K_2)^+ \;\geq\; \big(\tfrac{1}{2}S_1 + \tfrac{1}{2}S_2 - \tfrac{1}{2}K_1 - \tfrac{1}{2}K_2\big)^+,$$

*since $a^+ + b^+ \geq (a+b)^+$ for all real a, b. The basket of calls dominates the call on the basket state by state, hence in price under any martingale measure. This is the price-space form of the upper rung of Proposition **??**; its vol-space reading $\sigma_B \leq \bar{\sigma}$ uses the geometric bridge of Lemma **??** together with the monotonicity-and-concavity step of Remark **??**.*

Lemma 1.11 (Geometric-basket bridge). *Let $S_1 = S_2 = S$, let the legs be jointly lognormal with vols σ_1, σ_2 and correlation ρ (the reading of Definition **??**), and write $\sigma := \max(\sigma_1, \sigma_2)$. Then the at-the-money arithmetic-basket call price satisfies*

$$C_B \;=\; \mathbb{E}\big[(A_T - S)^+\big] \;=\; C(S, \sigma_B) + O\Big(S(\sigma\sqrt{T})^3\Big), \qquad A_T := \tfrac{1}{2}\big(S_1(T) + S_2(T)\big). \tag{1.8}$$

Proof. Write $G_T := \sqrt{S_1(T)S_2(T)}$ (the geometric basket) and $x_T := \ln\big(S_1(T)/S_2(T)\big)$, so that $A_T = G_T \cosh(x_T/2) \geq G_T$ and $\mathrm{Var}(x_T) = \Sigma_M^2 T$.

(i) The geometric call is exact Black–Scholes with a forward deficit. $\ln G_T$ is Gaussian with volatility exactly σ_B (this is Definition **??**) and $\mathbb{E}[G_T] = S\, e^{-\Sigma_M^2 T/8}$: the mean of $\ln G_T$ is $\ln S - \frac{1}{4}(\sigma_1^2 + \sigma_2^2)T$, and $\frac{1}{2}\sigma_B^2 - \frac{1}{4}(\sigma_1^2 + \sigma_2^2) = -\frac{1}{8}\Sigma_M^2$ by Lemma **??**. Hence $\mathbb{E}[(G_T - S)^+]$ is an exact Black–Scholes price at vol σ_B and forward $S\, e^{-\Sigma_M^2 T/8}$; lowering the forward from S by $S\Sigma_M^2 T/8 + O(S(\sigma^2 T)^2)$ lowers the ATM call by (delta $= \frac{1}{2} + O(\sigma\sqrt{T})$) times that:

$$\mathbb{E}[(G_T - S)^+] = C(S, \sigma_B) - \frac{S\Sigma_M^2 T}{16}\big(1 + O(\sigma\sqrt{T})\big).$$

(ii) The convexity premium of arithmetic over geometric. The payoff gap $(A_T - S)^+ - (G_T - S)^+$ equals $A_T - G_T$ on $\{G_T \geq S\}$, lies in $[0, A_T - G_T]$ on the band $\{A_T \geq S > G_T\}$ — whose probability is $O(\sigma\sqrt{T})$ while $A_T - G_T = O(S\sigma^2 T)$ there, a cubic-order contribution — and vanishes otherwise. On $\{G_T \geq S\}$, $A_T - G_T = G_T\big(\cosh(x_T/2) - 1\big) = \frac{1}{8} S\, x_T^2\big(1 + O(\sigma\sqrt{T})\big)$, and $\mathbb{E}\big[x_T^2\, \mathbf{1}\{G_T \geq S\}\big] = \frac{1}{2}\Sigma_M^2 T\big(1 + O(\sigma\sqrt{T})\big)$: condition x_T on $\ln G_T$ — for jointly Gaussian variables, $\mathbb{E}[x^2\, \mathbf{1}\{g > \mathrm{med}(g)\}] = \frac{1}{2}\mathbb{E}[x^2]$ exactly, and the mean and threshold offsets are $O(\sigma^2 T)$ relative to the $O(\sigma\sqrt{T})$ standard deviation. Hence the premium is $\dfrac{S\Sigma_M^2 T}{16}\big(1 + O(\sigma\sqrt{T})\big)$.

(iii) Cancellation. The two $O(S\sigma^2 T)$ terms — the forward deficit of the geometric basket and the convexity premium of the arithmetic over it — cancel at leading order, leaving (**??**). Numerically, at the anchor marks of Example **??** ($\sigma = 20\%$, $\rho = 0.50$, $T = 1$, $S = 100$) the exact gap is $6.9079 - 6.9013 = 0.0066$ per 100 of notional (an implied-vol gap of 0.017 vol points), well inside the cubic budget. $\square$

Proposition 1.12 (The CVC spread equals the vol spread at leading order)**.** *With equal weights $w_1 = w_2 = \frac{1}{2}$ and equal spots $S_1 = S_2 = S$ (hence $S_B = S$, and the common ATM strike equals S since $F = S$ under zero rates — flagged), the CVC spread (**??**) satisfies*

$$\boxed{\Pi \;=\; \frac{S\sqrt{T}}{\sqrt{2\pi}}\,\xi \;+\; O\Big(S(\sigma\sqrt{T})^3\Big), \qquad \sigma := \max(\sigma_1,\sigma_2),} \tag{1.9}$$

*with ξ the basket vol spread (**??**). Consequently, at fixed spot and expiry, Π is constant if and only if ξ is constant, to this order. The exact inequality $\Pi \geq 0$ is Proposition **??** and does* not *follow from the truncation: the expansion adds only that the leading term is the (non-negative) spread ξ times the ATM cash-vega scale; the truncated remainder carries no sign of its own.*

Proof. At the money (strike = forward = S under zero rates, flagged) the Black–Scholes call price at implied volatility σ is

$$C(S,\sigma) = S\big[N(\tfrac{1}{2}\sigma\sqrt{T}) - N(-\tfrac{1}{2}\sigma\sqrt{T})\big] = S\big[2\,N(\tfrac{1}{2}\sigma\sqrt{T}) - 1\big].$$

Since $N'(0) = 1/\sqrt{2\pi}$ and $N''(0) = 0$, the Taylor expansion is $N(u) = \frac{1}{2} + u/\sqrt{2\pi} + O(u^3)$; with $u = \frac{1}{2}\sigma\sqrt{T}$,

$$C(S,\sigma) = \frac{S\,\sigma\sqrt{T}}{\sqrt{2\pi}} + O\Big(S(\sigma\sqrt{T})^3\Big).$$

By Lemma **??**, $C_B = C(S,\sigma_B) + O(S(\sigma\sqrt{T})^3)$ — the geometric convention costs nothing at this order. Substituting into (**??**) with $w_1 = w_2 = \frac{1}{2}$ and $S_1 = S_2 = S_B = S$,

$$\Pi = \frac{S\sqrt{T}}{\sqrt{2\pi}}\Big[\tfrac{1}{2}(\sigma_1+\sigma_2) - \sigma_B\Big] + O(\cdot) = \frac{S\sqrt{T}}{\sqrt{2\pi}}\,\xi + O(\cdot)$$

by (**??**). The leading term is non-negative since $\xi \geq 0$ (Proposition **??**); the exact statement $\Pi \geq 0$ is Proposition **??**. □

Remark 1.13 (The price of dispersion). The CVC spread is the premium of the dispersion package: long the single-name calls, short the basket call. Its carrying value is, to leading order, the diversification benefit ξ times the ATM cash-vega scale $S\sqrt{T}/\sqrt{2\pi}$. At $S = 100$, $T = 1$ and the anchor spread $\xi = 0.0268$ of Example **??** below, this is $\Pi \approx 100 \times 0.3989 \times 0.0268 \approx 1.07$ per 100 of basket notional. The exact Black–Scholes difference *computed under the geometric convention* $C_B = C(S_B,\sigma_B)$ of Definition **??** is 1.06; the cubic remainder in (**??**) is worth less than half a cent at these vols, and pricing the true arithmetic-basket call instead moves Π by less than a cent more (Lemma **??**). Holding Π fixed at fixed spot and expiry is the same statement as holding ξ fixed — the sticky-CVC convention of Section **??**.

1.4 The exact correlation–volatility relation

Everything downstream rests on one identity: solve (**??**) for ρ and substitute the spread.

Proposition 1.14 (Exact ρ as a function of (σ_1,σ_2,ξ))**.** *With $\sigma_B = \frac{1}{2}(\sigma_1+\sigma_2) - \xi$,*

$$\boxed{\rho \;=\; 1 - 2\Big(\frac{1}{\sigma_1} + \frac{1}{\sigma_2}\Big)\xi + \frac{2\xi^2}{\sigma_1\sigma_2}.} \tag{1.10}$$

Proof. From (??), $\frac{1}{2}\rho\,\sigma_1\sigma_2 = \sigma_B^2 - \frac{1}{4}\sigma_1^2 - \frac{1}{4}\sigma_2^2$. Substitute $\sigma_B = \bar\sigma - \xi$ with $\bar\sigma = \frac{1}{2}(\sigma_1+\sigma_2)$, so $\sigma_B^2 = \frac{1}{4}(\sigma_1+\sigma_2)^2 - (\sigma_1+\sigma_2)\xi + \xi^2$. Then

$$\sigma_B^2 - \tfrac{1}{4}\sigma_1^2 - \tfrac{1}{4}\sigma_2^2 = \tfrac{1}{2}\sigma_1\sigma_2 - (\sigma_1+\sigma_2)\xi + \xi^2.$$

Multiplying by $2/(\sigma_1\sigma_2)$ gives (??). □

Corollary 1.15 (Equal-vol specialisation)**.** *If $\sigma_1 = \sigma_2 = \sigma$ then (??) gives the exact*

$$\rho \;=\; 1 - \frac{4\xi}{\sigma} + \frac{2\xi^2}{\sigma^2}, \qquad \textit{equivalently} \qquad 1-\rho \;=\; \frac{2\xi(2\sigma-\xi)}{\sigma^2}, \tag{1.11}$$

and at leading order in the small quantity ξ/σ, $1-\rho \approx 4\xi/\sigma$.

Theorem 1.16 (First-order spread)**.** *Conversely, with $\sigma_1 = \sigma_2 = \sigma$, expanding (??) about $1-\rho = 0$,*

$$\sigma_B = \sigma\sqrt{\tfrac{1}{2}(1+\rho)} = \sigma\sqrt{1-\tfrac{1}{2}(1-\rho)} = \sigma\Big(1 - \tfrac{1}{2}(1-\rho)\cdot\tfrac{1}{2} + O\big((1-\rho)^2\big)\Big), \tag{1.12}$$

hence $\xi = \sigma - \sigma_B = \frac{1}{4}\sigma(1-\rho) + O\big((1-\rho)^2\big)$. For unequal vols the same expansion gives $\xi \approx \frac{1}{2}\,\dfrac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}\,(1-\rho)$.

Proof. $\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \cdots$ with $x = \frac{1}{2}(1-\rho)$ gives $\sigma_B/\sigma = 1 - \frac{1}{4}(1-\rho) - \frac{1}{32}(1-\rho)^2 - \cdots$; subtract from 1 and multiply by σ. For unequal vols, write $\sigma_B^2 = \bar\sigma^2 - \frac{1}{2}\sigma_1\sigma_2(1-\rho)$ and expand the square root about $\bar\sigma$. □

Remark 1.17 (Validity). Equation (??) is exact for all $(\sigma_1, \sigma_2, \xi)$ with $\sigma_i > 0$ — exact, that is, as a relation among the *quoted vols* $(\sigma_1, \sigma_2, \sigma_B)$; its link to option prices goes through Lemma **??** and carries that lemma's $O(S(\sigma\sqrt{T})^3)$ tolerance. In vol space it is the backbone of everything in this and the next two chapters. The leading-order forms $1-\rho \approx 4\xi/\sigma$ and $\xi \approx \sigma(1-\rho)/4$ are equal-vol shortcuts valid near $\rho \approx 1$ (small ξ/σ); they are used only for intuition and quick estimates, never where the exact formula is needed.

The relation (??) makes ρ a determined function of the vols once ξ is held fixed. Its partial derivatives at fixed ξ are the channel through which every vol move touches correlation.

Proposition 1.18 (Vol sensitivities of ρ at fixed spread)**.** *Differentiating (??) at fixed ξ,*

$$\boxed{\left.\frac{\partial\rho}{\partial\sigma_1}\right|_\xi = \frac{2\xi}{\sigma_1^2}\Big(1-\frac{\xi}{\sigma_2}\Big), \qquad \left.\frac{\partial\rho}{\partial\sigma_2}\right|_\xi = \frac{2\xi}{\sigma_2^2}\Big(1-\frac{\xi}{\sigma_1}\Big).} \tag{1.13}$$

In the equal-vol case $\sigma_1 = \sigma_2 = \sigma$, a parallel move $\mathrm{d}\sigma_1 = \mathrm{d}\sigma_2 = \mathrm{d}\sigma$ gives the total

$$\boxed{\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_\xi = \frac{4\xi}{\sigma^2} - \frac{4\xi^2}{\sigma^3} = \frac{4\xi}{\sigma^2}\Big(1-\frac{\xi}{\sigma}\Big) \;>\; 0 \quad \textit{for } 0<\xi<\sigma, \qquad \textit{leading order: } +\frac{4\xi}{\sigma^2}.} \tag{1.14}$$

The strict inequality holds on the open interval $0<\xi<\sigma$ (equivalently $-1<\rho<1$) only: the slope vanishes at both *rungs, $\xi = 0$ ($\rho = 1$) and $\xi = \sigma$ ($\rho = -1$).*

Proof. Differentiate (??) term by term in σ_1 at fixed ξ: $+2\xi/\sigma_1^2 - 2\xi^2/(\sigma_1^2\sigma_2)$, which factors as stated; symmetrically for σ_2. Summing the two at $\sigma_1 = \sigma_2 = \sigma$ gives (??). □

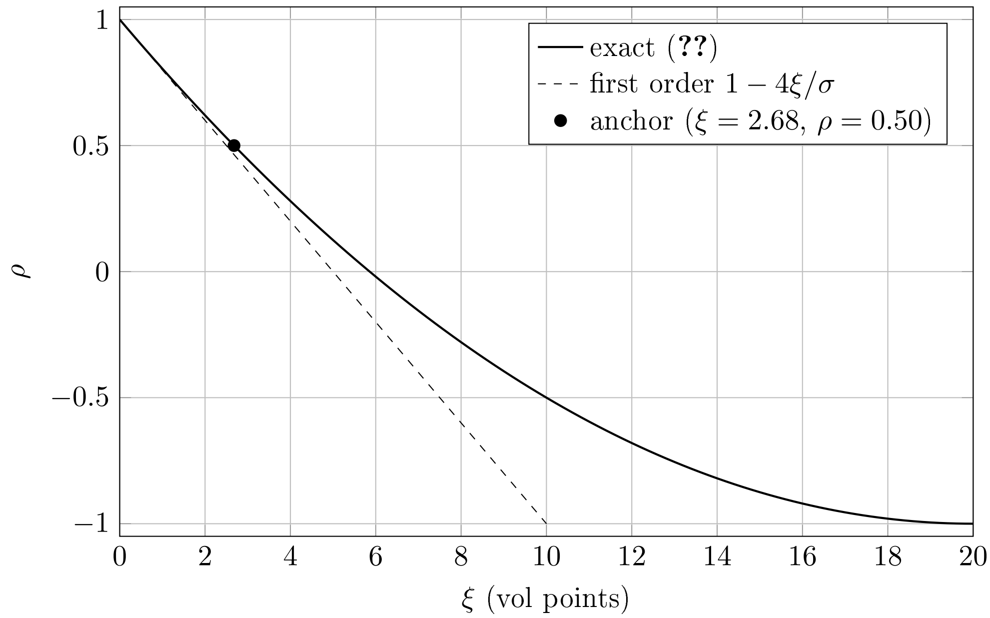

Figure 1.1: The exact correlation–volatility relation (**??**) at equal vols $\sigma_1 = \sigma_2 = 20\%$. The map runs from $\rho = 1$ at $\xi = 0$ down to $\rho = -1$ at $\xi = \sigma = 20$ points — exactly the two-name rungs of Proposition **??**. The dashed first-order line $1 - 4\xi/\sigma$ is tangent at $\rho = 1$ and overstates $1 - \rho$ away from it; at the anchor point of Example **??** the gap is already visible. The map is convex in ξ: each further point of spread costs less correlation than the last.

Remark 1.19 (Sign and degeneracy). The sign in (**??**) is *positive* throughout the interior of the ladder: with the spread fixed, raising σ shrinks the relative gap $1 - \rho \approx 4\xi/\sigma$, pushing ρ toward 1. The slope degenerates at both ends. At the top rung $\xi = 0$ ($\rho = 1$) the factor $4\xi/\sigma^2$ vanishes: a perfectly correlated pair stays perfectly correlated under fixed-ξ re-marks, and anything that divides by the slope (**??**) — as the shadow-Greek constructions of the next chapter do — must be guarded there. At the bottom rung the picture depends on the vols. In general $\partial\rho/\partial\sigma_1|_\xi > 0$ if and only if $\xi < \sigma_2$, which holds throughout the interior of the ladder (**??**) *including* the bottom rung when $\sigma_1 < \sigma_2$ (there $\xi = \sigma_1 < \sigma_2$, and the σ_1-sensitivity stays strictly positive; it is the partial with respect to the *higher*-vol name, $\partial\rho/\partial\sigma_2|_\xi \propto (1 - \xi/\sigma_1)$, that dies). Only in the equal-vol case do both partials vanish at the bottom rung: there $\xi \to \sigma = \sigma_2$ and the factor $(1 - \xi/\sigma_2)$ kills the sensitivity — near $\rho = -1$ a re-mark of the spread no longer moves correlation through the vols. The positive slope (**??**) is the single quantity the shadow Greeks of the next chapter are built from.

1.5 Cega: the correlation sensitivity

The value sensitivity to correlation is expressed through the position's basket vega — when the basket vol is the book's only correlation channel. Let V be the value of a correlation-sensitive book, viewed as a function of ρ with everything else held fixed. Differentiating (**??**),

$$\frac{\partial \sigma_B}{\partial \rho} = \frac{\sigma_1 \sigma_2}{4\sigma_B} > 0, \tag{1.15}$$

so, *for a book whose only ρ-dependence is through σ_B — basket-vol instruments only, no spread or exchange legs* — a position with basket vol sensitivity $\frac{\partial V}{\partial \sigma_B}$ (whose quoted vega is $\mathcal{V}_B = \frac{1}{100}\frac{\partial V}{\partial \sigma_B}$) has

$$\frac{\partial V}{\partial \rho} \;=\; \frac{\partial V}{\partial \sigma_B}\,\frac{\partial \sigma_B}{\partial \rho} \;=\; \frac{\partial V}{\partial \sigma_B}\,\frac{\sigma_1\sigma_2}{4\sigma_B}. \tag{1.16}$$

The hypothesis matters. The chapter itself supplies the second channel: by Lemma **??**, the Margrabe vol $\Sigma_M = \sqrt{\sigma_1^2+\sigma_2^2-2\rho\sigma_1\sigma_2}$ also depends on ρ, with $\frac{\partial \Sigma_M}{\partial \rho} = -\sigma_1\sigma_2/\Sigma_M$ (differentiate the definition of Σ_M^2) — *opposite* in sign to (**??**). For a general two-name book $V(\sigma_B(\rho), \Sigma_M(\rho))$, carrying both basket-vol and spread legs, the full chain rule is

$$\frac{\partial V}{\partial \rho} \;=\; \frac{\partial V}{\partial \sigma_B}\,\frac{\sigma_1\sigma_2}{4\sigma_B} \;-\; \frac{\partial V}{\partial \Sigma_M}\,\frac{\sigma_1\sigma_2}{\Sigma_M}. \tag{1.17}$$

The spread book of the exchange-option chapter lives in the second channel; (**??**) must never be applied to it.

Remark 1.20 (The ρ-to-vol translation is singular at the bottom rung). Equation (**??**) *diverges* as $\sigma_B \to 0$ — at equal vols, as $\rho \to -1$, the bottom rung of Proposition **??**, where $\sigma_B = \frac{1}{2}|\sigma_1 - \sigma_2| = 0$. At equal vols $\sigma_B = \sigma\sqrt{(1+\rho)/2}$, so $\frac{\partial \sigma_B}{\partial \rho} = \frac{1}{4}\sigma\sqrt{2/(1+\rho)}$: a blow-up of order $(1+\rho)^{-1/2}$. On the running example's own line ($\sigma = 20\%$) the mark is 0.0577 at the anchor $\rho = 0.50$, 0.224 at $\rho = -0.90$, and 0.707 at $\rho = -0.99$ — twelve times the anchor. A book marked near the PSD boundary therefore shows a finite $\frac{\partial V}{\partial \rho}$ but an unbounded correlation-to-vol conversion: every Cega-to-vega number in this section is valid only bounded away from the bottom rung, and in a decorrelation crisis it is the *translation*, not necessarily the risk, that explodes first. This is the mirror image of Remark **??**, where the fixed-ξ slope $\mathrm{d}\rho/\mathrm{d}\sigma|_\xi$ *vanishes* at the same rung.

Definition 1.21 (Cega)**.** Cega is the correlation sensitivity quoted *per correlation point* (the ratified convention of the notation contract):

$$\boxed{\text{Cega} \;:=\; \frac{1}{100}\,\frac{\partial V}{\partial \rho},} \tag{1.18}$$

the P&L of an absolute +0.01 move in ρ, exact by definition. For a basket-only book, (**??**) gives the desk form $\text{Cega} = \mathcal{V}_B\,\dfrac{\sigma_1\sigma_2}{4\sigma_B}$, with $\mathcal{V}_B = \frac{1}{100}\frac{\partial V}{\partial \sigma_B}$ the basket vega per vol point: the two normalisers (/100 on each side) cancel against each other.

Definition 1.22 (Gap-bump P&L ΔV_ω — local to this chapter)**.** The *gap-bump P&L* is the exact value change under the convex bump $\rho^* = \rho + (1-\rho)/10$ of the conventions list:

$$\Delta V_\omega \;:=\; V(\rho^*) - V(\rho). \tag{1.19}$$

It is a distinct object from Cega, and this volume never uses the bare ratified symbol Cega for it; the routed-sensitivities chapter quotes this same object under the locally scoped, subscripted desk name Cega_{gap}, with the conversion to the ratified per-point Cega stated there. To first order in the bump size,

$$\Delta V_\omega \;=\; \frac{\partial V}{\partial \rho}\,\frac{1-\rho}{10} \;+\; O\!\left(\left(\frac{1-\rho}{10}\right)^{2}\frac{\partial^2 V}{\partial \rho^2}\right) \;=\; 10\,(1-\rho)\,\text{Cega} \;+\; O\!\left(\left(\frac{1-\rho}{10}\right)^{2}\frac{\partial^2 V}{\partial \rho^2}\right). \tag{1.20}$$

Remark 1.23 (Reconciling the two quotes). Two quotes circulate on desks. The ratified symbol Cega is the per-correlation-point quote of Definition **??**; the gap-bump P&L ΔV_ω is a desk convention for a finite, convex move toward $\rho = 1$. The first-order conversion $\Delta V_\omega \approx 10(1-\rho)\,\mathrm{Cega}$ carries a factor $10(1-\rho)$ — equal to 5 at $\rho = 0.50$ — so the two are never interchangeable without it. Three cautions. (i) The bump is *one-sided*, always toward $\rho = 1$: it never measures the decorrelation tail the dispersion desk actually fears, and for a convexity-carrying correlation book it understates that tail; a symmetric $\pm$ bump pair is the honest convexity quote. (ii) The linearisation error $O\big(((1-\rho)/10)^2\,\partial^2 V/\partial\rho^2\big)$ in (**??**) grows toward low ρ, where the bump is largest ($(1-\rho)/10 = 0.2$ at $\rho = -1$) and where $\frac{\partial \sigma_B}{\partial \rho}$ varies fastest (Remark **??**). (iii) Inverting (**??**) as $\frac{\partial V}{\partial \rho} \approx 10\,\Delta V_\omega/(1-\rho)$ is exact only to first order in the bump; the exact per-point object is $\frac{\partial V}{\partial \rho} = 100\,\mathrm{Cega}$, by definition. Whenever a number travels between chapters it travels as Cega, on the contract basis.

Remark 1.24 (Who is long, who is short). By (**??**), anything long basket vol is long correlation: a long basket (or index) straddle has $\frac{\partial V}{\partial \sigma_B} > 0$ and hence Cega > 0. The dispersion package of Section **??** — long single-name vol, short basket vol — has Cega < 0: it collects the CVC spread and pays out when the names decorrelate. Cega is the desk's scorecard for that side.

1.6 Two re-marking conventions: sticky-ρ and sticky-CVC

The identity (**??**) is one equation in the four quantities $(\sigma_1, \sigma_2, \rho, \sigma_B)$; equivalently, by (**??**), in $(\sigma_1, \sigma_2, \rho, \xi)$. A quote set pins them at one instant, but when the single-name vols move the desk must decide what its marks *do*. Two conventions reproduce the current σ_B and differ only in their dynamics.

Definition 1.25 (sticky-ρ)**.** Under sticky-ρ, correlation is held fixed as vols move: $\mathrm{d}\rho = 0$, and ξ floats. The basket vol moves only through the vol arguments of (**??**):

$$\mathrm{d}\sigma_B \;=\; \frac{\partial \sigma_B}{\partial \sigma_1}\bigg|_\rho \mathrm{d}\sigma_1 + \frac{\partial \sigma_B}{\partial \sigma_2}\bigg|_\rho \mathrm{d}\sigma_2, \qquad \frac{\partial \sigma_B}{\partial \sigma_i}\bigg|_\rho = \frac{\sigma_i + \rho\,\sigma_j}{4\sigma_B}, \quad j \neq i. \tag{1.21}$$

(The partial follows in one line from (**??**) squared: $2\sigma_B\;\partial\sigma_B/\partial\sigma_i|_\rho = \frac{1}{2}\sigma_i + \frac{1}{2}\rho\,\sigma_j$ at fixed ρ.)

Definition 1.26 (sticky-CVC)**.** Under sticky-CVC, the CVC spread — equivalently, by Proposition **??**, the basket vol spread ξ at fixed spot and expiry — is held invariant: $\mathrm{d}\xi = 0$. The basket vol is pinned to the single-name average minus a fixed gap and ρ becomes the determined function $\rho(\sigma_1, \sigma_2, \xi)$ of (**??**):

$$\sigma_B = \tfrac{1}{2}(\sigma_1 + \sigma_2) - \xi, \qquad \mathrm{d}\sigma_B = \tfrac{1}{2}\,\mathrm{d}\sigma_1 + \tfrac{1}{2}\,\mathrm{d}\sigma_2, \qquad \mathrm{d}\rho = \frac{\partial \rho}{\partial \sigma_1}\bigg|_\xi \mathrm{d}\sigma_1 + \frac{\partial \rho}{\partial \sigma_2}\bigg|_\xi \mathrm{d}\sigma_2. \tag{1.22}$$

Remark 1.27 (The pair is exhaustive here, and the difference is the risk). For this corpus the two conventions are the whole menu: the shadow Greeks of the next chapter are exactly the difference between the sticky-CVC and sticky-ρ values of the corresponding total Greek, and they vanish identically under sticky-ρ (with $\mathrm{d}\rho = 0$ the correlation channel carries nothing). The choice between the two is the model risk of the two-name book: same marks today, different P&L tomorrow.

The size of that difference is visible already at first order. Take the anchor marks of Example **??** ($\sigma_1 = \sigma_2 = 20\%$, $\rho = 0.50$) and bump both single-name vols by +1 vol point. Under sticky-CVC, (**??**) moves the basket vol one for one: $\mathrm{d}\sigma_B = +1.00$ point. Under sticky-ρ, (**??**) gives, at equal vols, $\mathrm{d}\sigma_B/\mathrm{d}\sigma|_\rho = \sigma(1+\rho)/(2\sigma_B) = \sigma_B/\sigma = \sqrt{(1+\rho)/2} = 0.866$, so $\mathrm{d}\sigma_B = +0.87$ points. The missing 0.13 points of basket vol is what the sticky-CVC correlation re-mark supplies (correlation rises per (**??**)), and 0.13 points of basket vega P&L on every parallel vol move is precisely the kind of risk the direct Greeks at fixed ρ never show. Quantifying it is the business of the shadow Greeks.

1.7 The anchor example

Example 1.28 (Anchor numbers — the running example of Chapters 1–3)**.** Take $\sigma_1 = \sigma_2 = \sigma = 20\%$ and $\rho = 0.50$, expiry $T = 1$, and a book marked at Cega $= +0.2$ per correlation point (Definition **??**). The correlation mark is the whole threaded declaration: Chapters 2–3 consume $\frac{\partial V}{\partial \rho} = 20$ and the channel slope computed below, nothing else — the routed-sensitivities chapter re-declares the same mark as the differential parameter of its declared book class, a book carried through its correlation channel alone. For the vega translation of this section — and only there — read the mark through a *hypothetical basket-only carrier* (sole correlation channel σ_B, e.g. net long the basket straddle against single-name vol), so that (**??**) applies to it.

Basket vol and spread. Exactly, $\sigma_B = 0.20\sqrt{\frac{1}{2}(1+0.5)} = 0.20\sqrt{0.75} = 0.17321$ and $\xi = 0.20 - 0.17321 = 0.02679$ (2.68 vol points). The first-order $\xi \approx \sigma(1-\rho)/4 = 0.20 \times 0.50/4 = 0.025$ (2.5 points) is within 7% of exact and serves for estimates; the exact $\xi = 0.02679$ is used for the book.

Consistency check against (**??**)*.* At $\xi = 0.02679$: $\rho = 1 - 4(0.02679)/0.20 + 2(0.02679)^2/0.20^2 = 1 - 0.5358 + 0.0359 = 0.500$. The relation closes.

CVC spread. At $S = 100$ (with $F = S$ under zero rates, flagged), $\Pi \approx S\sqrt{T}\,\xi/\sqrt{2\pi} = 100 \times 0.3989 \times 0.02679 \approx 1.07$ per 100 of basket notional (Remark **??**); exact Black–Scholes under the geometric convention $C_B = C(S_B, \sigma_B)$ gives 1.06, and the true arithmetic-basket price is within a cent of that (Lemma **??**).

Correlation sensitivities. From (**??**), $\frac{\partial \sigma_B}{\partial \rho} = \sigma^2/(4\sigma_B) = 0.04/0.69282 = 0.05774$: one full unit of ρ is worth 5.77 basket vol points, so one correlation point (+0.01) moves σ_B by 0.058 points. By Definition **??**, exactly,

$$\frac{\partial V}{\partial \rho} \;=\; 100\,\mathrm{Cega} \;=\; 20, \qquad \frac{\partial V}{\partial \sigma_B} \;=\; \frac{\partial V}{\partial \rho}\Big/\frac{\partial \sigma_B}{\partial \rho} = \frac{20}{0.05774} = 346.4, \qquad \mathcal{V}_B = +3.46$$

per vol point — the second step is (**??**), legitimate for the basket-only carrier and for it alone; the number is this section's translation illustration, not a threaded mark, and no worked number in Chapters 2–3 consumes it. A *basket-only* book carrying Cega $= +0.2$ at these marks is implicitly long about $3\frac{1}{2}$ basket vega points. Under the desk's gap bump ($\rho^* = 0.55$), (**??**) gives $\Delta V_\omega \approx 10\,(1-\rho)\,\mathrm{Cega} = +1$ — first order in the bump size; any convexity $\partial^2 V/\partial \rho^2$ the book carries is not in this number.

The channel slope. From (**??**),

$$\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi} = \frac{4(0.02679)(0.20-0.02679)}{0.20^3} = \frac{4 \times 0.02679 \times 0.17321}{0.008} = +2.321:$$

under sticky-CVC, a +1 vol-point parallel move in the single names re-marks correlation by +2.3 correlation points. This slope, consumed against $\frac{\partial V}{\partial \rho} = 20$, is the input to the shadow vega and shadow delta of the next chapter; the same anchor reappears in the exchange-option chapter, where Lemma **??** gives the companion Margrabe vol $\Sigma_M = \sigma\sqrt{2(1-\rho)} = 0.20$ at $\rho = 0.50$.

The chapter's yield is a small set of exact objects — the identity (**??**), the spread (**??**) with its ladder (**??**), the relation (**??**) with its fixed-ξ slopes (**??**), and the Cega normalisation (**??**) — plus one modelling choice, sticky-ρ versus sticky-CVC. The next chapter takes the choice seriously: it routes vol and spot moves through $\rho(\sigma_1, \sigma_2, \xi)$ and prices what the direct Greeks leave out.

Quantity	Source	Anchor value	Units / basis
$\sigma_1 = \sigma_2 = \sigma$	given	20%	vol (absolute 0.20)
ρ	given	0.50	pure number
σ_B	(??)	17.32	vol points
ξ exact (first order)	(??), Thm ??	2.68 (2.50)	vol points
Π at $S = 100$, $T = 1$	(??)	≈ 1.07	per 100 notional
$\frac{\partial \sigma_B}{\partial \rho}$	(??)	0.0577	abs. vol per unit ρ (0.058 pts / corr pt)
Cega	given	+0.2	per correlation point (contract basis)
$\frac{\partial V}{\partial \rho} = 100\,\mathrm{Cega}$	(??)	20	per unit ρ
ΔV_ω	(??)	+1	gap bump $(1-\rho)/10$; first order
$\mathcal{V}_B$	(??)	+3.46	per vol point (basket-only carrier; illustration, not threade
$\mathrm{d}\rho/\mathrm{d}\sigma\|_\xi$	(??)	+2.321	unit ρ per unit σ (= 2.32 corr pts / vol pt)
Σ_M	Lemma ??	20%	vol (absolute 0.20)

Table 1.1: The anchor example, carried forward. Chapters 2 and 3 reuse these numbers on the stated bases: the routed-sensitivities chapter builds ShadowVega and ShadowDelta from $\frac{\partial V}{\partial \rho} = 100\,\mathrm{Cega} = 20$ and the channel slope +2.321 — the correlation mark and the map, the only rows the thread consumes; its declared book carries the mark through the correlation channel alone. The $\mathcal{V}_B$ row is the basket-only translation of Section **??**, an illustration Chapters 2–3 do not consume. The exchange-option chapter prices the spread book at $\Sigma_M = 20\%$ against the same correlation mark, but computes that book's own $\frac{\partial V}{\partial \rho}$ through the Σ_M channel of (**??**); it must not consume the basket book's 20.

Chapter 2

Routed Sensitivities II: Shadow Greeks and the Correlation Update Rule

A long basket (or index) straddle is *long correlation*: the basket volatility rises when the constituents correlate, so $\frac{\partial \sigma_B}{\partial \rho} > 0$ and the position gains as ρ marks up. The short-correlation side of the same trade is the dispersion book — long single-name volatility, short index volatility. Whichever side of it a desk holds, the position carries a risk that the direct Greeks do not show: when the single-name volatilities move, the implied correlation may be forced to move with them by the market's quoting convention, and the value change routed through that forced move is booked nowhere on a fixed-ρ risk report. This chapter derives those missing sensitivities — the *shadow Greeks* — and the finite re-marking rule that books them consistently.

The machinery is not new. Volume I, Chapter 3 states one routed-derivative operator: for an ordered chain of C^2 intermediates over a driver, the total first derivative is the gradient–velocity contraction and the total second derivative is the Hessian–velocity–velocity plus gradient–acceleration contraction, each intermediate's own total derivatives supplied by the same formula one level down. A specialization supplies exactly three items: the driver, the ordered links with their partial derivatives, and the partial derivatives of the value function. Volume I, Chapter 3 supplied the list (F, σ); this chapter supplies (F_i, σ_i, ρ) — the same operator, one link deeper:

$$S \longrightarrow F \longrightarrow \sigma \longrightarrow \rho \longrightarrow V.$$

Correlation is not a function of spot until a quoting convention closes the system. The convention studied here, sticky-CVC, declares the basket vol spread invariant; the implicit function theorem then determines ρ as a C^2 function of the single-name volatilities, appending one live link to the Volume I route. Under the alternative convention, sticky-ρ, the appended link is inert and every result of this chapter collapses to the Volume I route name by name. The shadow Greeks are, by definition, the difference between the two completed routes; the model risk of the chapter is the choice between them, and the shadow Greeks are its price.

The standing assumption of the corpus holds throughout: zero rates and zero dividends, the forward map carried generally as $F_i(S_i)$ and the collapse $F_i = S_i$ flagged at every point of use. One worked example — $\sigma = 20\%$, $\rho = 0.50$, spread $\xi = 2.68$ vol points exact (2.5 at first order), $\mathrm{Cega}_{\mathrm{gap}} = +1$, the book carried throughout as the constant-$\mathrm{Cega}_{\mathrm{gap}}$ class of Section **??** (one book model, declared once, used in every worked number) — is threaded through the entire chapter so that every number reconciles with every other.

2.1 The two-name setting: basket volatility, the spread, and Cega

Every symbol is fixed before it is used.

The basket object is Chapter 1's. The moment-matching basket volatility σ_B of the equal-weight two-name basket $S_B = \frac{1}{2}(S_1 + S_2)$ is Definition **??**, Eq. (**??**):

$$\sigma_B \;=\; \sqrt{\tfrac{1}{4}\sigma_1^2 + \tfrac{1}{4}\sigma_2^2 + \tfrac{1}{2}\rho\,\sigma_1\sigma_2}.$$

Equation (**??**) is the ratified *moment-matching definition* of σ_B, not the exact implied volatility of the arithmetic basket $\frac{1}{2}(S_1 + S_2)$, which is not lognormal; the arithmetic-versus-lognormal gap is quantified in Lemma **??** and absorbed by the error term of Proposition **??**.

The constituent volatilities in (**??**) are not free symbols: per the bridging identity of the notation contract, they are the single-name at-the-money volatilities,

$$\sigma_i \;\equiv\; \sigma_{\mathrm{ATM},i}(\ell_{F,i}, T), \qquad i = 1, 2, \tag{2.1}$$

evaluated at the current forward, where $\ell_{F,i} = \ln(F_i/F_{\mathrm{ref},i})$; the per-name symbols $\ell_{F,i}$ and $F_{\mathrm{ref},i}$ are the ones ratified by the contract's C1 note (the name index is the C1 note's, not a local extension of this chapter). Every derivative taken through σ_i in this chapter therefore inherits the Volume I surface dynamics — in particular the per-name ATM slope $\mathrm{vol.slope}_i = \frac{\partial \sigma_{\mathrm{ATM},i}}{\partial \ell_{F,i}}$ — through this identity. This is the single load-bearing hypothesis of the chapter: the correlation convention is a relation among *at-the-money* volatilities, so the only spot sensitivity that reaches ρ travels through $\sigma_{\mathrm{ATM},i}$.

The basket vol spread ξ is that of Definition **??**: with $\bar\sigma = \frac{1}{2}(\sigma_1 + \sigma_2)$ the maximal ($\rho = 1$) basket vol of (**??**), the boxed (**??**) sets $\xi := \bar\sigma - \sigma_B \geq 0$ — the diversification benefit of the basket, in vol points.

Remark 2.1 (No-arbitrage reading of $\xi \geq 0$). $\xi \geq 0$ is the upper rung of the two-name no-arbitrage ladder, Proposition **??** (**??**): a basket cannot be more volatile than the weighted average of its constituents, with equality iff $\rho = 1$. Thus ξ is a clean, sign-definite "how far from perfectly correlated" coordinate, and it is the natural state variable for the dynamics studied here. A mark that implies $\xi < 0$ is a free-money flag, not a market view; the clamp in the update rule of Section **??** enforces the bound. The price counterpart is exact for the *arithmetic-basket* form of the spread: with the basket leg priced as the arithmetic-basket call $\mathbb{E}\big(\frac{1}{2}(S_{1,T} + S_{2,T}) - S_B\big)^+$, payoff convexity gives the model-free bound $\frac{1}{2}C(S_1, \sigma_1) + \frac{1}{2}C(S_2, \sigma_2) - \mathbb{E}\big(\frac{1}{2}(S_{1,T} + S_{2,T}) - S_B\big)^+ \geq 0$ with no expansion (Proposition **??**; proof of Proposition **??**). The spread Π of Definition **??** subtracts the moment-matched lognormal call $C(S_B, \sigma_B)$ instead, so it inherits non-negativity only up to the moment-match error of Lemma **??**: $\Pi \geq -O\big(S(\sigma\sqrt{T})^3\big)$, not an exact sign.

Definition 2.2 (CVC spread: basket of calls versus call on the basket)**.** Fix an expiry $T > 0$; under zero rates, let $C(S, \sigma)$ be the at-the-money price (strike equal to the forward, which under zero rates is $F = S$, flagged) of a European call on an underlying of level S and implied volatility σ. The *CVC spread* is the price difference between a basket of single-name at-the-money calls and the at-the-money call on the basket,

$$\Pi \;:=\; w_1\, C(S_1, \sigma_1) + w_2\, C(S_2, \sigma_2) - C(S_B, \sigma_B). \tag{2.2}$$

Π is a locally scoped symbol of this chapter.

Lemma 2.3 (Moment matching absorbs the arithmetic/lognormal gap)**.** *With equal spots $S_1 = S_2 = S$ and lognormal legs carrying the stated vols and correlation, the at-the-money (strike equal to the common forward S) call on the arithmetic basket $\frac{1}{2}(S_1+S_2)$ and the Black–Scholes call $C(S,\sigma_B)$ on a lognormal underlying of volatility σ_B agree to $O\big(S(\sigma\sqrt{T})^3\big)$: both equal $S\sigma_B\sqrt{T}/\sqrt{2\pi} + O\big(S(\sigma\sqrt{T})^3\big)$.*

Proof. For any underlying B_T with forward $\mathbb{E}[B_T] = S$, the ATM call is $\mathbb{E}(B_T - S)^+ = \frac{1}{2}\mathbb{E}|B_T - S|$ (add and subtract $\frac{1}{2}\mathbb{E}(B_T - S) = 0$). For the arithmetic basket $B_T = \frac{1}{2}(S_{1,T} + S_{2,T})$ with lognormal legs, the variance is

$$\mathrm{Var}(B_T) = \tfrac{1}{4}S^2\big[(e^{\sigma_1^2 T} - 1) + (e^{\sigma_2^2 T} - 1) + 2(e^{\rho\sigma_1\sigma_2 T} - 1)\big] = S^2\sigma_B^2 T\,\big(1 + O(\sigma^2 T)\big)$$

by (**??**), and to the same relative order each leg is $S(1 + \sigma_i\sqrt{T}Z_i)$ with (Z_1, Z_2) jointly standard normal at correlation ρ, so $B_T - S$ is Gaussian up to an $O(\sigma^2 T)$-relative residual whose contribution to $\mathbb{E}|B_T - S|$ enters at the stated order. Hence $\frac{1}{2}\mathbb{E}|B_T - S| = \frac{1}{2}\sqrt{2/\pi}\,\mathrm{sd}(B_T)\,(1 + O(\sigma^2 T)) = S\sigma_B\sqrt{T}/\sqrt{2\pi} + O\big(S(\sigma\sqrt{T})^3\big)$. The lognormal call $C(S,\sigma_B)$ has the same expansion by the ATM Taylor expansion in the proof of Proposition **??**. □

Pricing $C(S_B, \sigma_B)$ in (**??**) as a lognormal call on the arithmetic level with the moment-matched vol is therefore legitimate exactly to the order carried below — this is the load-bearing step of the CVC–ξ equivalence, and it is an approximation with a stated order, not an identity.

Proposition 2.4 (The CVC spread equals the vol spread at leading order)**.** *With equal weights $w_1 = w_2 = \frac{1}{2}$ and equal spots $S_1 = S_2 = S$ (hence $S_B = S$), the CVC spread (**??**) satisfies*

$$\Pi \;=\; \frac{S\sqrt{T}}{\sqrt{2\pi}}\,\xi \;+\; O\big(S(\sigma\sqrt{T})^3\big), \qquad \sigma := \max(\sigma_1,\sigma_2), \tag{2.3}$$

*with ξ the basket vol spread (**??**). Consequently, at fixed spot and expiry, Π is constant if and only if ξ is constant, to this order. The leading term is non-negative since $\xi \geq 0$; the exact model-free inequality holds for the arithmetic-basket form of the spread (see the proof), while Π itself is non-negative only up to the $O\big(S(\sigma\sqrt{T})^3\big)$ moment-match error.*

Proof. At the money (strike equal to the forward; under zero rates, $F = S$, flagged) the Black–Scholes call price at implied volatility σ is

$$C(S,\sigma) = S\big[N(\tfrac{1}{2}\sigma\sqrt{T}) - N(-\tfrac{1}{2}\sigma\sqrt{T})\big] = S\big[2N(\tfrac{1}{2}\sigma\sqrt{T}) - 1\big],$$

with N the standard normal distribution function. Since $N'(0) = \varphi(0) = 1/\sqrt{2\pi}$ and $N''(0) = 0$, the Taylor expansion is $N(u) = \frac{1}{2} + u/\sqrt{2\pi} + O(u^3)$; with $u = \frac{1}{2}\sigma\sqrt{T}$,

$$C(S,\sigma) = \frac{S\sigma\sqrt{T}}{\sqrt{2\pi}} + O\big((\sigma\sqrt{T})^3\big)S.$$

Substituting into (**??**) with $w_1 = w_2 = \frac{1}{2}$ and $S_1 = S_2 = S_B = S$,

$$\Pi = \frac{S\sqrt{T}}{\sqrt{2\pi}}\Big[\tfrac{1}{2}(\sigma_1 + \sigma_2) - \sigma_B\Big] + O(\cdot) = \frac{S\sqrt{T}}{\sqrt{2\pi}}\,\xi + O(\cdot)$$

by (**??**), the identification of $C(S_B, \sigma_B)$ with the arithmetic-basket call being Lemma **??**, absorbed by the same error term. For the sign: with all three strikes at the common forward S, the payoffs satisfy

$$\big(\tfrac{1}{2}(S_{1,T} + S_{2,T}) - S\big)^+ \;\leq\; \tfrac{1}{2}(S_{1,T} - S)^+ + \tfrac{1}{2}(S_{2,T} - S)^+$$

pointwise, by convexity of $x \mapsto (x-K)^+$; taking expectations under any joint law reproducing the marks gives, with no expansion,

$$\tfrac{1}{2}C(S,\sigma_1) + \tfrac{1}{2}C(S,\sigma_2) \;\geq\; \mathbb{E}\big(\tfrac{1}{2}(S_{1,T}+S_{2,T}) - S\big)^+.$$

This is exact model-free non-negativity for the *arithmetic-basket* CVC spread, whose basket leg is the arithmetic-basket call. The spread Π of Definition **??** subtracts the moment-matched lognormal call $C(S,\sigma_B)$ instead, which equals the arithmetic-basket call only to $O\big(S(\sigma\sqrt{T})^3\big)$ (Lemma **??**), and that difference is not sign-pinned; hence for Π as defined, convexity delivers only $\Pi \geq -O\big(S(\sigma\sqrt{T})^3\big)$ — for tiny ξ the error term can dominate the leading term. The exact sign statement lives in vol space: $\xi \geq 0$ holds because $\rho \leq 1$ (Definition **??**), and the leading-order identity (**??**) ties the price and vol sign statements together only at that order. □

2.1.1 Conventions used throughout

- **Vol points.** Vols and the spread ξ are quoted in vol points; 1 vol point = 0.01. A per-unit-σ derivative is divided by 100 to express it *per vol point*. This is the source of every "/100".

- **Correlation bump (the "/10").** Correlation is bumped not by a fixed absolute amount but by a fixed fraction $\frac{1}{10}$ of its remaining distance to 1: $\rho^* = \rho + (1-\rho)/10$. This is a convex move toward 1 (it can never leave $[-1,1]$). The bump $(1-\rho)/10$ is a fixed fraction of the remaining gap, not a 0.10 absolute shift in ρ.

- **Spot normaliser (dollar delta).** A spot derivative $\frac{\partial V}{\partial S}$ is put on a cash footing by multiplying by the spot level: the *dollar delta* $S\,\frac{\partial V}{\partial S}$ is the cash value of the equivalent stock position, the value change for a unit relative move $\mathrm{d}S/S = 1$. The P&L of a 1% move is this divided by 100.

- **ATM vol slope.** vol.slope $= \frac{\partial \sigma_{\mathrm{ATM}}}{\partial \ell_F}$ is the slope of the at-the-money volatility in the forward log-level — how the ATM vol moves as the forward moves. It is the slope of the *one* ATM point as the forward moves, never the slope of the smile across strikes (that is $\frac{\partial \sigma_{\mathrm{smile}}}{\partial m}$, which plays no role in this chapter because the correlation convention sees only ATM vols, per (**??**)). For an equity surface vol.slope < 0: forward down, ATM vol up. Under zero rates, $F = S$ (flagged), the forward log-level and log-spot move one for one, $\mathrm{d}\ell_F = \mathrm{d}\ln S$ at fixed F_{ref}, so the slope may be read against log-spot without adjustment. On a term-structure footing the slope steepens at short tenors, empirically roughly as $1/\sqrt{T}$ — a stylised fact quoted for orientation, not a derived property, and not used quantitatively here; the worked example keeps $T = 1$.

- **Vega per vol point.** $\mathcal{V}_B = \frac{1}{100}\frac{\partial V}{\partial \sigma_B}$ is the basket vega per vol point. "Shadow" will always mean the extra piece routed through $\rho(\sigma)$, additive to and on the same normalised basis as the corresponding direct Greek: the shadow delta carries the spot normaliser S, the shadow vega the vol normaliser $1/100$. The one footing exception is the shadow gamma, which is the spot gradient of the *dollar* shadow delta and sits one factor of S away from the raw Γ_{adj}; the explicit conversion is stated at Definition **??** and additivity is only ever claimed after converting.

2.1.2 Correlation sensitivity: the Cega

The value sensitivity to correlation is quoted in two conventions: per correlation point — the contract's ratified Cega of Definition **??** — and per gap bump, the desk quotation this chapter works in under the locally scoped name $\mathrm{Cega}_{\mathrm{gap}}$ (Definition **??**). Both are expressed here through the position's basket vega, via Chapter 1's ρ-to-vol translation: by (**??**), $\frac{\partial \sigma_B}{\partial \rho} = \sigma_1\sigma_2/(4\sigma_B) > 0$, and by the chain rule (**??**) a position with basket vol sensitivity $\frac{\partial V}{\partial \sigma_B}$ (quoted vega $\mathcal{V}_B = \frac{1}{100}\frac{\partial V}{\partial \sigma_B}$) has $\frac{\partial V}{\partial \rho} = 100\,\mathcal{V}_B\,\sigma_1\sigma_2/(4\sigma_B)$. Equation (**??**) presumes the book reaches ρ only through σ_B: basket legs plus ρ-independent single-name legs. Books with direct ρ exposure not routed through σ_B — spread and exchange options, whose exercise variance carries $-2\rho\sigma_1\sigma_2$ — are outside this chapter's book class and are treated in their own chapter.

Definition 2.5 (Gap-bump Cega (locally scoped))**.** $\mathrm{Cega}_{\mathrm{gap}}$ is this chapter's desk quotation name for the gap-bump P&L of Chapter 1: $\mathrm{Cega}_{\mathrm{gap}} := \Delta V_\omega = V(\rho^*) - V(\rho)$ of Definition **??**, (**??**), under the gap bump $\rho^* = \rho + (1-\rho)/10$. $\mathrm{Cega}_{\mathrm{gap}}$ is a locally scoped symbol of this chapter. It is *not* the contract's ratified Cega, which is quoted per correlation point; the conversion between the two is stated once in Remark **??**, and no redefinition of the ratified symbol is made or implied. Wherever $\mathrm{Cega}_{\mathrm{gap}}$ and ρ are marked, the book's raw correlation slope is recovered as

$$\frac{\partial V}{\partial \rho} \;\approx\; \frac{10\,\mathrm{Cega}_{\mathrm{gap}}}{1-\rho}, \tag{2.4}$$

with equality exact only for books linear in ρ over the bump (Remark **??**).

Remark 2.6 (Units: the ratified Cega versus the local $\mathrm{Cega}_{\mathrm{gap}}$)*.* A *correlation point* is 0.01 of ρ; the contract's ratified Cega is the per-correlation-point sensitivity, $\mathrm{Cega} = \frac{1}{100}\frac{\partial V}{\partial \rho}$. The two quotations convert, to the first order of (**??**), by

$$\mathrm{Cega}_{\mathrm{gap}} \;\approx\; 10\,(1-\rho)\,\mathrm{Cega}, \tag{2.5}$$

stated once here and used wherever a per-point number is needed. The gap bump $(1-\rho)/10$ at the chapter's anchor $\rho = 0.50$ is 0.05, i.e. five correlation points, so $\mathrm{Cega}_{\mathrm{gap}} = +1$ there corresponds to $\mathrm{Cega} = +0.2$ per correlation point. The gap-bump quotation is the desk convention because it respects the boundary at $\rho = 1$: the same book re-quoted at higher ρ shows a smaller $\mathrm{Cega}_{\mathrm{gap}}$ even at unchanged $\frac{\partial V}{\partial \rho}$, which is exactly how much re-marking room remains. Both quotations appear on risk reports; this chapter works in $\mathrm{Cega}_{\mathrm{gap}}$ throughout and keeps the ratified per-point Cega untouched — re-ratifying the gap bump as the contract's Cega would require a FORMALIS amendment, and none is claimed.

Remark 2.7 (When the recovery (**??**) is exact)*.* Equation (**??**) inverts a finite bump into a derivative, so it is exact only for books linear in ρ over the bump. For the constant-$\mathrm{Cega}_{\mathrm{gap}}$ class of Section **??** — the book model of the worked thread — the identity $\frac{\partial V}{\partial \rho} = 10\,\mathrm{Cega}_{\mathrm{gap}}/(1-\rho)$ is adopted as the *definition* of the class parameter $\mathrm{Cega}_{\mathrm{gap}}$ (the differential quotation): for that book the finite gap-bump value differs from the parameter by bump convexity, $10\ln(10/9) \approx 1.054$ per unit, a $\approx 5\%$ quotation nuance. The threaded example declares $\mathrm{Cega}_{\mathrm{gap}} = +1$ as the differential parameter, so every closed form below that uses (**??**) as an identity does so under this declaration, not by assuming the bump away.

2.2 The exact correlation–volatility relation

Everything downstream rests on one identity, established once in Chapter 1 by solving (**??**) for ρ and substituting the spread: the boxed exact relation of Proposition **??**, Eq. (**??**),

$$\rho \;=\; 1 \;-\; 2\Big(\frac{1}{\sigma_1} + \frac{1}{\sigma_2}\Big)\xi \;+\; \frac{2\xi^2}{\sigma_1\sigma_2}, \qquad \sigma_B = \tfrac{1}{2}(\sigma_1+\sigma_2) - \xi.$$

Corollary 2.8 (Admissibility factorisation)**.** *Equation (??) factorises at both boundaries:*

$$1-\rho \;=\; \frac{2\xi\,(\sigma_1+\sigma_2-\xi)}{\sigma_1\sigma_2}, \qquad 1+\rho \;=\; \frac{2\,(\sigma_1-\xi)(\sigma_2-\xi)}{\sigma_1\sigma_2}. \tag{2.6}$$

Consequently every admissible mark ($\sigma_i>0$, $\rho\in[-1,1]$) satisfies

$$0 \;\le\; \xi \;\le\; \min(\sigma_1,\sigma_2), \tag{2.7}$$

with $\xi=0$ iff $\rho=1$ and $\xi=\min(\sigma_1,\sigma_2)$ iff $\rho=-1$: the sensitivity-degeneracy locus $\xi=\min(\sigma_1,\sigma_2)$ of Proposition ?? is exactly the $\rho=-1$ boundary.

Proof. Both identities in (??) are one-line rearrangements of (??). The band (??), with its attainment cases, is the ξ form of the two-name rungs, Proposition ?? (??); it is not re-proved here. Conversely, a ξ inside the band produces $\rho\in[-1,1]$ by (??), since both numerators are then non-negative. ☐

The equal-vol specialisation is Corollary ??: for $\sigma_1=\sigma_2=\sigma$, (??) gives the exact $1-\rho=2\xi(2\sigma-\xi)/\sigma^2$, and at leading order in the small quantity ξ/σ, $1-\rho\approx 4\xi/\sigma$. The converse first-order expansion of the spread is Theorem ??: expanding (??) about $1-\rho=0$ (Eq. (??)) gives $\xi=\frac{1}{4}\sigma(1-\rho)+O((1-\rho)^2)$ at equal vols and $\xi\approx\frac{1}{2}\,\sigma_1\sigma_2(1-\rho)/(\sigma_1+\sigma_2)$ for unequal vols.

Remark 2.9 (Validity). Equation (??) is exact for all (σ_1,σ_2,ξ) with $\sigma_i>0$ and is the backbone of everything below. The leading-order forms $1-\rho\approx 4\xi/\sigma$ and $\xi\approx\sigma(1-\rho)/4$ are equal-vol shortcuts valid near $\rho\approx 1$ (small ξ/σ); they are used only for intuition and quick estimates, never where the exact formula is needed. The worked thread below quantifies the error at $\rho=0.5$: about 7% — acceptable for a limit check, not for a mark.

Example 2.10 (Anchor numbers for the threaded example)**.** $\sigma_1=\sigma_2=\sigma=20\%$, $\rho=0.50$, both names quoted at their ATM points per (??). Exactly, $\sigma_B=0.20\sqrt{0.75}=0.173205$ and $\xi=0.20-0.173205=0.026795$ (2.68 vol points). The first-order $\xi\approx\sigma(1-\rho)/4=0.20\times 0.50/4=0.025$ (2.5 points) is within 7% of exact and serves for estimates; the book carries the exact $\xi=0.026795$. These numbers ($\sigma=20\%$, $\rho=0.5$, exact $\xi=2.68$ points, $\mathrm{Cega}_{\mathrm{gap}}=+1$) run through the whole chapter.

2.3 Sticky-CVC as a convention-completed route

The convention is stated precisely — in which variables the spread is held fixed, under which moves, and with which consequence for ρ — and identified as a convention-completed route in the sense of Volume I, Chapter 3.

Assumption 2.11 (sticky-CVC: the CVC spread is held constant)**.** As the volatility surface moves, the CVC spread (Definition ??) is held *invariant*; equivalently — to the leading order in $\sigma\sqrt{T}$ at which Proposition ?? identifies the two — at fixed spot and expiry the basket vol spread $\xi=\frac{1}{2}(\sigma_1+\sigma_2)-\sigma_B$ is held invariant. What the contract adopts as sticky-CVC, and what everything below uses, is the $\xi=$ const form; the identification with the traded option spread Π carries the $O((\sigma\sqrt{T})^3)$ caveat of Proposition ??. Concretely:

(i) The state variables that move are the single-name ATM vols (σ_1,σ_2) — driven either directly (a vega shock) or indirectly by spot through the per-name ATM slopes $\mathrm{vol.slope}_i$ (Section ??).

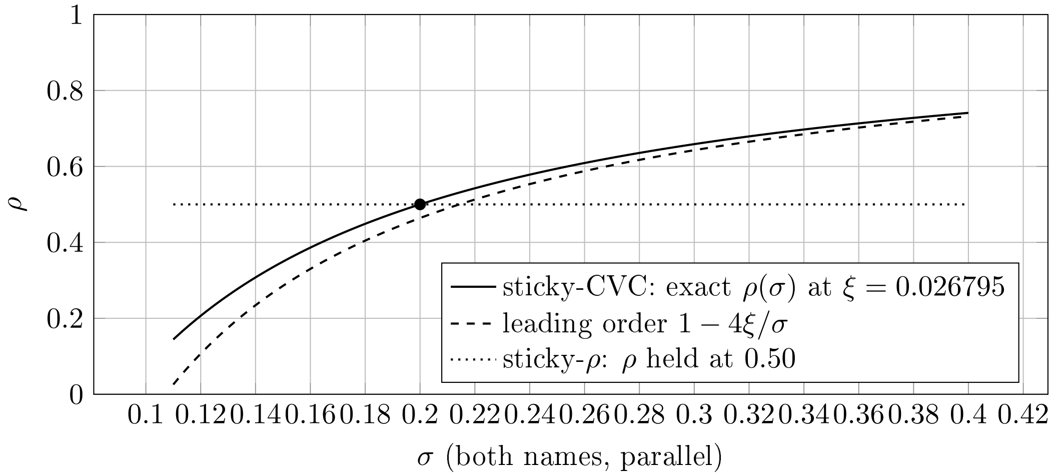

Figure 2.1: The correlation forced by the sticky-CVC convention as both single-name ATM vols move in parallel, at the frozen spread $\xi = 0.026795$ of Example **??** (dot: the anchor $\sigma = 20\%$, $\rho = 0.50$). The exact map (**??**) (solid) is concave and increasing: vols up, correlation up. The leading-order form (dashed) is accurate near $\rho \approx 1$ and understates ρ at the anchor by the 7% of Remark **??**. Under sticky-ρ (dotted) correlation does not respond at all; the vertical distance between the solid and dotted curves is what the shadow Greeks of Section **??** price.

(ii) Under any such move, $\mathrm{d}\xi = 0$.

(iii) Consequently the basket vol is pinned to the single-name average minus a fixed gap, and ρ is a determined function of the vols:

$$\sigma_B = \tfrac{1}{2}(\sigma_1 + \sigma_2) - \xi, \qquad \xi = \text{const}, \qquad \rho = \rho(\sigma_1, \sigma_2, \xi) \text{ via } (\textbf{??}). \tag{2.8}$$

Remark 2.12 (The convention closes the route). This is precisely a quoting convention in the sense of the convention-completed-route definition of Volume I, Chapter 3: the invariant is $g(\sigma_1, \sigma_2, \rho) = \bar{\sigma} - \sigma_B(\sigma_1, \sigma_2, \rho)$, and by (**??**),

$$\frac{\partial g}{\partial \rho} = -\frac{\partial \sigma_B}{\partial \rho} = -\frac{\sigma_1 \sigma_2}{4\sigma_B} \neq 0,$$

so the implicit function theorem determines $\rho = \rho(\sigma_1, \sigma_2, \xi)$ locally as a C^2 function of the vols — here globally and in closed form, by Proposition **??**. The correlation is thereby appended as one more link to the Volume I route $S_i \to F_i \to \sigma_i$, and every routed derivative of this chapter is an instance of the routed-derivative operator of Volume I, Chapter 3 applied to the completed route (F_i, σ_i, ρ). Under sticky-ρ the invariant is ρ itself, the appended link is inert, and the completed route collapses to the Volume I route name by name.

What the convention implies for surface motion. Differentiate $\sigma_B = \frac{1}{2}(\sigma_1 + \sigma_2) - \xi$ with $\mathrm{d}\xi = 0$:

$$\mathrm{d}\sigma_B = \tfrac{1}{2}\,\mathrm{d}\sigma_1 + \tfrac{1}{2}\,\mathrm{d}\sigma_2. \tag{2.9}$$

A parallel +1 vol-point shift $\mathrm{d}\sigma_1 = \mathrm{d}\sigma_2 = +0.01$ thus raises σ_B by exactly $+0.01$. To keep ξ fixed while σ_B tracks the average, implied correlation must rise. The next proposition quantifies this — it is the live link derivative that the completed route transmits.

Proposition 2.13 (The correlation link derivatives)**.** *Under Assumption **??**, the partial derivatives of $\rho(\sigma_1, \sigma_2, \xi)$ at fixed ξ are*

$$\left.\frac{\partial \rho}{\partial \sigma_1}\right|_{\xi} = \frac{2\xi}{\sigma_1^2}\Big(1 - \frac{\xi}{\sigma_2}\Big), \qquad \left.\frac{\partial \rho}{\partial \sigma_2}\right|_{\xi} = \frac{2\xi}{\sigma_2^2}\Big(1 - \frac{\xi}{\sigma_1}\Big), \tag{2.10}$$

each strictly positive for $0 < \xi < \sigma_j$ (j the other name). In the symmetric case $\sigma_1 = \sigma_2 = \sigma$ moving in parallel ($\mathrm{d}\sigma_1 = \mathrm{d}\sigma_2 = \mathrm{d}\sigma$), the total slope is their sum,

$$\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi} = \frac{4\xi}{\sigma^2}\Big(1 - \frac{\xi}{\sigma}\Big) = \frac{4\xi(\sigma - \xi)}{\sigma^3} \ > \ 0, \qquad \textit{leading order: } + \frac{4\xi}{\sigma^2}. \tag{2.11}$$

Proof. Differentiate (**??**) in σ_1 at fixed (σ_2, ξ):

$$\left.\frac{\partial \rho}{\partial \sigma_1}\right|_{\xi} = \frac{2\xi}{\sigma_1^2} - \frac{2\xi^2}{\sigma_1^2 \sigma_2} = \frac{2\xi}{\sigma_1^2}\Big(1 - \frac{\xi}{\sigma_2}\Big),$$

and symmetrically in σ_2. Positivity is immediate from $0 < \xi < \sigma_j$. For the symmetric parallel move set $\sigma_1 = \sigma_2 = \sigma$ and sum the two partials; equivalently, differentiate (**??**) directly: $\frac{\mathrm{d}\rho}{\mathrm{d}\sigma} = 4\xi/\sigma^2 - 4\xi^2/\sigma^3$. □

Remark 2.14 (Sign)*.* The slope (**??**) is *positive*: with the spread fixed, raising σ shrinks the relative gap $(1 - \rho) \approx 4\xi/\sigma$, pushing ρ toward 1. Intuitively, ξ is an absolute number of vol points; at a higher vol level the same absolute diversification benefit is a smaller relative one, and a smaller relative benefit means higher correlation. The degeneracy at $\xi \to \sigma_j$ — where the factor $(1 - \xi/\sigma_j)$ kills the sensitivity — is, by Corollary **??**, the $\rho = -1$ admissibility boundary itself; it is picked up again in the boundary conditions of Section **??**.

Remark 2.15 (The two conventions, and what the difference costs)*.* Two re-marking conventions reproduce the current σ_B and differ only in their dynamics; one fixes ρ, the other fixes ξ, and the remaining quantity moves:

- sticky-ρ: $\mathrm{d}\rho = 0$, ξ varies. Then σ_B moves only through the vol terms in (**??**) and the shadow Greeks of Section **??** vanish identically.
- sticky-CVC: $\mathrm{d}\xi = 0$, with ρ determined by (**??**).

Per the hook of Volume I, Chapter 3, the shadow Greeks are *defined* as the difference between the two completed routes; the pair is exhaustive for this corpus, and the choice between them is the model risk of the chapter. A desk that risk-manages under sticky-ρ while the market re-marks under sticky-CVC will find the difference in its unexplained P&L — that is where to look first when a vol move produces correlation P&L that the direct $\mathrm{Cega}_{\mathrm{gap}}$ bump did not predict.

2.4 Shadow vega and shadow delta

Once Assumption **??** makes ρ a function of the vols, the direct Greeks — direct delta $\frac{\partial V}{\partial S_i}$ and direct vega $\frac{\partial V}{\partial \sigma_B}$, both computed at fixed ρ — omit the value that moves *through* correlation. The omitted pieces are the terms $\frac{\partial V}{\partial \rho} \cdot$ (routed derivative of ρ); these are the shadow Greeks, and per the hook of Volume I, Chapter 3 they are exactly the difference between the sticky-CVC- and sticky-ρ-completed routes.

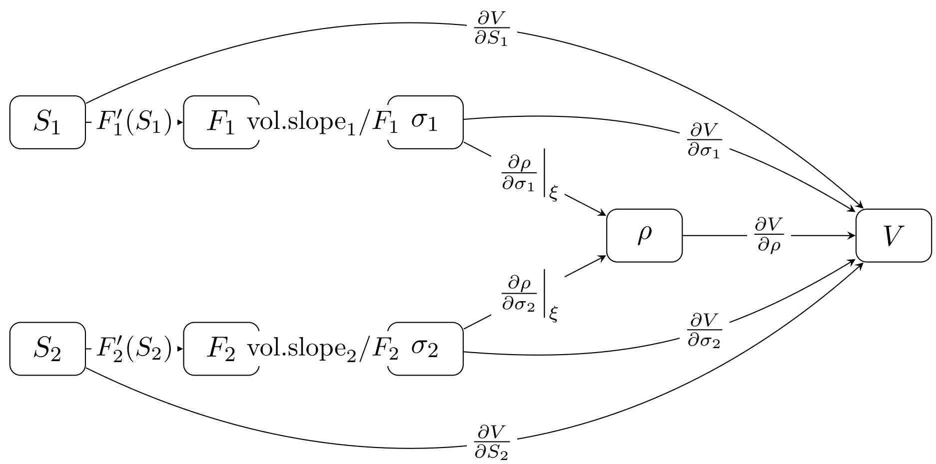

Figure 2.2: The convention-completed route of this chapter: the Volume I spot–volatility route, per name ($S_i \to F_i \to \sigma_i$, each $\sigma_i \equiv \sigma_{\mathrm{ATM},i}(\ell_{F,i}, T)$ per (**??**), link derivatives vol.slope$_i/F_i$ through the forward), with the correlation link appended by sticky-CVC (Remark **??**). Under zero rates, $F_i = S_i$ and $F_i'(S_i) = 1$ (flagged). Under sticky-ρ the two edges into ρ transmit nothing and the diagram collapses to the Volume I route name by name. The direct edges $\frac{\partial V}{\partial S_i}$, $\frac{\partial V}{\partial \sigma_i}$ carry the direct Greeks; the path through ρ carries the shadow Greeks.

2.4.1 Instantiating the interface

The routed-derivative operator of Volume I, Chapter 3 requires three items of a specialization:

(a) *Driver.* The spot S_i of a given name (for the symmetric worked thread: a common level S moving both names in parallel).

(b) *Ordered links.* (F_i, σ_i, ρ): the forward $F_i(S_i)$ with derivative $F_i'(S_i)$; the single-name ATM vol $\sigma_i = \sigma_{\mathrm{ATM},i}(\ell_{F,i}, T)$ with $\frac{\partial \sigma_i}{\partial F_i} = \text{vol.slope}_i/F_i$ (the Volume I moneyness link, specialised to the ATM point: the smile coordinate m plays no role because the convention sees only ATM vols, per (**??**)); and the correlation $\rho(\sigma_1, \sigma_2, \xi)$ with link derivatives (**??**).

(c) *Value partials.* $\frac{\partial V}{\partial \rho}$ from the book's marked Cega$_{\mathrm{gap}}$ via (**??**) (under the declaration of Remark **??**), alongside the direct $\frac{\partial V}{\partial S_i}$ and $\frac{\partial V}{\partial \sigma_i}$, which stay in their own (direct) buckets.

The operator then outputs the routed correlation derivative

$$\frac{\mathrm{d}\rho}{\mathrm{d}S} = \sum_{i=1}^{2} \frac{\partial \rho}{\partial \sigma_i}\bigg|_\xi \frac{\mathrm{d}\sigma_i}{\mathrm{d}S}, \qquad \frac{\mathrm{d}\sigma_i}{\mathrm{d}S} = \frac{\text{vol.slope}_i}{F_i} F_i'(S_i) \frac{\mathrm{d}S_i}{\mathrm{d}S}, \tag{2.12}$$

the chain carried through the forward as always: the correlation link composes with the forward links, it does not bypass them. Under zero rates, $F_i = S_i$, so $F_i'(S_i) = 1$ and the factor collapses to vol.slope$_i/S_i$ (flagged). In the symmetric parallel thread ($S_1 = S_2 = S$, $\sigma_1 = \sigma_2 = \sigma$, vol.slope$_1$ = vol.slope$_2$ = vol.slope, $\frac{\mathrm{d}S_i}{\mathrm{d}S} = 1$), the sum (**??**) collapses to

$$\frac{\mathrm{d}\rho}{\mathrm{d}S} = \frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\bigg|_\xi \cdot \frac{\text{vol.slope}}{S} \qquad \text{(under zero rates, } F = S\text{, flagged)}, \tag{2.13}$$

with $\frac{d\rho}{d\sigma}\Big|_\xi$ the parallel slope (??).

2.4.2 Shadow vega

Bumped: σ (+1 vol point, both names in parallel). **Held fixed:** the spread ξ (Assumption ??), spot, time. **Chain:** $\sigma \to \rho \to V$, via (??) then (??).

Definition 2.16 (Shadow vega). The *per-vol-point value change routed through correlation,*

$$\mathrm{ShadowVega} \;:=\; \frac{1}{100}\,\frac{\partial V}{\partial \rho}\,\frac{d\rho}{d\sigma}\Big|_\xi . \tag{2.14}$$

This is the normalised first-order difference between the two completed routes stated in the hook of Volume I, Chapter 3, now with every factor evaluable in closed form.

Proposition 2.17 (Shadow vega: exact and shortcut). *In the equal-vol case, using* $\frac{\partial V}{\partial \rho} = 10\,\mathrm{Cega_{gap}}/(1-\rho)$ *from (??) (exact under the constant-*$\mathrm{Cega_{gap}}$ *declaration of Remark ??) and the exact slope* $\frac{d\rho}{d\sigma}\Big|_\xi = 4\xi(\sigma-\xi)/\sigma^3$ *from (??),*

$$\mathrm{ShadowVega} \;=\; \frac{1}{100}\cdot\frac{10\,\mathrm{Cega_{gap}}}{1-\rho}\cdot\frac{4\xi(\sigma-\xi)}{\sigma^3} \;>\; 0 \quad \textit{for } \mathrm{Cega_{gap}} > 0. \tag{2.15}$$

At leading order in ξ/σ *this collapses to*

$$\mathrm{ShadowVega} \;\approx\; +\frac{\mathrm{Cega_{gap}}}{10\,\sigma}. \tag{2.16}$$

Proof. Insert $\frac{\partial V}{\partial \rho}$ and $\frac{d\rho}{d\sigma}\Big|_\xi$ into (??) and apply the 1/100 normaliser; that is (??). For the shortcut replace $1-\rho \to 4\xi/\sigma$ and $\sigma-\xi \to \sigma$: the two factors of $4\xi/\sigma$ cancel, leaving $\frac{dV}{d\sigma} \to 10\,\mathrm{Cega_{gap}}/\sigma$ and $\mathrm{ShadowVega} \to \mathrm{Cega_{gap}}/(10\sigma)$. The cancellation is *leading order in* ξ/σ, not exact: it uses the approximate $1-\rho$ in the denominator while the book's true $\frac{\partial V}{\partial \rho}$ uses the actual $1-\rho$. □

Intuition. Vol up ⇒ (spread fixed) correlation up ⇒ a long-correlation book ($\mathrm{Cega_{gap}} > 0$) gains. Hence the sign is *positive*: the book carries *extra long vega* through the correlation channel, additive to whatever direct (fixed-ρ) vega it marks. For the threaded book the correlation channel is the whole declaration — the direct vega is a separate mark, not derivable from $\mathrm{Cega_{gap}}$, and is not restated in the thread. One caution on bases: the one-for-one move (??) is the sticky-CVC *total* and already contains the correlation re-mark, so it must never be used to state the direct term — doing so double-counts the channel (Example ??).

Example 2.18 (Shadow vega at the anchor). $\mathrm{Cega_{gap}} = +1$, $\sigma = 20\%$, $\rho = 0.5$, $\xi = 0.026795$ (Example ??). The exact (??) uses the true gap $1-\rho = 0.5$ and the exact slope

$$\frac{d\rho}{d\sigma}\Big|_\xi = \frac{4(0.026795)(0.20-0.026795)}{0.20^3} = 2.321,$$

giving

$$\mathrm{ShadowVega} = \frac{1}{100}\cdot\frac{10}{0.5}\cdot 2.321 = \frac{1}{100}\cdot 20\cdot 2.321 = +0.464 \text{ per vol point.}$$

The shortcut (??) gives $\mathrm{Cega_{gap}}/(10\sigma) = 1/(10\times 0.20) = +0.500$. The two differ by $+0.036$ ($\approx 7.7\%$): the shortcut over-states because at $\rho = 0.5$ the leading-order $4\xi/\sigma = 0.536$ exceeds the true $1-\rho = 0.50$; near $\rho \approx 1$ the gap closes. A $+1$ vol-point parallel move thus adds about $+0.46$ of value through the correlation channel, *on top of* the direct vega in σ_B.

For scale — a hypothetical full re-mark, *not* part of the reconciling thread: the threaded book is declared through its correlation channel alone, so it quotes no direct basket vega, and its per-vol-point number for a parallel move *is* the shadow vega. If instead the same $\frac{\partial V}{\partial \rho} = 20$ were carried by the *basket-only* carrier of Chapter 1's translation illustration (V a function of σ_B alone, a book *outside* this chapter's class per Proposition **??**), then $\frac{\partial \sigma_B}{\partial \rho} = \sigma_1\sigma_2/(4\sigma_B) = 0.04/0.69282 = 0.05774$ would translate the mark into a direct basket vega of $\mathcal{V}_B = \frac{1}{100}\frac{\partial V}{\partial \rho}/\frac{\partial \sigma_B}{\partial \rho} = \frac{1}{100} \times 20/0.05774 = 3.464$ per *basket* vol point. That book reconciles differently: at fixed ρ a parallel $+1$-point single-name move lifts σ_B by only $\sqrt{(1+\rho)/2} = 0.866$ points (Chapter 1's sticky-ρ partials), direct P&L $3.464 \times 0.866 = 3.00$; the correlation re-mark supplies the missing 0.134 basket points, worth $3.464 \times 0.134 = 0.46$ — the shadow vega again; and the sticky-CVC total is $3.00 + 0.46 = 3.46 = \mathcal{V}_B \times 1$, the one-for-one move of (**??**). Adding the shadow vega *on top of* $\mathcal{V}_B$ quoted against that one-for-one move — "3.464+0.464 = 3.93" — would double-count the channel, a 13% over-statement of the vega and a mis-sized hedge. No worked number in this chapter's thread is sourced from the basket-only carrier.

2.4.3 Shadow delta

The shadow vega routed a vol move into ρ; the shadow delta routes a *spot* move into ρ. The CVC invariant is a relation among at-the-money vols, so ρ is determined by the ATM single-name vols through (**??**), and the only spot sensitivity that reaches ρ is the response of the ATM vol to the forward: the per-name slope $\text{vol.slope}_i = \frac{\partial \sigma_{\text{ATM},i}}{\partial \ell_{F,i}}$, transmitted through the forward link as in (**??**). For an equity surface $\text{vol.slope} < 0$ — forward down, ATM vol up.

Remark 2.19 (The slope is an empirical input). vol.slope is the realised move of the ATM vol per unit of forward log-level; it carries the entire spot-to-correlation channel. If the ATM vol does not respond to the forward ($\text{vol.slope} = 0$), the shadow delta vanishes — see the limits in Section **??**. The reference numerical engine of the corpus encodes exactly that convention ($\sigma_{\text{ATM}} = \text{const}$, i.e. $\text{vol.slope} = 0$): every shadow-delta and shadow-gamma figure in this chapter is therefore *invisible* to that engine, and no worked number below is sourced from it; all are derived analytically here under the stated vol.slope.

Bumped: spot S (both names in parallel). **Held fixed:** ξ. **Chain:** $S \to F \to \sigma \to \rho \to V$, via $\text{vol.slope}/F$, then (**??**), then (**??**).

Definition 2.20 (Shadow delta). The *dollar (cash) delta routed through correlation*, on the spot normaliser S:

$$\text{ShadowDelta} := S \, \frac{\partial V}{\partial \rho} \, \frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\bigg|_{\xi} \, \frac{\mathrm{d}\sigma}{\mathrm{d}S}. \tag{2.17}$$

Proposition 2.21 (Closed form; the normaliser-ratio identity). *With the routed vol derivative* $\frac{\mathrm{d}\sigma}{\mathrm{d}S} = (\text{vol.slope}/F)\,F'(S)$*, which under zero rates,* $F = S$*, collapses to* $\text{vol.slope}/S$ *(flagged), the factor* S *from the normaliser cancels the* $1/S$*, leaving*

$$\text{ShadowDelta} = \text{vol.slope}\,\frac{\partial V}{\partial \rho}\,\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\bigg|_{\xi} = \text{vol.slope}\,\frac{\mathrm{d}V}{\mathrm{d}\sigma}\bigg|_{\xi} = 100\,\text{vol.slope} \cdot \text{ShadowVega}. \tag{2.18}$$

Proof. Insert $\frac{\mathrm{d}\sigma}{\mathrm{d}S} = \text{vol.slope}/S$ (zero-rate collapse $F = S$, flagged) into (**??**):

$$\text{ShadowDelta} = S\,\frac{\mathrm{d}V}{\mathrm{d}\sigma}\bigg|_{\xi} \cdot \frac{\text{vol.slope}}{S} = \text{vol.slope}\,\frac{\mathrm{d}V}{\mathrm{d}\sigma}\bigg|_{\xi} = 100\,\text{vol.slope} \cdot \text{ShadowVega},$$

since ShadowVega $= \frac{1}{100}\,\frac{\mathrm{d}V}{\mathrm{d}\sigma}\big|_{\xi}$ by (**??**). □

The factor 100 is the ratio of the two normalisers — the dollar delta is per unit relative move while the shadow vega is per vol point — and (**??**) is the normaliser-ratio identity of the notation contract. The leading-order shortcut is

$$\text{ShadowDelta} \;\approx\; 100\,\text{vol.slope}\cdot\frac{\text{Cega}_{\text{gap}}}{10\,\sigma} \;=\; \frac{10\,\text{vol.slope}\cdot\text{Cega}_{\text{gap}}}{\sigma}. \tag{2.19}$$

Remark 2.22 (Vanna in disguise). Structurally the shadow delta is a cross-Greek running $S \to \sigma \to \rho$: it lives where vanna ($\frac{\partial^2 V}{\partial S\,\partial\sigma}$) lives on the risk report and behaves like it — largest where $|\text{vol.slope}|$ is large (short tenors, per the empirical $1/\sqrt{T}$ steepening flagged in Section **??** — a stylised fact, not a derived scaling) and where Cega_{gap} is large (low ρ, wide dispersion). The shadow delta is additive to the dollar delta, the shadow vega to the direct fixed-ρ vega, each on the basis of its direct Greek.

Intuition. Spot down ⇒ (negative slope) vols up ⇒ (spread fixed) correlation up ⇒ a long-correlation book ($\text{Cega}_{\text{gap}} > 0$) gains on the down move. Gaining on down moves is, by definition, a *negative* delta. So with vol.slope < 0 the shadow delta is negative: the book is effectively *long downside* — a sell-off helps, a rally hurts — and the stock hedge that neutralises it is a *buy*.

Example 2.23 (Shadow delta at the anchor)**.** $\text{Cega}_{\text{gap}} = +1$, $\sigma = 20\%$, $\rho = 0.5$, $\xi = 0.026795$, vol.slope $= -0.25$ (a 25 vol-point ATM fall per unit forward log-level, both names; an illustrative short-tenor equity slope — steeper than the vol.slope $= -0.04$ used for the single-name illustration of Volume I, Chapter 3, and declared per name here), $T = 1$. Using the exact ShadowVega $= +0.464$ from Example **??**,

$$\text{ShadowDelta} = 100\,\text{vol.slope}\cdot\text{ShadowVega} = 100\times(-0.25)\times 0.464 = -11.6.$$

The shortcut (**??**) gives $10\times(-0.25)\times 1/0.20 = -12.5$. The sign is *negative*: the long-correlation book carries a dollar shadow delta of about -11.6 — a short-stock exposure through the correlation channel, additive to the direct dollar delta. It gains when spot *falls* and loses when spot *rises*; left unhedged, it shows up as delta P&L the direct report cannot explain.

Remark 2.24 (Additive corrections). Both shadow Greeks are $\frac{\partial V}{\partial\rho}\times$ (routed derivative of ρ) terms added to a direct Greek at fixed units. The totals are the dollar delta + ShadowDelta and the direct fixed-ρ vega + ShadowVega (per parallel vol point), each summed on the basis of its direct Greek — the direct vega stated at fixed ρ, never as $\mathcal{V}_B$ against the one-for-one sticky-CVC basket move (**??**), which already contains the correlation re-mark (Example **??**). This is the total-derivative structure of the Volume I routed delta, extended one further link to ρ — the first-order formula of the routed-derivative operator applied to the completed route, nothing more.

2.5 Shadow gamma: the second-order route

The shadow delta (**??**) depends on spot through the ATM vol $\sigma = \sigma_{\text{ATM}}(\ell_F, T)$ and, in turn, through $\rho = \rho(\sigma)$. Its spot derivative is the *shadow gamma* — the second-order output of the routed-derivative operator on the completed route, i.e. the Hessian–velocity–velocity plus gradient–acceleration contraction of Volume I, Chapter 3, restricted to the correlation channel.

Definition 2.25 (Shadow gamma)**.** The spot derivative of the dollar shadow delta,

$$\text{ShadowGamma} \;:=\; \frac{\mathrm{dShadowDelta}}{\mathrm{d}S}, \tag{2.20}$$

a locally scoped name of this chapter. Its footing must be stated precisely, because it is *not* that of the contract's Γ_{adj}: ShadowDelta is a cash number, so ShadowGamma carries units of value per unit spot (dollar-delta change per unit spot; its integral over a spot move is dollar-delta change), whereas Γ_{adj} is a raw second derivative with units of value per spot squared. The raw-footing shadow gamma, the object additive to Γ_{adj}, is

$$\Gamma^{\text{sh}} \;:=\; \frac{1}{S}\Big(\text{ShadowGamma} - \frac{\text{ShadowDelta}}{S}\Big), \tag{2.21}$$

by the identity $S\,\frac{\partial^2 V}{\partial S^2} = \frac{\mathrm{d}}{\mathrm{d}S}\big(S\frac{\partial V}{\partial S}\big) - \frac{\partial V}{\partial S}$ applied to the correlation channel (exact for the book class of Proposition **??**, whose mixed cross terms vanish). Adding ShadowGamma to Γ_{adj} without this conversion mixes normalisers and is a booking error.

The second-order route consumes the curvature of the correlation link.

Lemma 2.26 (Concavity of the correlation map)**.** *In the equal-vol parallel case, at fixed ξ,*

$$\left.\frac{\mathrm{d}^2\rho}{\mathrm{d}\sigma^2}\right|_{\xi} \;=\; -\frac{8\xi}{\sigma^3} + \frac{12\xi^2}{\sigma^4} \;=\; -\frac{4\xi(2\sigma - 3\xi)}{\sigma^4} \;<\; 0 \quad \textit{for } \xi < \tfrac{2}{3}\sigma. \tag{2.22}$$

Proof. Differentiate the exact parallel slope $\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi} = 4\xi/\sigma^2 - 4\xi^2/\sigma^3$ once more in σ at fixed ξ. □

The map $\sigma \mapsto \rho$ at fixed spread is increasing and *concave*: each further vol point buys less correlation than the last, because the same absolute spread keeps shrinking in relative terms at a decreasing rate. This concavity fixes the sign of the shadow gamma.

Proposition 2.27 (Shadow gamma: closed form)**.** *Assume the equal-vol parallel setting with* vol.slope *locally constant in spot, and write $R = \left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi}$. In full generality,*

$$\text{ShadowGamma} \;=\; \frac{\text{vol.slope}^2}{S}\left[\frac{\partial^2 V}{\partial \rho^2}\,R^2 + \frac{\partial V}{\partial \rho}\,\left.\frac{\mathrm{d}^2\rho}{\mathrm{d}\sigma^2}\right|_{\xi}\right] \;+\; \text{vol.slope}\,R\left(\frac{\partial^2 V}{\partial S\,\partial\rho} + \frac{\text{vol.slope}}{S}\,\frac{\partial^2 V}{\partial\sigma\,\partial\rho}\right) \tag{2.23}$$

(under zero rates, $F = S$, flagged, so that $\frac{\mathrm{d}\sigma}{\mathrm{d}S} = \text{vol.slope}/S$). Assume further the book class of this chapter*: $\frac{\partial V}{\partial \rho}$ is a function of ρ alone, so the mixed cross-partials vanish, $\frac{\partial^2 V}{\partial S\,\partial\rho} = \frac{\partial^2 V}{\partial\sigma\,\partial\rho} = 0$. Then the last bracket drops and*

$$\text{ShadowGamma} \;=\; \frac{\text{vol.slope}^2}{S}\left[\frac{\partial^2 V}{\partial\rho^2}\left(\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi}\right)^2 + \frac{\partial V}{\partial\rho}\,\left.\frac{\mathrm{d}^2\rho}{\mathrm{d}\sigma^2}\right|_{\xi}\right]. \tag{2.24}$$

For a book whose Cega_{gap} *is locally constant in ρ — so that $\frac{\partial V}{\partial\rho} = 10\,\text{Cega}_{\text{gap}}/(1-\rho)$ holds as an identity in ρ (Remark **??**), placing the book in the class by construction, and differentiates to $\frac{\partial^2 V}{\partial\rho^2} = 10\,\text{Cega}_{\text{gap}}/(1-\rho)^2$ — this becomes*

$$\text{ShadowGamma} \;=\; \frac{10\,\text{Cega}_{\text{gap}}\,\text{vol.slope}^2}{S\,(1-\rho)}\left[\frac{\big(\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi}\big)^2}{1-\rho} + \left.\frac{\mathrm{d}^2\rho}{\mathrm{d}\sigma^2}\right|_{\xi}\right], \tag{2.25}$$

and at leading order in ξ/σ,

$$\text{ShadowGamma} \;\approx\; -\frac{10\,\text{Cega}_{\text{gap}}\,\text{vol.slope}^2}{S\,\sigma^2} \;<\; 0 \quad \textit{for } \text{Cega}_{\text{gap}} > 0. \tag{2.26}$$

Proof. By (??), ShadowDelta = vol.slope $\frac{\partial V}{\partial \rho} R$ carries no explicit S: the spot dependence enters through $\sigma(S)$, with $\frac{\mathrm{d}\sigma}{\mathrm{d}S} = \text{vol.slope}/S$ (zero-rate collapse $F = S$, flagged). Spot reaches $\frac{\partial V}{\partial \rho}$ through all three of its arguments — directly, through σ, and through $\rho(\sigma)$ — and reaches R through σ:

$$\frac{\mathrm{d}}{\mathrm{d}S}\left[\frac{\partial V}{\partial \rho}\right] = \frac{\partial^2 V}{\partial S\,\partial \rho} + \frac{\partial^2 V}{\partial \sigma\,\partial \rho}\,\frac{\text{vol.slope}}{S} + \frac{\partial^2 V}{\partial \rho^2}\,R\,\frac{\text{vol.slope}}{S}, \qquad \frac{\mathrm{d}R}{\mathrm{d}S} = \left.\frac{\mathrm{d}^2\rho}{\mathrm{d}\sigma^2}\right|_{\xi}\frac{\text{vol.slope}}{S}.$$

With vol.slope constant, the product rule gives (??). Under the class hypothesis the two mixed cross-partials vanish — for the constant-Cega_{gap} book this holds by construction, since $\frac{\partial V}{\partial \rho} = 10\,\text{Cega}_{\text{gap}}/(1-\rho)$ depends on nothing but ρ — leaving (??); substituting the constant-Cega_{gap} forms of $\frac{\partial V}{\partial \rho}$ and $\frac{\partial^2 V}{\partial \rho^2}$ gives (??). A basket straddle is *not* in the class ($\frac{\partial V}{\partial \rho} = \frac{\partial V}{\partial \sigma_B}\frac{\partial \sigma_B}{\partial \rho}$ inherits direct σ_B- and spot-dependence), and for such books the mixed terms of (??) must be carried. For the shortcut substitute the leading-order $\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi} \approx 4\xi/\sigma^2$, $\mathrm{d}^2\rho/\mathrm{d}\sigma^2\big|_{\xi} \approx -8\xi/\sigma^3$, and $1-\rho \approx 4\xi/\sigma$: the bracket becomes $(16\xi^2/\sigma^4)/(4\xi/\sigma) - 8\xi/\sigma^3 = 4\xi/\sigma^3 - 8\xi/\sigma^3 = -4\xi/\sigma^3$, the prefactor becomes $10\,\text{Cega}_{\text{gap}}\,\text{vol.slope}^2\sigma/(4\xi S)$, and the product is (??). □

Remark 2.28 (Where the sign comes from — and for which books it holds). The two contributions to the leading-order bracket are $+4\xi/\sigma^3$ from *value convexity* (under constant Cega_{gap}, the book's $\frac{\partial V}{\partial \rho}$ steepens as ρ rises toward the boundary) and $-8\xi/\sigma^3$ from *map concavity* (Lemma ??). For the constant-Cega_{gap} class the concavity term dominates two to one and fixes the sign: the shadow gamma is *negative* for $\text{Cega}_{\text{gap}} > 0$. The two-to-one ratio is a property of that class — it comes from $\frac{\partial^2 V}{\partial \rho^2} = 10\,\text{Cega}_{\text{gap}}/(1-\rho)^2$ — not a universal law. What *is* slope-independent is the $\text{vol.slope}^2 > 0$ prefactor: whatever the bracket's sign, it is unchanged by the orientation of the surface. Outside the class the sign is conditional: it holds whenever $\frac{\partial^2 V}{\partial \rho^2} R^2$ does not exceed $-\frac{\partial V}{\partial \rho}\;\mathrm{d}^2\rho/\mathrm{d}\sigma^2\big|_{\xi}$. For a long basket straddle $\frac{\partial^2 V}{\partial \rho^2} < 0$ (vega roughly constant while $\frac{\partial \sigma_B}{\partial \rho}$ falls as ρ rises), so the value term is itself negative and the two bracket contributions *reinforce* — though for the straddle the mixed terms of (??) are also live and must be signed before the total is called; the two-to-one story in any case disappears. Conversely, a book with a large enough positive $\frac{\partial^2 V}{\partial \rho^2}$ flips the shadow gamma positive. Short-gamma behaviour in the correlation channel of a long-correlation book is therefore the constant-Cega_{gap} and long-straddle result, not a theorem for every $\text{Cega}_{\text{gap}} > 0$ book.

Example 2.29 (Shadow gamma at the anchor). $\text{Cega}_{\text{gap}} = +1$, $\sigma = 20\%$, $\rho = 0.5$, $\xi = 0.026795$, $\text{vol.slope} = -0.25$, $T = 1$, and $S = 100$; the book is the constant-Cega_{gap} thread book, in class per Proposition ??, so the mixed terms of (??) are zero. With the exact slope $\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi} = 2.320$ (Example ??) and $\mathrm{d}^2\rho/\mathrm{d}\sigma^2\big|_{\xi} = -4(0.026795)(0.40 - 0.080385)/0.20^4 = -21.41$ (Lemma ??), the bracket of (??) is

$$\frac{2.320^2}{0.5} - 21.41 = 10.76 - 21.41 = -10.65,$$

and the prefactor is $10 \times 1 \times 0.0625/(100 \times 0.5) = 0.0125$, so

$$\text{ShadowGamma} = 0.0125 \times (-10.65) = -0.133.$$

The shortcut (**??**) gives $-10 \times 1 \times 0.0625/(100 \times 0.04) = -0.156$. Both are negative; the shortcut overstates the magnitude at $\rho = 0.5$, exactly as for the shadow vega, because $4\xi/\sigma = 0.536$ exceeds the true $1 - \rho = 0.50$.

Remark 2.30 (Relation to the shadow delta). Combining the two shortcuts, ShadowGamma $\approx$ $-$vol.slope $\cdot$ ShadowDelta$/(S\sigma)$. For equity vol.slope < 0, so $-$vol.slope$/(S\sigma) > 0$ and the shadow gamma carries the sign of the shadow delta: a long-correlation book ($\mathrm{Cega}_{\mathrm{gap}} > 0$) with a negative shadow delta also has a negative shadow gamma — its dollar shadow delta grows more negative as spot rises. Short-gamma behaviour confined to the correlation channel: the hedge against it bleeds in exactly the way a short-gamma stock hedge does, and the desk should budget the re-hedging cost accordingly.

Remark 2.31 (Spot-dependent vol slope). If vol.slope = vol.slope(S), the product rule applied to (**??**) adds one term to (**??**):

$$\text{ShadowGamma} = \frac{\text{vol.slope}^2}{S}\left[\frac{\partial^2 V}{\partial\rho^2}\Big(\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\Big|_{\xi}\Big)^2 + \frac{\partial V}{\partial\rho}\,\frac{\mathrm{d}^2\rho}{\mathrm{d}\sigma^2}\Big|_{\xi}\right] \;+\; \frac{\mathrm{dvol.slope}}{\mathrm{d}S}\,\frac{\text{ShadowDelta}}{\text{vol.slope}},$$

the last term being the spot-curvature of the slope itself (in the Volume I notation, it carries $\frac{\partial^2\sigma_{\mathrm{ATM}}}{\partial\ell_F^2}$ through the forward). It vanishes whenever the ATM slope is locally constant in spot, the regime assumed in Proposition **??** and in the worked thread.

Remark 2.32 (The mixed cross terms are shadow, not direct). The full second-order expansion of the completed route generates, besides the two terms of (**??**), mixed terms in which one differentiation is direct and one is routed: $\big(\frac{\partial^2 V}{\partial S\,\partial\rho} + \frac{\partial^2 V}{\partial\sigma\,\partial\rho}\,\frac{\mathrm{d}\sigma}{\mathrm{d}S}\big)\frac{\mathrm{d}\rho}{\mathrm{d}S}$ contributions, the last bracket of (**??**). Each carries a factor $\frac{\mathrm{d}\rho}{\mathrm{d}S}$, the forced motion of ρ; under sticky-ρ that factor is zero, so these terms *vanish* with the convention and are shadow terms by the chapter's own definition. They do *not* belong to the direct vanna/volga buckets: those are fixed-ρ objects of the Volume I engine and cannot contain a term proportional to $\mathrm{d}\rho$. (A term proportional to $\mathrm{d}\rho$ cannot "survive under sticky-ρ"; only the purely direct cross-partials at fixed ρ do, and those are already in the Volume I buckets.) The mixed terms drop out of the worked thread only because the constant-$\mathrm{Cega}_{\mathrm{gap}}$ book satisfies the class hypothesis of Proposition **??**; for books outside the class — the basket straddle among them — they must be carried explicitly via (**??**), or a second-order correlation P&L term is booked nowhere. With that accounting, the shadow gamma (through the raw-footing conversion (**??**)) is exactly the difference between the two conventions at second order, consistent with the definition of the shadow Greeks as a difference of completed routes.

2.6 The correlation update rule

The shadow Greeks give the differential sensitivity; the update rule is the finite, non-perturbative re-mark of ρ when the vols change. It is Assumption **??** plus Proposition **??** applied discretely.

Proposition 2.33 (sticky-CVC correlation re-mark)**.** *Given current marks $(\sigma_1, \sigma_2, \rho)$:*

1. ***Freeze the spread.*** *Compute once $\xi = \frac{1}{2}(\sigma_1 + \sigma_2) - \sigma_B(\sigma_1, \sigma_2, \rho)$ from* (**??**) *(or invert* (**??**)*), and hold it.*

2. ***On a vol move to (σ_1', σ_2'), re-set correlation to***

$$\rho' \;=\; 1 \;-\; 2\Big(\frac{1}{\sigma_1'} + \frac{1}{\sigma_2'}\Big)\xi \;+\; \frac{2\xi^2}{\sigma_1'\sigma_2'} \tag{2.27}$$

(this is (??) at the new vols with the old ξ), then ***clamp*** $\rho' \leftarrow \max\bigl(\min(\rho', 1), -1\bigr)$. *Equal-vol shortcut:* $\rho' \approx 1 - 4\xi/\sigma'$.

3. ***Freeze-violation event.*** *A binding clamp on either side means, by Corollary ??, that the frozen ξ has left the admissible band $[0, \min(\sigma_1', \sigma_2')]$: the convention's invariant can no longer be maintained. The post-clamp mark implies a spread different from the frozen one, so the event must be booked as a $\mathrm{d}\xi \neq 0$ re-mark on the separate P&L line of the boundary conditions below — never absorbed silently into the correlation mark.*

Proof. By Assumption ??, ξ is invariant; by Proposition ??, ρ' is the unique value consistent with $(\sigma_1', \sigma_2', \xi)$ through the basket identity (??). Re-evaluating (??) at the new vols with the old ξ is therefore the unique re-set consistent with the convention wherever the frozen ξ remains admissible, $0 \le \xi \le \min(\sigma_1', \sigma_2')$ (Corollary ??). The clamps are guards, not part of the convention. The upper one is nearly vacuous: for a frozen $0 < \xi < \sigma_1' + \sigma_2'$, $1 - \rho' = 2\xi(\sigma_1' + \sigma_2' - \xi)/(\sigma_1'\sigma_2') > 0$ automatically, so $\rho' < 1$ without any clamp; it can bind only in the extreme collapse $\xi \ge \sigma_1' + \sigma_2'$. The operative guard is the *lower* one: a vol fall that lands the frozen ξ strictly between min and max of the new vols makes $(\sigma_1' - \xi)(\sigma_2' - \xi) < 0$ in (??) and (??) would otherwise emit an inadmissible $\rho' < -1$ as a mark. In either case a binding clamp is the freeze-violation event of step 3: the frozen-ξ invariant is inconsistent with admissibility, and the excess is booked as $\mathrm{d}\xi \neq 0$, not hidden in the clamp. □

Why re-set rather than bump. Equation (??) is exact for *any* size of move — exact, that is, against the $\xi = $ const convention itself, which is what the contract's sticky-CVC adopts. Its identification with the traded CVC option spread of Definition ?? is leading order in $\sigma\sqrt{T}$ (Proposition ??), so for long-dated or high-vol baskets the $O((\sigma\sqrt{T})^3)$ term should be re-checked before the re-set is trusted as a mark rather than a limit check. For small moves it linearises to

$$\Delta\rho \approx \sum_i \left.\frac{\partial\rho}{\partial\sigma_i}\right|_{\xi} \Delta\sigma_i, \tag{2.28}$$

with the link derivatives (??). This is the bridge between the two halves of the chapter: the shadow Greeks of Section ?? are the linearisation of (??), so the first-order P&L of the re-mark equals the shadow-Greek P&L, and the second-order correction is priced by the same $\mathrm{d}^2\rho/\mathrm{d}\sigma^2$-plus-value-convexity machinery as the shadow gamma of Section ?? — for a vol move, without the vol.slope^2 factor (a *shadow volga*; see Example ??); for a spot move, the shadow gamma itself. The equal-vol sum $\left.\frac{\partial\rho}{\partial\sigma_1}\right|_{\xi} + \left.\frac{\partial\rho}{\partial\sigma_2}\right|_{\xi} = (4\xi/\sigma^2)(1 - \xi/\sigma)$ recovers (??).

Remark 2.34 (Regularity). $\left.\frac{\partial\rho}{\partial\sigma_i}\right|_{\xi} > 0$ if and only if $\xi < \sigma_j$ (j the other name). For $\xi < \sigma_2$, raising σ_1 raises ρ, consistent with (??). By Corollary ?? the locus $\xi = \min(\sigma_1, \sigma_2)$ where the sensitivity dies is precisely the $\rho = -1$ admissibility boundary, so "the re-mark stops responding" and "the mark is about to leave $[-1, 1]$" are the same event.

Boundary conditions.

- **Upper clamp** $\rho' \le 1$ (equivalently $\xi \ge 0$): the no-arbitrage bound of Remark ??. For a frozen $\xi > 0$ it is automatic (proof of Proposition ??) — it can bind only for a $\xi = 0$ book sitting at $\rho = 1$ or in the extreme collapse $\xi \ge \sigma_1' + \sigma_2'$. At the clamp the book pins: further vol rises produce no correlation P&L, and the shadow Greeks computed just below the boundary overstate the remaining risk — expiry-style pin behaviour, in correlation space.

- **Lower clamp** $\rho' \geq -1$ (equivalently $\xi \leq \min(\sigma_1', \sigma_2')$, Corollary **??**): the boundary that actually binds in a vol collapse. The convention freezes ξ as an *absolute* number of vol points, so a vol fall drives $\sigma_B = \bar{\sigma}' - \xi$ toward zero and ρ' toward -1: in the equal-vol thread, $\sigma' = 3\%$ at the frozen $\xi = 0.026795$ already gives $\rho' = -0.98$, and $\rho' = -1$ is reached exactly at $\sigma' = \xi$ ($\sigma_B = 0$). Below that the frozen ξ is inadmissible and the update is the freeze-violation event of Proposition **??**: re-mark ξ, book $\mathrm{d}\xi \neq 0$.

- **Degeneracy at $\xi \to \sigma_j$:** the factor $(1 - \xi/\sigma_j)$ in (**??**) tends to 0, so $\left.\frac{\partial \rho}{\partial \sigma_i}\right|_{\xi} \to 0$ and the re-mark becomes insensitive to σ_i. Wide-dispersion, low-correlation books sit in this regime; their sticky-CVC risk dies out even as their $\mathrm{Cega}_{\mathrm{gap}}$ is largest. By Corollary **??** this locus is the $\rho = -1$ boundary itself: the degeneracy and the lower admissibility bound are one phenomenon, seen from the sensitivity and from the mark.

- **P&L decomposition:** the re-mark (**??**) is attributable to the vol move at fixed ξ; a variation $\mathrm{d}\xi \neq 0$ (a genuine re-pricing of the dispersion premium, including every freeze-violation event above) is a separate, additional P&L line. Keeping the two lines separate is what makes the unexplained residual chaseable.

Example 2.35 (The update rule, tying the chapter together)**.** Start at the anchor: $\sigma_1 = \sigma_2 = \sigma = 20\%$, $\rho = 0.50$, exact frozen spread $\xi = 0.026795$ (Example **??**).

+1 point. $\sigma' = 21\%$, ξ frozen, equal-vol re-set (**??**):

$$\rho' = 1 - \frac{4(0.026795)}{0.21} + \frac{2(0.026795)^2}{0.21^2} = 1 - 0.5103 + 0.0326 = 0.522,$$

so $\Delta\rho = +0.022$: +1 vol point $\Rightarrow$ +2.2 correlation points, correlation up as (**??**) requires. (The leading-order $\Delta\rho \approx (4\xi/\sigma^2)\Delta\sigma$ with the rounded $\xi = 2.5$ points gives +0.025.)

+2 points, reconciled with the shadow vega. The threaded book is the constant-$\mathrm{Cega}_{\mathrm{gap}}$ book of Proposition **??**, declared once for the whole chapter: $\frac{\partial V}{\partial \rho} = 10\,\mathrm{Cega}_{\mathrm{gap}}/(1-\rho)$ as an identity in ρ, i.e. $V(\rho) = -10\,\mathrm{Cega}_{\mathrm{gap}} \ln(1-\rho) + \text{const}$, with $\mathrm{Cega}_{\mathrm{gap}} = +1$ the differential parameter of Remark **??**; hence $\frac{\partial^2 V}{\partial \rho^2} = 10\,\mathrm{Cega}_{\mathrm{gap}}/(1-\rho)^2 = 40$ at the anchor, not zero. Bump to $\sigma' = 22\%$. Exact re-set: $1 - \rho' = 2(0.026795)(0.44 - 0.026795)/0.22^2 = 0.4575$, so $\rho' = 0.5425$, $\Delta\rho = +0.0425$. The exact finite re-mark for this book is the integral of its correlation slope, not the anchor slope times the step:

$$\Delta V \;=\; \int_{\rho}^{\rho'} \frac{10\,\mathrm{Cega}_{\mathrm{gap}}}{1-u}\,\mathrm{d}u \;=\; 10\,\mathrm{Cega}_{\mathrm{gap}} \ln\frac{1-\rho}{1-\rho'} \;=\; 10 \ln\frac{0.5}{0.4575} \;=\; +0.888 \;\approx\; +0.89$$

(the anchor-slope linearisation $20 \times 0.0425 = +0.85$ would under-state it by the value convexity $\frac{1}{2}\frac{\partial^2 V}{\partial \rho^2}(\Delta\rho)^2 = \frac{1}{2} \times 40 \times 0.0425^2 = +0.036$ plus higher order — for this book +0.85 is first-order in V, not exact). Three individually correct P&L numbers for the move arise:

(a) **Leading order:** shortcut $\mathrm{ShadowVega} = +0.500$ (Eq. (**??**)), $\times 2 = +1.00$.

(b) **Exact instantaneous** (the differential shadow vega): exact $\mathrm{ShadowVega} = +0.464$ (Eq. (**??**), Example **??**), $\times 2 = +0.93$.

(c) **Exact finite re-set:** $\Delta V = +0.89$.

Two gaps, two distinct causes:

$$\underbrace{1.00 - 0.93 = +0.07}_{\text{leading-order substitution bias (survives at zero step)}} \qquad \underbrace{0.93 - 0.89 = +0.04}_{\text{genuine } O(\Delta\sigma^2) \text{ curvature}},$$

total +0.11. The first gap is the 7.7% shortcut bias of Example **??** and does not shrink with the step; the second is exactly what the second-order machinery prices — *both* terms of the bracket of Proposition **??**, here driven by a vol move rather than a spot move:

$$\tfrac{1}{2}\left[\frac{\partial^2 V}{\partial\rho^2}\Big(\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\Big|_\xi\Big)^2 + \frac{\partial V}{\partial\rho}\,\frac{\mathrm{d}^2\rho}{\mathrm{d}\sigma^2}\Big|_\xi\right](\Delta\sigma)^2 = \tfrac{1}{2}\big[40\times 2.3205^2 + 20\times(-21.41)\big](0.02)^2 = \tfrac{1}{2}\,(215.4 - 428.2)(0.0004) = -0.043,$$

the same value-convexity (+0.043) and map-concavity (−0.086) pieces as the shadow-gamma bracket of Example **??**, without the vol.slope^2 prefactor because the driver here is a vol move, not a spot move — a *shadow volga*, not the shadow gamma itself. It accounts for the −0.04 to within an $O(\Delta\sigma^3)$ residual of +0.002. A desk that books the exact re-set and explains with the exact shadow Greeks chases only that last residual.

2.7 Coherence, signs, limits, and desk usage

2.7.1 How the three results cohere

(1) sticky-CVC **frames the move.** Assumption **??** fixes ξ, which pins $\sigma_B = \frac{1}{2}(\sigma_1 + \sigma_2) - \xi$ and forces $\rho = \rho(\sigma_1, \sigma_2, \xi)$ via (**??**), rising in the vols (Proposition **??**). In route language: the convention completes the route, appending the correlation link to the Volume I chain (Remark **??**).

(2) $\rho(\sigma)$ **is the channel.** Its positive slope $\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\big|_\xi > 0$ is precisely what creates the shadow Greeks: ShadowVega $= \frac{1}{100}\,\frac{10\,\mathrm{Cega_{gap}}}{1-\rho}\,\frac{4\xi(\sigma-\xi)}{\sigma^3}$ ($\approx \mathrm{Cega_{gap}}/(10\sigma)$ near $\rho = 1$) and ShadowDelta $=$ 100 vol.slope · ShadowVega, each additive to its direct Greek. At the anchor: ShadowVega $=$ +0.464, ShadowDelta $= -11.6$, ShadowGamma $= -0.133$.

(3) **The update rule books it.** Proposition **??** is (1)+(2) applied *discretely*: freeze ξ, re-evaluate ρ at the new vols, clamp to $[-1, 1]$ (a binding clamp being a freeze-violation event, booked as $\mathrm{d}\xi \neq 0$). Its linearisation (**??**) is the intraday first-order form, and the resulting P&L is precisely the shadow-Greek number (Example **??**), with the second-order residual priced by the same $\mathrm{d}^2\rho/\mathrm{d}\sigma^2$-plus-value-convexity curvature machinery as the shadow gamma — the shadow volga of Example **??** for a vol move; the shadow gamma itself, with its vol.slope^2 prefactor, for a spot move.

2.7.2 The risk report at the anchor

The threaded example assembles into one table — the way the book should be read. All numbers at $\sigma = 20\%$, $\rho = 0.50$, $\xi = 0.026795$, $\mathrm{Cega_{gap}} = +1$, vol.slope $= -0.25$, $T = 1$, $S = 100$; under zero rates, $F = S$ (flagged), throughout the spot-routed lines.

Bucket	Direct (fixed ρ)	Shadow (sticky-CVC)	Total
Vega (per vol pt, parallel)	—	ShadowVega $= +0.464$	$+0.464$ (additive to direct f
Dollar delta (corr channel)	—	ShadowDelta $= -11.6$	-11.6 (additive to direct do
Gamma (corr channel, dollar-Δ/spot)	—	ShadowGamma $= -0.133$	raw equiv. $\Gamma^{\mathrm{sh}} = -1.7 \times 10$
$\mathrm{Cega}_{\mathrm{gap}}$ (gap bump, local)	$+1$	—	$+1$

The direct fixed-ρ vega, the direct dollar delta and the direct total gamma Γ_{adj} of the two single-name legs are the Volume I objects — computed name by name along the route $S_i \to F_i \to \sigma_i \to V$ with the full smile machinery — and are not restated here: the threaded book is declared through its correlation channel alone, so it pins none of them, and in particular no direct vega number (such as the basket-only $\mathcal{V}_B = 3.464$ of the hypothetical aside in Example **??**) belongs in this table; the shadow lines are what this chapter adds on top, and they are additive on the stated normalised bases. The gamma line requires its stated conversion: ShadowGamma is dollar-delta change per unit spot (Definition **??**), while Γ_{adj} is a raw $\frac{\partial^2 V}{\partial S^2}$; the number additive to Γ_{adj} is $\Gamma^{\mathrm{sh}} = (\mathrm{ShadowGamma} - \mathrm{ShadowDelta}/S)/S = (-0.133 + 0.116)/100 = -1.7 \times 10^{-4}$ per (**??**). Summing -0.133 directly onto Γ_{adj} would mix normalisers.

2.7.3 Signs and limits

Signs. $\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi} > 0$ by (**??**); ShadowVega > 0 for $\mathrm{Cega}_{\mathrm{gap}} > 0$ by (**??**); ShadowDelta $= 100\,\mathrm{vol.slope} \cdot$ ShadowVega < 0 for $\mathrm{Cega}_{\mathrm{gap}} > 0$ under vol.slope < 0 by (**??**); ShadowGamma < 0 for $\mathrm{Cega}_{\mathrm{gap}} > 0$ regardless of the sign of vol.slope, by Proposition **??** — a statement about the constant-$\mathrm{Cega}_{\mathrm{gap}}$ class; for books outside it (the basket straddle included, whose mixed terms in (**??**) are live) the sign is conditional, per Remark **??**.

Limit $\rho \to 1$. Then $\xi \to 0$, and the clamped re-mark of Proposition **??** yields zero incremental P&L for an up-move in σ, since ρ sits at the boundary. The differential shadow vega has a finite limit: in (**??**) the factors $\left.\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\right|_{\xi} \sim 4\xi/\sigma^2$ and $1/(1-\rho) \sim \sigma/(4\xi)$ are reciprocal, so

$$\lim_{\rho\to 1} \mathrm{ShadowVega} = \frac{\mathrm{Cega}_{\mathrm{gap}}}{10\,\sigma}, \qquad \lim_{\rho\to 1} \mathrm{ShadowDelta} = \frac{10\,\mathrm{vol.slope} \cdot \mathrm{Cega}_{\mathrm{gap}}}{\sigma}.$$

The differential and the clamped-discrete views disagree at the boundary — the differential Greek is one-sided there, and risk systems should report it as such.

Limit vol.slope $\to 0$. ShadowDelta $= 100\,\mathrm{vol.slope} \cdot$ ShadowVega $\to 0$ and ShadowGamma $\to 0$ (it carries $\mathrm{vol.slope}^2$); the shadow vega survives, since a vega shock reaches ρ without touching spot. This is the regime the reference numerical engine encodes ($\sigma_{\mathrm{ATM}} = \mathrm{const}$): any spot-channel shadow number produced under that convention is identically zero and must be flagged as "computed under $\sigma_{\mathrm{ATM}} = \mathrm{const}$", never quoted as a property of the market.

Constraints. Admissibility is the band $0 \le \xi \le \min(\sigma_1, \sigma_2)$ of Corollary **??**. Its lower end in ξ ($\rho \le 1$, i.e. $\sigma_B \le \bar{\sigma}$) is the no-arbitrage bound of Remark **??**; its upper end in ξ ($\rho \ge -1$) is the one that binds in a vol collapse at frozen ξ and is guarded by the lower clamp of Proposition **??**, booked as a freeze-violation event when it binds. The sensitivity degeneracy at $\xi \to \sigma_j$ (Remark **??**) is that same boundary seen from the Greeks; an exogenous variation $\mathrm{d}\xi \neq 0$ lies outside Assumption **??** and contributes value changes additional to those above.

2.7.4 What the model assumes away

The chapter's results are exact within their assumptions; a desk should know which frictions those assumptions exclude before running the book on them.

- **Zero rates and dividends.** Carried throughout; every $F = S$ (per name, $F_i = S_i$) collapse is flagged at point of use. With carry, the forward links $F_i'(S_i)$ in (**??**) are no longer unity and the shadow delta picks up the same forward-map corrections as the Volume I total delta; dividend surprises hit $\ell_{F,i}$ directly and re-mark σ_i, hence ρ, without any spot move.

- **Correlation is a mark, not a screen price.** ρ here is implied from basket versus single-name vol quotes; there is no liquid two-name correlation market to close the loop. The update rule is a *marking* discipline — its P&L is real only to the extent the basket and single-name vol markets it is implied from are tradable at the marks. Bid/ask on the vol legs translates directly into a band around ξ, hence around every shadow number.

- **Hedging the shadow delta** means carrying stock against a correlation view; the hedge is cheap to run, but it converts correlation model risk (sticky-ρ vs sticky-CVC) into realised delta P&L. The negative shadow gamma of Section **??** means the hedge bleeds on large two-way spot moves — budget it like any short-gamma position.

- **Equal weights, two names, ATM.** The formulas use the equal-weight two-name basket and ATM quotes; the CVC–ξ equivalence (Proposition **??**) is leading-order in $\sigma\sqrt{T}$ and degrades for long-dated or high-vol baskets, where the $O((\sigma\sqrt{T})^3)$ term is worth checking before the invariant is trusted.

- **The vol-crush regime.** Freezing ξ as an *absolute* number of vol points is the fragile leg of the convention when vols collapse: economically the diversification benefit should shrink with the vol level, but the frozen-absolute ξ instead drives $\sigma_B = \bar{\sigma}' - \xi$ toward zero and ρ' toward -1, hitting the lower clamp at $\sigma' = \xi$ in the equal-vol case (boundary conditions of Section **??**). A desk running sticky-CVC through a crush should either re-mark ξ explicitly (booked as $d\xi \neq 0$ on its own P&L line) or switch to a relative spread $\xi/\bar{\sigma}$, which degrades gracefully; the frozen-absolute convention stops being defensible once the vols are within a small multiple of ξ.

The choice sticky-ρ versus sticky-CVC is the residual model risk, and it is irreducible within this chapter: both conventions reproduce today's marks and differ only in dynamics. The shadow Greeks are exactly the difference between the two views — they vanish under sticky-ρ — so a book that carries them explicitly is carrying its convention risk where it can be seen, hedged, and argued about, rather than in the unexplained residual.

Chapter 3

Spread and Exchange Options: The Absolute Spread under Margrabe

The payoff is $|S_1 - S_2|$ at expiry: a straddle on the spread between two tradables. Before any formula, the risk profile. The position is *long Margrabe volatility* $\Sigma_M = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$ — the volatility of the ratio S_1/S_2 — and therefore *short correlation*: as $\rho \to 1$ with equal vols, $\Sigma_M \to 0$ and the optionality dies. It dies expensively: as $\Sigma_M \to 0$ the price collapses while gamma and the correlation sensitivity per unit of position blow up like $1/\Sigma_M$ at finite T. Theta does *not* join them: the decay is the rank-one gamma times the Margrabe variance, and the Σ_M^2 in the variance beats the $1/\Sigma_M$ in the gamma, so theta *vanishes* proportionally to Σ_M (Remark **??**) — a nearly worthless position pinned on the spread, with unbounded gamma and correlation risk but almost no decay left to collect against them. There is no free money in that pairing: the realised spread variance dies like Σ_M^2, so the harvestable gamma P&L vanishes with the theta; what stays unbounded is the exposure to the *marks*. The short-correlation tail, not just the expiry tail, is where this book gets hurt. It is the mirror image of the basket straddle of Chapters 1–2, which is long correlation; a book running both is a dispersion-style position, and the correlation risk nets across them. The deltas are bounded, the entire gamma lives in a single direction (the spread), and the vega *per leg* can be short even though the payoff is an option: raising the low-vol name's volatility can cheapen the position when correlation is high.

Hedging is delta in both names against a rank-one gamma; the residual risks are the ones the spot hedge cannot touch: Σ_M-vega, and the correlation mark. Correlation is not directly tradable here, so how the desk *re-marks* ρ when single-name vols move is a real P&L convention, not a formality. This chapter prices the payoff in a manifestly symmetric form, derives the spot Greeks, and then computes the vol derivatives under both re-marking conventions of this volume: sticky-ρ ($\mathrm{d}\rho = 0$) and sticky-CVC ($\mathrm{d}\xi = 0$, the basket vol spread held invariant). The comparison is exact: at equal vols the sticky-CVC sensitivity is a precise fraction of the sticky-ρ one, $1/\big(1 + \sqrt{(1+\rho)/2}\big) \in [1/2, 1]$; with mismatched vols the ratio can leave that band, and sticky-CVC can *amplify* the vega in magnitude (Remark **??**). The chapter closes the worked correlation example threaded through this volume ($\sigma = 20\%$, $\rho = 0.50$, $\xi = 0.0268$): the same anchor numbers, pushed through the spread option, reproduce the shadow-Greek machinery of Chapter 2 with the correlation exposure flipped in sign.

3.1 Setup and notation

Assumption 3.1 (Margrabe setup)**.** Let $S_1, S_2 > 0$ be two tradables and let $T > 0$ be the time to expiry. Under the standing assumption of zero rates and zero dividends the forwards collapse to spot, $F_i = S_i$ (flagged here at point of use; S_i is written throughout this chapter). The model inputs are constant single-name volatilities $\sigma_1, \sigma_2 > 0$ and a constant instantaneous correlation $\rho \in [-1, 1]$. Assume further $\Sigma_M > 0$ (see Definition **??**), excluding the degenerate point $\rho = 1$ with $\sigma_1 = \sigma_2$: there the ratio S_1/S_2 is riskless, the payoff is worth its intrinsic $V = |S_1 - S_2|$, and that value is recovered from (**??**) by direct limit as $\Sigma_M \to 0$. Every expression below carrying $1/\omega_M$ or $1/\Sigma_M$ — the gamma matrix, theta, both vega sets, the correlation sensitivity — is read on this domain. (Under sticky-CVC with $\xi > 0$ the exclusion is automatic: see the note after Proposition **??**.)

Remark 3.2 (Where the surface enters). The constants σ_i are not free parameters: by the bridging identity C1 they are the single-name ATM volatilities, $\sigma_i \equiv \sigma_{\text{ATM},i}(\ell_{F,i}, T)$ with $\ell_{F,i}$ the i-th name's forward log-level, evaluated at the current forwards ($F_i = S_i$ under zero rates, flagged). The Margrabe model freezes them over the pricing horizon and carries no smile — σ_{smile} risk on either name is outside this model. The surface dynamics of Volume I re-enter exactly through the question this chapter answers — how the marks move when the $\sigma_{\text{ATM},i}$ move — and the sticky-ρ and sticky-CVC conventions decide it.

Definition 3.3 (Margrabe effective volatility)**.** The effective volatility entering the exchange-option formula is

$$\Sigma_M := \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}. \tag{3.1}$$

Define also

$$\omega_M := \Sigma_M \sqrt{T}, \tag{3.2}$$

the root total Margrabe variance over the horizon.

Definition 3.4 (Symmetric normal variables)**.** Define

$$a := \frac{\ln(S_1/S_2)}{\omega_M} + \frac{\omega_M}{2}, \qquad b := \frac{\ln(S_2/S_1)}{\omega_M} + \frac{\omega_M}{2}. \tag{3.3}$$

Equivalently $b = -\ln(S_1/S_2)/\omega_M + \omega_M/2$, so the relabelling $1 \leftrightarrow 2$ exchanges a and b. φ and N denote the standard normal pdf and cdf.

Two facts fix the scale. At $S_1 = S_2$, $a = b = \omega_M/2$, and the price below reduces to $2S\big(2N(\omega_M/2) - 1\big) \approx 0.8\, S\, \omega_M$ for small ω_M — the ATM-straddle approximation, with ω_M in the role of total vol. And Σ_M at equal vols is $\sigma\sqrt{2(1-\rho)}$: at $\sigma = 20\%$, $\rho = 0.5$, that is exactly 20% again. These two lines price the anchor example of Section **??** to first order.

3.2 Symmetric price of the payoff $|S_1 - S_2|$

Proposition 3.5 (Symmetric Margrabe formula)**.** *The time-t price of the payoff $|S_1 - S_2|$ at expiry is*

$$V = S_1\big(2N(a) - 1\big) + S_2\big(2N(b) - 1\big). \tag{3.4}$$

Proof. Decompose

$$|S_1 - S_2| = (S_1 - S_2)^+ + (S_2 - S_1)^+. \tag{3.5}$$

The first term is the Margrabe exchange option. Take S_2 as numéraire: $(S_1 - S_2)^+ = S_2\,(S_1/S_2 - 1)^+$, so its price is S_2 times the price of a call struck at $K = 1$ on the ratio forward $F = S_1/S_2$, a martingale under the S_2-numéraire measure with volatility Σ_M — the vol of $\ln(S_1/S_2)$ from (**??**). This is the change-of-numéraire derivation of Margrabe (1978, *The value of an option to exchange* *one asset for another*, J. Finance 33(1), 177–186); zero rates make the reduction frictionless. The contract objects d_1, d_2 (owner I.1, with $m = \ln(K/F)$), evaluated at $(K, F, \sigma) = (1,\, S_1/S_2,\, \Sigma_M)$ — so $m = -\ln(S_1/S_2)$ — read

$$d_1 = -\frac{m}{\Sigma_M\sqrt{T}} + \frac{\Sigma_M\sqrt{T}}{2} = \frac{\ln(S_1/S_2)}{\omega_M} + \frac{\omega_M}{2} = a, \qquad d_2 = d_1 - \omega_M = a - \omega_M; \tag{3.6}$$

these are the same contract symbols, not local redefinitions. The Black formula then gives

$$C_{12} := \text{price of } (S_1 - S_2)^+ = S_2\big[\tfrac{S_1}{S_2}N(d_1) - N(d_2)\big] = S_1N(a) - S_2N(a - \omega_M). \tag{3.7}$$

Relabelling $1 \leftrightarrow 2$ (which exchanges a and b by Definition **??**), the second exchange option prices as

$$C_{21} := \text{price of } (S_2 - S_1)^+ = S_2N(b) - S_1N(b - \omega_M). \tag{3.8}$$

From (**??**),

$$a - \omega_M = -b, \qquad b - \omega_M = -a. \tag{3.9}$$

Hence

$$V = C_{12} + C_{21} = S_1\big[N(a) - N(-a)\big] + S_2\big[N(b) - N(-b)\big], \tag{3.10}$$

and $N(x) - N(-x) = 2N(x) - 1$ gives (**??**). □

Lemma 3.6 (Symmetry identity; the constant $\varkappa$)**.** *The two weighted densities coincide:*

$$\varkappa := S_1\varphi(a) = S_2\varphi(b) > 0. \tag{3.11}$$

Proof. By (**??**),

$$a^2 - b^2 = \Big(\frac{\ln(S_1/S_2)}{\omega_M} + \frac{\omega_M}{2}\Big)^2 - \Big(\frac{\ln(S_2/S_1)}{\omega_M} + \frac{\omega_M}{2}\Big)^2 = 2\ln(S_1/S_2). \tag{3.12}$$

Therefore

$$\frac{\varphi(a)}{\varphi(b)} = \exp\Big(-\tfrac{1}{2}\big(a^2 - b^2\big)\Big) = e^{-\ln(S_1/S_2)} = \frac{S_2}{S_1}, \tag{3.13}$$

which is exactly (**??**). □

The constant $\varkappa$ is the workhorse of everything that follows: every second- order spot Greek and every vega below is a rational multiple of it. It plays the role that $S\varphi(d_1)$ plays for a vanilla.

3.3 Spot Greeks

One inline flag before the hedge ratios (the load-bearing point for the $F = S$ collapse): under the standing zero rates $F_i = S_i$, so the forward Greeks derived below *are* the spot hedge ratios the desk trades. With non-zero carry each delta would pick up the corresponding forward factor, and the hedge would be sized in forwards, not spot.

Proposition 3.7 (Spot deltas)**.**

$$\frac{\partial V}{\partial S_1} = 2N(a) - 1, \qquad \frac{\partial V}{\partial S_2} = 2N(b) - 1. \tag{3.14}$$

Proof. Write $V = C_{12} + C_{21}$ as in the proof of Proposition **??**; by (**??**), $C_{12} = S_1 N(a) - S_2 N(-b)$ and $C_{21} = S_2 N(b) - S_1 N(-a)$. Since ω_M carries no spot dependence, (**??**) gives

$$\frac{\partial a}{\partial S_1} = \frac{1}{S_1 \omega_M}, \quad \frac{\partial a}{\partial S_2} = -\frac{1}{S_2 \omega_M}, \quad \frac{\partial b}{\partial S_1} = -\frac{1}{S_1 \omega_M}, \quad \frac{\partial b}{\partial S_2} = \frac{1}{S_2 \omega_M}, \tag{3.15}$$

in particular $\frac{\partial b}{\partial S_1} = -\frac{\partial a}{\partial S_1}$. Differentiating,

$$\begin{aligned} \frac{\partial C_{12}}{\partial S_1} &= N(a) + \big[S_1 \varphi(a) - S_2 \varphi(b)\big] \frac{\partial a}{\partial S_1} = N(a), \\ \frac{\partial C_{21}}{\partial S_1} &= -N(-a) + \big[S_2 \varphi(b) - S_1 \varphi(a)\big] \frac{\partial b}{\partial S_1} = -N(-a), \end{aligned} \tag{3.16}$$

where in the first line $\frac{\partial}{\partial S_1} N(-b) = \varphi(b)\, \frac{\partial a}{\partial S_1}$ (using $\varphi(-b) = \varphi(b)$ and $\frac{\partial(-b)}{\partial S_1} = \frac{\partial a}{\partial S_1}$) and both density brackets vanish by Lemma **??**: $S_1 \varphi(a) = S_2 \varphi(b) = \varkappa$. This is the Margrabe analogue of the vanilla cancellation $F\varphi(d_1) = K\varphi(d_2)$; equivalently it is Euler's theorem for the degree-one homogeneous $C_{12}(S_1, S_2)$. Hence

$$\frac{\partial V}{\partial S_1} = N(a) - N(-a) = 2N(a) - 1. \tag{3.17}$$

The $1 \leftrightarrow 2$ symmetry gives the second identity. □

Both deltas live in $(-1, 1)$, and at $S_1 = S_2$ both equal $2N(\omega_M/2) - 1 > 0$: the at-the-money absolute spread is slightly long *both* names, because each exchange option's convexity is centred there.

The second-order block needs a matrix symbol. The ratified Γ of the contract table is the single-name scalar (owner I.1) and stays untouched; $\boldsymbol{\Gamma}$ (bold, scope: this chapter) denotes the 2×2 matrix, whose diagonal entries are the two names' contract gammas. If II.4–5 want the matrix too, ratifying a multi-name gamma glyph is a FORMALIS amendment, not a local recast.

Proposition 3.8 (Spot gamma matrix)**.** *The full spot gamma matrix is*

$$\boldsymbol{\Gamma} := \begin{pmatrix} \frac{\partial^2 V}{\partial S_1^2} & \frac{\partial^2 V}{\partial S_1\, \partial S_2} \\ \frac{\partial^2 V}{\partial S_2\, \partial S_1} & \frac{\partial^2 V}{\partial S_2^2} \end{pmatrix} = \frac{2\varkappa}{\omega_M} \begin{pmatrix} \dfrac{1}{S_1^2} & -\dfrac{1}{S_1 S_2} \\ -\dfrac{1}{S_1 S_2} & \dfrac{1}{S_2^2} \end{pmatrix}. \tag{3.18}$$

Proof. Differentiate (**??**) using (**??**) and Lemma **??**:

$$\frac{\partial^2 V}{\partial S_1^2} = 2\varphi(a)\, \frac{1}{S_1 \omega_M} = \frac{2\varkappa}{S_1^2\, \omega_M}, \qquad \frac{\partial^2 V}{\partial S_1\, \partial S_2} = 2\varphi(a) \Big(-\frac{1}{S_2 \omega_M}\Big) = -\frac{2\varkappa}{S_1 S_2\, \omega_M}. \tag{3.19}$$

The other mixed partial is computed, not assumed: differentiating the second identity of (**??**) in S_1,

$$\frac{\partial^2 V}{\partial S_2\, \partial S_1} = 2\varphi(b)\, \frac{\partial b}{\partial S_1} = 2\varphi(b) \Big(-\frac{1}{S_1 \omega_M}\Big) = -\frac{2\varkappa}{S_1 S_2\, \omega_M}, \tag{3.20}$$

by $S_2 \varphi(b) = \varkappa$ — confirming the symmetry of the mixed partials directly (as Schwarz requires for the smooth V). Finally $\frac{\partial^2 V}{\partial S_2^2} = 2\varphi(b)/(S_2 \omega_M) = 2\varkappa/(S_2^2\, \omega_M)$ by the $1 \leftrightarrow 2$ relabelling. □

Remark 3.9 (Rank-one gamma and what it means for the hedge). The matrix (**??**) factors as $\mathbf{\Gamma} = (2\varkappa/\omega_M)\, vv^{\mathsf{T}}$ with $v = (1/S_1, -1/S_2)^{\mathsf{T}}$: it is rank one, with all convexity in the spread direction $\mathrm{d}S_1/S_1 - \mathrm{d}S_2/S_2$ and none in the parallel direction $\mathrm{d}S_1/S_1 + \mathrm{d}S_2/S_2$. The delta hedge therefore rebalances only against relative moves; a parallel market move generates no gamma P&L. The frictions concentrate accordingly: the position pins where $S_1 = S_2$ near expiry ($\omega_M \to 0$ blows up $2\varkappa/\omega_M$ along the spread exactly as an expiring ATM straddle does), and the cross-gamma term means the rebalance always trades the two names against each other — a pairs order whose cost scales with the wider of the two bid/asks. The $\omega_M \to 0$ blow-up has a second door besides expiry: dispersion collapse, $\rho \to 1$ at equal vols, where $\Sigma_M \to 0$ makes gamma and the correlation sensitivity per unit explode at *finite* T while the price itself vanishes. Theta behaves differently at this door: by (**??**) below, $\theta = -\varkappa\,\Sigma_M/\sqrt{T}$ — the gamma's $1/\omega_M$ is multiplied by the Margrabe variance Σ_M^2, and the product *vanishes* proportionally to Σ_M. The two doors are not symmetric: at expiry ($T \to 0$ at fixed Σ_M) gamma *and* theta blow up together, like the expiring ATM straddle's; in dispersion collapse gamma and the correlation sensitivity blow up while the decay dies — unbounded convexity with nothing to collect on it, and no arbitrage in it either, since the realised spread variance dies like Σ_M^2 and the harvestable gamma P&L vanishes with the theta. What remains unbounded is the exposure to the marks — the short-correlation tail flagged in the introduction, sitting exactly at the boundary excluded by Assumption **??**.

For theta, fix the sign convention: $\theta := \frac{\partial V}{\partial t} = -\frac{\partial V}{\partial T}$ at frozen (Σ_M, S_1, S_2). The direct computation from (**??**) is two lines: $\frac{\partial V}{\partial T} = \frac{\partial V}{\partial \omega_M}\frac{\partial \omega_M}{\partial T} = 2\varkappa \cdot \Sigma_M/(2\sqrt{T}) = \varkappa\Sigma_M/\sqrt{T}$. Equivalently, V solves the two-asset zero-rate pricing PDE

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sum_{i,j}\rho_{ij}\,\sigma_i\sigma_j S_i S_j\,\frac{\partial^2 V}{\partial S_i\,\partial S_j} = 0, \qquad \rho_{11} = \rho_{22} = 1, \quad \rho_{12} = \rho_{21} = \rho \tag{3.21}$$

(indexed ρ_{ij} local to this remark), so the gamma/theta balance reads

$$\theta = -\tfrac{1}{2}\sum_{i,j}\rho_{ij}\,\sigma_i\sigma_j S_i S_j\,\mathbf{\Gamma}_{ij} = -\frac{\varkappa}{\omega_M}\,\Sigma_M^2 = -\frac{\varkappa\,\Sigma_M}{\sqrt{T}}, \tag{3.22}$$

the middle equality by substituting (**??**): $\sum \rho_{ij}\sigma_i\sigma_j S_i S_j \mathbf{\Gamma}_{ij} = (2\varkappa/\omega_M)(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2) = 2\varkappa\Sigma_M^2/\omega_M$. The two routes agree; the decay the desk pays is the rank-one gamma times the Margrabe variance, nothing else.

3.4 Vol derivatives under sticky-ρ

The first convention holds ρ fixed as vols move: $\mathrm{d}\rho = 0$, the sticky-ρ mark of this volume. All derivatives in this section are partials at fixed ρ in the sense of the derivative-notation convention.

Proposition 3.10 (Margrabe vol sensitivities at fixed correlation)**.**

$$\left.\frac{\partial \Sigma_M}{\partial \sigma_1}\right|_{\rho} = \frac{\sigma_1 - \rho\sigma_2}{\Sigma_M}, \qquad \left.\frac{\partial \Sigma_M}{\partial \sigma_2}\right|_{\rho} = \frac{\sigma_2 - \rho\sigma_1}{\Sigma_M}. \tag{3.23}$$

Proof. Differentiate (**??**) at fixed ρ: $2\Sigma_M\left.\frac{\partial \Sigma_M}{\partial \sigma_1}\right|_{\rho} = 2\sigma_1 - 2\rho\sigma_2$, and similarly for σ_2. □

Proposition 3.11 (Price sensitivity to the Margrabe vol)**.** *The price* (**??**) *depends on* $(\sigma_1, \sigma_2, \rho)$ *only through* Σ_M*, and*

$$\frac{\partial V}{\partial \Sigma_M} = 2\varkappa\sqrt{T} > 0. \tag{3.24}$$

Proof. Write $x := \ln(S_1/S_2)$ (local to this proof), so $a = x/\omega_M + \omega_M/2$ and $b = -x/\omega_M + \omega_M/2$, and

$$\frac{\partial a}{\partial \omega_M} = -\frac{x}{\omega_M^2} + \frac{1}{2}, \qquad \frac{\partial b}{\partial \omega_M} = \frac{x}{\omega_M^2} + \frac{1}{2}. \tag{3.25}$$

Differentiating (**??**) in ω_M and using $S_1\varphi(a) = S_2\varphi(b) = \varkappa$ from Lemma **??**,

$$\frac{\partial V}{\partial \omega_M} = 2S_1\varphi(a)\frac{\partial a}{\partial \omega_M} + 2S_2\varphi(b)\frac{\partial b}{\partial \omega_M} = 2\varkappa\Big[\Big(-\frac{x}{\omega_M^2} + \frac{1}{2}\Big) + \Big(\frac{x}{\omega_M^2} + \frac{1}{2}\Big)\Big] = 2\varkappa. \tag{3.26}$$

Since $\omega_M = \Sigma_M\sqrt{T}$, the chain rule gives $\frac{\partial V}{\partial \Sigma_M} = 2\varkappa\sqrt{T}$. □

The reading is the vanilla one: an ATM option has vega $S\varphi(d_1)\sqrt{T}$; here each of the two exchange options contributes one unit of $\varkappa\sqrt{T}$, and the moneyness terms cancel by the symmetry (**??**).

Proposition 3.12 (Vegas under sticky-ρ)**.**

$$\frac{\partial V}{\partial \sigma_1}\bigg|_\rho = 2\varkappa\sqrt{T}\,\frac{\sigma_1 - \rho\sigma_2}{\Sigma_M}, \qquad \frac{\partial V}{\partial \sigma_2}\bigg|_\rho = 2\varkappa\sqrt{T}\,\frac{\sigma_2 - \rho\sigma_1}{\Sigma_M}. \tag{3.27}$$

Proof. Combine (**??**) with (**??**). □

Remark 3.13 (The per-leg vega can be short)*.* Because $\frac{\partial V}{\partial \Sigma_M} = 2\varkappa\sqrt{T} > 0$ and $\Sigma_M > 0$, the sign of the leg vega (**??**) is the sign of $\sigma_1 - \rho\sigma_2$. The book is therefore *short* σ_1-vega whenever $\rho > 0$ and $\sigma_1 < \rho\sigma_2$: even though $|S_1 - S_2|$ is an option, raising the low-vol leg's volatility *lowers* its value. Intuitively, moving σ_1 up toward the variance-minimising point $\sigma_1 = \rho\sigma_2$ of (**??**) shrinks the Margrabe volatility the payoff is long. Only the parallel vega is sign-safe:

$$\frac{\partial V}{\partial \sigma_1}\bigg|_\rho + \frac{\partial V}{\partial \sigma_2}\bigg|_\rho = 2\varkappa\sqrt{T}\,\frac{(\sigma_1 + \sigma_2)(1 - \rho)}{\Sigma_M} \;\geq\; 0. \tag{3.28}$$

On a real book this is not a curiosity: a desk long the absolute spread on a high-correlation pair with mismatched vols will buy back single-name vol on the low-vol leg thinking it is covering a long-vega position, and discover the mark moves against it.

3.5 sticky-CVC: fixed vol spread

The second convention freezes the basket vol spread

$$\xi := \tfrac{1}{2}(\sigma_1 + \sigma_2) - \sigma_B = \bar{\sigma} - \sigma_B, \tag{3.29}$$

where the equal-weight basket volatility is (constraint C1: the σ_i here are the single-name ATM vols $\sigma_{\mathrm{ATM},i}$)

$$\sigma_B = \sqrt{\tfrac{1}{4}\sigma_1^2 + \tfrac{1}{4}\sigma_2^2 + \tfrac{1}{2}\rho\,\sigma_1\sigma_2}. \tag{3.30}$$

Under sticky-CVC, $\mathrm{d}\xi = 0$ and ρ becomes a determined function of $(\sigma_1, \sigma_2, \xi)$. Derivatives at fixed ξ are *routed* through $\rho(\sigma_1, \sigma_2, \xi)$ and are written with the total-d notation. A determined function must land in $[-1, 1]$ to describe a joint law at all; Corollary **??** below pins down exactly where it does, and *every* fixed-ξ statement in this section and the next is read on that domain.

Proposition 3.14 (Exact correlation map). *With $\sigma_B = \frac{1}{2}(\sigma_1+\sigma_2)-\xi$,*

$$\rho = 1 - 2\Big(\frac{1}{\sigma_1}+\frac{1}{\sigma_2}\Big)\xi + \frac{2\xi^2}{\sigma_1\sigma_2}. \tag{3.31}$$

Proof. From (**??**),

$$\tfrac{1}{2}\rho\,\sigma_1\sigma_2 = \sigma_B^2 - \tfrac{1}{4}\sigma_1^2 - \tfrac{1}{4}\sigma_2^2. \tag{3.32}$$

Substituting $\sigma_B = \frac{1}{2}(\sigma_1+\sigma_2)-\xi$ gives $\sigma_B^2 = \frac{1}{4}(\sigma_1+\sigma_2)^2 - (\sigma_1+\sigma_2)\xi + \xi^2$, hence

$$\sigma_B^2 - \tfrac{1}{4}\sigma_1^2 - \tfrac{1}{4}\sigma_2^2 = \tfrac{1}{2}\sigma_1\sigma_2 - (\sigma_1+\sigma_2)\xi + \xi^2. \tag{3.33}$$

Multiplying by $2/(\sigma_1\sigma_2)$ gives (**??**). □

Corollary 3.15 (Admissibility domain of sticky-CVC). *The map* (**??**) *satisfies $\rho \in [-1,1]$ if and only if $\xi \in [0, \min(\sigma_1,\sigma_2)]$ (on the branch connected to $\xi = 0$). This is the validity domain of the* sticky-CVC *convention.*

Proof. One line each way, by factorising (**??**):

$$1-\rho = \frac{2\xi\,(\sigma_1+\sigma_2-\xi)}{\sigma_1\sigma_2}, \qquad 1+\rho = \frac{2\,(\sigma_1-\xi)(\sigma_2-\xi)}{\sigma_1\sigma_2}. \tag{3.34}$$

With the contract domain $\xi \ge 0$: $\rho \le 1$ iff $\xi \le \sigma_1+\sigma_2$, automatic on the stated domain; and $\rho \ge -1$ iff $(\sigma_1-\xi)(\sigma_2-\xi) \ge 0$, i.e. $\xi \le \min(\sigma_1,\sigma_2)$ or $\xi \ge \max(\sigma_1,\sigma_2)$, of which only the first branch contains $\xi = 0$ ($\rho = 1$) and is reachable by moving vols continuously at fixed ξ. □

The boundary matters on a live book. At fixed ξ, once either single-name vol falls to ξ, the map hits $\rho = -1$ exactly — and any further vol decline leaves *no* admissible correlation: no valid joint law, and the routed derivatives below are meaningless. That is precisely the scenario the convention is meant to govern (vols moving with ξ frozen), so the failure mode is not exotic: a desk re-marking a wide-spread pair under sticky-CVC can silently push implied ρ through -1. At $\min_i \sigma_i \downarrow \xi$ the convention has a hard stop; the mark must fall back to sticky-ρ or re-anchor ξ.

Proposition 3.16 (Margrabe variance at fixed ξ). *Under the map* (**??**),

$$\Sigma_M^2 = (\sigma_1-\sigma_2)^2 + 4\xi(\sigma_1+\sigma_2-\xi). \tag{3.35}$$

Proof. Substitute (**??**) into $\Sigma_M^2 = \sigma_1^2+\sigma_2^2-2\rho\sigma_1\sigma_2$:

$$\Sigma_M^2 = \sigma_1^2+\sigma_2^2-2\sigma_1\sigma_2+4\xi(\sigma_1+\sigma_2)-4\xi^2 = (\sigma_1-\sigma_2)^2+4\xi(\sigma_1+\sigma_2-\xi). \tag{3.36}$$

□

This is the structural payoff of the sticky-CVC convention: with the CVC spread frozen, the correlation disappears from the Margrabe variance entirely, replaced by the two observables $(\sigma_1-\sigma_2)$ and ξ. Further, on the admissibility domain with $\xi > 0$, (**??**) gives $\Sigma_M^2 \ge 4\xi(\sigma_1+\sigma_2-\xi) > 0$, so the degenerate point excluded by Assumption **??** is unreachable automatically — a strictly positive CVC spread keeps the Margrabe vol strictly alive. The vol derivatives fall out by inspection.

Corollary 3.17 (Margrabe vol sensitivities at fixed ξ).

$$\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\bigg|_{\xi} = \frac{\sigma_1-\sigma_2+2\xi}{\Sigma_M}, \qquad \frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_2}\bigg|_{\xi} = \frac{\sigma_2-\sigma_1+2\xi}{\Sigma_M}. \tag{3.37}$$

Proof. Differentiate (**??**) at fixed ξ: $2\Sigma_M \left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\right|_\xi = 2(\sigma_1 - \sigma_2) + 4\xi$, and similarly for σ_2. □

Corollary 3.18 (Parallel move at fixed ξ)**.** *For a parallel move* $\mathrm{d}\sigma_1 = \mathrm{d}\sigma_2 = \mathrm{d}\sigma$ *at fixed* ξ*,*

$$\left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma}\right|_{\xi,\,parallel} = \left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\right|_\xi + \left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_2}\right|_\xi = \frac{4\xi}{\Sigma_M}. \tag{3.38}$$

Proof. Add the two expressions in (**??**). □

Proposition 3.19 (Vegas under sticky-CVC)**.**

$$\left.\frac{\mathrm{d}V}{\mathrm{d}\sigma_1}\right|_\xi = 2\varkappa\sqrt{T}\,\frac{\sigma_1 - \sigma_2 + 2\xi}{\Sigma_M}, \qquad \left.\frac{\mathrm{d}V}{\mathrm{d}\sigma_2}\right|_\xi = 2\varkappa\sqrt{T}\,\frac{\sigma_2 - \sigma_1 + 2\xi}{\Sigma_M}, \tag{3.39}$$

and for the parallel move

$$\left.\frac{\mathrm{d}V}{\mathrm{d}\sigma}\right|_{\xi,\,parallel} = 2\varkappa\sqrt{T}\,\frac{4\xi}{\Sigma_M} = \frac{8\,\varkappa\,\xi\sqrt{T}}{\Sigma_M}. \tag{3.40}$$

Proof. Combine (**??**) with (**??**) and (**??**). □

The sign structure flips relative to (**??**): under sticky-CVC both leg vegas are positive whenever $\xi > |\sigma_1 - \sigma_2|/2$ — in particular always at equal vols — because part of any single-name vol rise is absorbed into a higher implied correlation (the map (**??**) has $\left.\frac{\partial\rho}{\partial\sigma_i}\right|_\xi > 0$ for $\xi < \sigma_j$, which holds on the admissibility domain of Corollary **??**), and the position's short-correlation exposure eats part of the long-Σ_M gain.

3.6 Comparison of the two conventions

Proposition 3.20 (Chain-rule decomposition of the sticky-CVC sensitivity)**.** *Holding* ξ *fixed,*

$$\left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\right|_\xi = \left.\frac{\partial\Sigma_M}{\partial\sigma_1}\right|_\rho + \frac{\partial\Sigma_M}{\partial\rho}\left.\frac{\partial\rho}{\partial\sigma_1}\right|_\xi. \tag{3.41}$$

Moreover,

$$\frac{\partial\Sigma_M}{\partial\rho} = -\frac{\sigma_1\sigma_2}{\Sigma_M}, \qquad \left.\frac{\partial\rho}{\partial\sigma_1}\right|_\xi = \frac{2\xi}{\sigma_1^2}\Big(1 - \frac{\xi}{\sigma_2}\Big), \tag{3.42}$$

so

$$\left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\right|_\xi = \left.\frac{\partial\Sigma_M}{\partial\sigma_1}\right|_\rho - \frac{2\xi(\sigma_2 - \xi)}{\sigma_1\,\Sigma_M}. \tag{3.43}$$

Proof. Equation (**??**) is the chain rule applied to $\Sigma_M\big(\sigma_1, \sigma_2, \rho(\sigma_1, \sigma_2, \xi)\big)$ at fixed ξ. Differentiating (**??**) in ρ gives $2\Sigma_M \frac{\partial\Sigma_M}{\partial\rho} = -2\sigma_1\sigma_2$, the first identity in (**??**). Differentiating (**??**) with respect to σ_1 at fixed (ξ, σ_2),

$$\left.\frac{\partial\rho}{\partial\sigma_1}\right|_\xi = \frac{2\xi}{\sigma_1^2} - \frac{2\xi^2}{\sigma_1^2\sigma_2} = \frac{2\xi}{\sigma_1^2}\Big(1 - \frac{\xi}{\sigma_2}\Big). \tag{3.44}$$

Substituting (**??**) into (**??**) gives (**??**). □

Remark 3.21 (Sign). On the admissibility domain $\xi \le \min(\sigma_1, \sigma_2)$ of Corollary ??, $\xi \le \sigma_2$, and the correction term in (??) is negative — strictly, for $0 < \xi < \sigma_2$, i.e. everywhere in the interior — so

$$\left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\right|_\xi < \left.\frac{\partial\Sigma_M}{\partial\sigma_1}\right|_\rho. \tag{3.45}$$

sticky-CVC damps the σ_1-sensitivity of the Margrabe volatility relative to sticky-ρ. The mechanism is the one identified in Chapter 2 with the sign reversed: the routed term $\frac{\partial\Sigma_M}{\partial\rho}\left.\frac{\partial\rho}{\partial\sigma_1}\right|_\xi$ is a shadow-vega channel, and for this payoff $\frac{\partial\Sigma_M}{\partial\rho} < 0$ while the basket has $\frac{\partial\sigma_B}{\partial\rho} > 0$.

Proposition 3.22 (Exact ratio of the two sensitivities)**.** *On the domain $\sigma_1 \ne \rho\sigma_2$ — away from the variance-minimising point where the* sticky-ρ *sensitivity (??) vanishes and no ratio exists — define (chapter scope)*

$$\mathcal{R} := \frac{\left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\right|_\xi}{\left.\frac{\partial\Sigma_M}{\partial\sigma_1}\right|_\rho}. \tag{3.46}$$

Then

$$\mathcal{R} = \frac{\sigma_1 - \sigma_2 + 2\xi}{\sigma_1 - \rho\sigma_2} = \frac{1}{1 + \dfrac{2\xi(\sigma_2 - \xi)}{\sigma_1(\sigma_1 - \sigma_2 + 2\xi)}}. \tag{3.47}$$

Proof. The first expression is the ratio of (??) to (??): the common factor $1/\Sigma_M$ cancels. For the second, use (??) to write

$$\sigma_1 - \rho\sigma_2 = \sigma_1 - \sigma_2 + 2\xi + \frac{2\xi(\sigma_2 - \xi)}{\sigma_1}, \tag{3.48}$$

and divide. □

Corollary 3.23 (Equal-vol specialisation)**.** *If $\sigma_1 = \sigma_2 = \sigma$, then*

$$\mathcal{R} = \frac{2\xi}{\sigma(1-\rho)}. \tag{3.49}$$

Using the exact equal-vol identity

$$1 - \rho = \frac{2\xi(2\sigma - \xi)}{\sigma^2}, \tag{3.50}$$

this becomes

$$\mathcal{R} = \frac{\sigma}{2\sigma - \xi} = \frac{1}{2 - \xi/\sigma}, \tag{3.51}$$

and, since $\xi = \sigma\big(1 - \sqrt{(1+\rho)/2}\big)$ at equal vols,

$$\mathcal{R} = \frac{1}{1 + \sqrt{\dfrac{1+\rho}{2}}}. \tag{3.52}$$

In particular $\frac{1}{2} \le \mathcal{R} \le 1$.

Proof. At $\sigma_1 = \sigma_2 = \sigma$ the first expression in (??) reads $2\xi/(\sigma - \rho\sigma)$, which is (??). Identity (??) is (??) at equal vols, rearranged. Then

$$\mathcal{R} = \frac{2\xi}{\sigma} \cdot \frac{\sigma^2}{2\xi(2\sigma - \xi)} = \frac{\sigma}{2\sigma - \xi}, \tag{3.53}$$

which is (??). Since $\xi/\sigma = 1 - \sqrt{(1+\rho)/2}$, $2 - \xi/\sigma = 1 + \sqrt{(1+\rho)/2}$, hence (??). As ρ runs from 1 down to -1 the square root runs from 1 down to 0, giving the bounds. □

Remark 3.24 (The bounds are an equal-vol fact; mismatched vols can amplify). The band $[1/2, 1]$ of Corollary **??** does *not* survive mismatched vols; only the signed inequality (**??**) is general on the admissibility domain. In magnitude, sticky-CVC can amplify. A worked counterexample: $\sigma_1 = 20\%$, $\sigma_2 = 30\%$, $\rho = 0.80$ gives $\xi = 0.0123$, $\Sigma_M = 0.1844$, sticky-ρ sensitivity $(\sigma_1 - \rho\sigma_2)/\Sigma_M = -0.217$, sticky-CVC sensitivity $(\sigma_1 - \sigma_2 + 2\xi)/\Sigma_M = -0.409$, hence $\mathcal{R} = 1.89$: the sticky-CVC vega is nearly *twice* the sticky-ρ vega in magnitude. (Both routes, (**??**) and (**??**), give the same number; the signed ordering $-0.409 < -0.217$ still respects (**??**).) The mechanism is the one Remark **??** flagged: near the variance-minimising point $\sigma_1 = \rho\sigma_2$ the sticky-ρ leg vega passes through zero, so $\mathcal{R}$ blows up on either side of it, and $\mathcal{R} < 0$ is attainable when $\sigma_2 - \sigma_1$ slightly exceeds 2ξ while $\sigma_1 > \rho\sigma_2$ — the two conventions then disagree on the very sign of the leg vega. The desk conclusion: on a high-correlation, mismatched-vol book — exactly the book of Remark **??** — size the vol hedge from (**??**) directly, never from an equal-vol ratio.

Remark 3.25 (Limits). At equal vols, $\rho \to 1$ gives $\mathcal{R} \to \frac{1}{2}$: near perfect correlation the sticky-CVC vega is asymptotically *half* the sticky-ρ vega. And $\rho \to -1$ gives $\mathcal{R} \to 1$: at very low correlation the two conventions coincide for this comparison. The $\rho \to -1$ end is not an interior limit: at equal vols it is $\xi \uparrow \sigma$, exactly the admissibility boundary of Corollary **??**, where the convention runs out of correlation to sell. The practical reading: the more correlated the pair, the more of a single-name vol move the sticky-CVC mark routes into implied correlation rather than into the Margrabe vol, and the smaller the effective vega the desk should carry against single-name vol moves. Marking a high-correlation spread book with sticky-ρ vegas when the market re-marks correlation CVC-style overstates the vol hedge by a factor approaching two.

Corollary 3.26 (The same ratio compares the vegas). *Since* $\frac{\partial V}{\partial \Sigma_M} = 2\varkappa\sqrt{T} > 0$,

$$\frac{\left.\frac{\mathrm{d}V}{\mathrm{d}\sigma_1}\right|_\xi}{\left.\frac{\partial V}{\partial\sigma_1}\right|_\rho} = \frac{2\varkappa\sqrt{T}\ \left.\frac{\mathrm{d}\Sigma_M}{\mathrm{d}\sigma_1}\right|_\xi}{2\varkappa\sqrt{T}\ \left.\frac{\partial\Sigma_M}{\partial\sigma_1}\right|_\rho} = \mathcal{R}. \tag{3.54}$$

3.7 Closing the correlation thread: the anchor example

The volume's worked correlation example — $\sigma_1 = \sigma_2 = \sigma = 20\%$, $\rho = 0.50$, frozen spread $\xi = \sigma\big(1 - \sqrt{(1+\rho)/2}\big) = 0.20\,(1 - \sqrt{0.75}\,) = 0.02679$ (about 2.7 vol points), Cega $= +0.20$ per correlation point in the contract normalisation ($\text{Cega}_{\text{gap}} = +1$ in the gap-bump quote of Chapter 2's constant-Cega_{gap} thread book, whose declared identity is $\frac{\partial V}{\partial \rho} = 10\,\text{Cega}_{\text{gap}}/(1-\rho) = 20$) — has so far run through the basket straddle: its shadow vega ($+0.464$ per vol point), its shadow delta (-11.6 dollar basis at vol.slope $= -0.25$), and the correlation update rule. The same numbers now run through the absolute spread option, closing the loop. All figures below are re-derived from the closed forms of this chapter; none is inherited from a numerical engine, and in particular none relies on $\sigma_{\text{ATM}} = \text{const}$ — the ATM dynamics enter explicitly through vol.slope where used.

Example 3.27 (Anchor pricing and Greeks). Take $S_1 = S_2 = 100$ (under zero rates, $F_i = S_i$ — flagged), $T = 1$, and the anchor vols. First the Margrabe vol: at equal vols $\Sigma_M = \sigma\sqrt{2(1-\rho)} = 0.20\sqrt{1} = 20\%$ exactly, so $\omega_M = \Sigma_M\sqrt{T} = 0.20$. Then $a = b = \omega_M/2 = 0.10$, and:

- **Price:** $V = 200\,\big(2N(0.10) - 1\big) = 200 \times 0.07966 = 15.93$. (Ballpark check: $0.8\,S\,\omega_M = 0.8 \times 100 \times 0.20 = 16$.)
- **Symmetry constant:** $\varkappa = 100\,\varphi(0.10) = 39.70$, so $\frac{\partial V}{\partial \Sigma_M} = 2\varkappa\sqrt{T} = 79.39$, i.e. 0.794 per vol point of Σ_M.

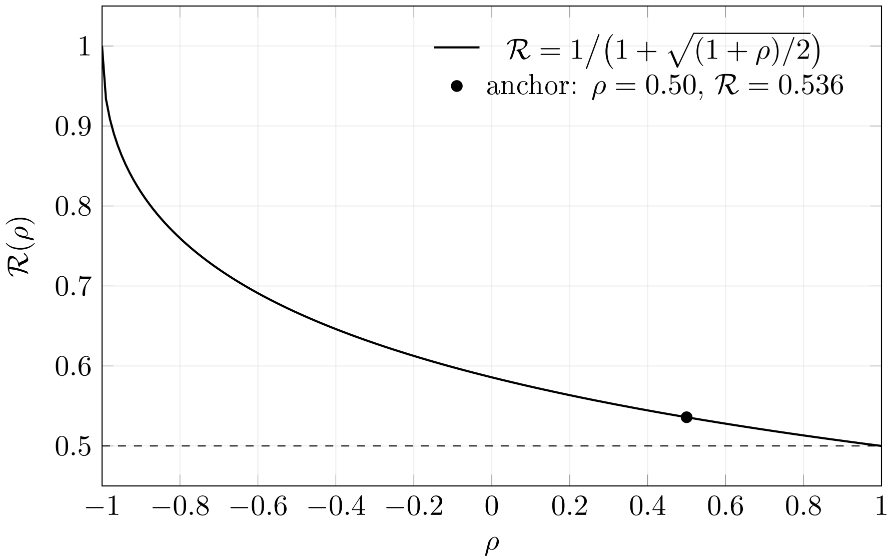

Figure 3.1: Equal-vol ratio $\mathcal{R}$ of the sticky-CVC to the sticky-ρ σ_1-sensitivity of Σ_M, as a function of ρ (Corollary **??**). *At equal vols* the ratio is bounded in $[1/2, 1]$: there sticky-CVC never amplifies and at most halves the vega. With mismatched vols the band fails and sticky-CVC can amplify (Remark **??**). The marked point is the anchor example of Section **??**.

- **Deltas:** $\frac{\partial V}{\partial S_1} = \frac{\partial V}{\partial S_2} = 2N(0.10) - 1 = +0.0797$ on each leg.
- **Gamma matrix:** $2\varkappa/\omega_M = 396.95$, so $\mathbf{\Gamma}_{11} = \mathbf{\Gamma}_{22} = +0.0397$ and $\mathbf{\Gamma}_{12} = -0.0397$ — rank one, all in the spread.
- sticky-ρ **vegas:** each leg $2\varkappa\sqrt{T}\,\sigma(1-\rho)/\Sigma_M = 79.39 \times 0.5 = 39.70$, i.e. 0.397 per vol point per leg; parallel 0.794 per vol point.
- sticky-CVC **vegas:** each leg $2\varkappa\sqrt{T} \cdot 2\xi/\Sigma_M = 79.39 \times 0.2679 = 21.27$, i.e. 0.213 per vol point per leg; parallel $8\varkappa\xi\sqrt{T}/\Sigma_M = 42.55$, i.e. 0.425 per vol point.
- **Ratio:** $\mathcal{R} = 0.213/0.397 = 0.5359 = 1/\big(1+\sqrt{0.75}\,\big)$, on the nose of Corollary **??** and Figure **??**. The sticky-CVC vega is 53.6% of the sticky-ρ vega at the anchor.

Example 3.28 (Closing the shadow-Greek loop)**.** The gap between the two conventions in Example **??** is exactly the shadow vega of Chapter 2, computed for this payoff.

- **Correlation exposure.** From $\frac{\partial \Sigma_M}{\partial \rho} = -\sigma_1\sigma_2/\Sigma_M = -0.04/0.20 = -0.20$,

$$\frac{\partial V}{\partial \rho} = 2\varkappa\sqrt{T}\,\frac{\partial \Sigma_M}{\partial \rho} = 79.39 \times (-0.20) = -15.88, \qquad \text{Cega} = \tfrac{1}{100}\frac{\partial V}{\partial \rho} = -0.159 \tag{3.55}$$

  per correlation point. Negative, as the payoff-and-risk paragraph promised: the absolute spread is short correlation, the anti-basket.

- **Correlation channel.** The anchor map gives, per leg, $\frac{\partial \rho}{\partial \sigma_1}\Big|_{\xi} = (2\xi/\sigma^2)(1-\xi/\sigma) = 1.160$, hence $\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\Big|_{\xi} = 2.321$ for the parallel move — the same channel slope that drove the basket book's shadow Greeks.

- **Shadow vega.** Per the contract normalisation,

$$\mathrm{ShadowVega} = \tfrac{1}{100}\,\frac{\partial V}{\partial \rho}\,\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\bigg|_{\xi} = \tfrac{1}{100} \times (-15.88) \times 2.321 = -0.368 \tag{3.56}$$

  per vol point. Cross-check against Example ??: the parallel sticky-CVC vega minus the parallel sticky-ρ vega is $0.425 - 0.794 = -0.368$ per vol point. The shadow vega *is* the convention gap, to the digit — the additivity promised by the chain rule (??).

- **Shadow delta.** With the anchor's ATM slope vol.slope $= -0.25$ (the surface dynamics of Volume I; a -25 vol-point ATM move per unit of ℓ_F — explicitly *not* $\sigma_{\mathrm{ATM}} = \mathrm{const}$), the normaliser-ratio identity gives

$$\mathrm{ShadowDelta} = 100 \cdot \mathrm{vol.slope} \cdot \mathrm{ShadowVega} = 100 \times (-0.25) \times (-0.368) = +9.2 \tag{3.57}$$

  on the dollar basis: spot down $\Rightarrow$ ATM vols up $\Rightarrow$ (under sticky-CVC) implied correlation up $\Rightarrow$ Σ_M down $\Rightarrow$ the spread option loses. The position thus carries $+9.2$ of routed dollar delta *on top of* its direct delta (shadow Greeks are additive per the contract's normaliser convention); to be flat through the routed channel the hedge must be *short* a further 9.2 of dollar delta.

The thread is now closed. The basket straddle of Chapter 2 carried ShadowVega $= +0.464$ and ShadowDelta $= -11.6$; the absolute spread at the same anchor carries ShadowVega $= -0.368$ and ShadowDelta $= +9.2$. The basket figures pass the same arithmetic as (??): with the basket's $\frac{\partial V}{\partial \rho} = +20$ (Cega $+0.20$ per correlation point, $+1$ gap-bump), $\frac{1}{100} \times 20 \times 2.321 = +0.464$ and $100 \times (-0.25) \times 0.464 = -11.6$, consistent to the digit. Same surface, same correlation channel $\frac{\mathrm{d}\rho}{\mathrm{d}\sigma}\big|_{\xi} = 2.321$, opposite signs — because $\frac{\partial \sigma_B}{\partial \rho} > 0$ while $\frac{\partial \Sigma_M}{\partial \rho} < 0$. A desk running the dispersion pair (long the spread option, short the basket straddle, or vice versa) nets the two shadow-Greek columns; a desk marking one leg sticky-ρ and the other sticky-CVC books a phantom vega of order half its true exposure. That inconsistency, not the model, is the free-money leak to guard against.

3.8 How the results cohere

(1) **Symmetric pricing.** The payoff $|S_1 - S_2|$ is the sum of the two exchange options, and its price admits the symmetric representation (??). The spot deltas (??) and the rank-one gamma matrix (??) follow in closed form from that symmetry, with the constant $\varkappa = S_1\varphi(a) = S_2\varphi(b)$ carrying all the second-order structure.

(2) **Two re-marking conventions.** Under sticky-ρ the relevant sensitivity is the direct partial (??); under sticky-CVC the exact correlation map (??) collapses the Margrabe variance to (??) and the routed derivative is (??). Multiplying either by $\frac{\partial V}{\partial \Sigma_M} = 2\varkappa\sqrt{T}$ gives the corresponding vegas, (??) and (??).

(3) **The comparison is exact.** Proposition ?? isolates the sticky-CVC correction as the routed chain-rule term $\frac{\partial \Sigma_M}{\partial \rho}\,\frac{\partial \rho}{\partial \sigma_1}\big|_{\xi}$, negative on the admissibility domain $\xi \le \min(\sigma_1, \sigma_2)$ of Corollary ??; Proposition ?? gives the exact ratio (where the sticky-ρ sensitivity is non-zero, $\sigma_1 \neq \rho\sigma_2$), which at equal vols is $\mathcal{R} = 1/\big(1 + \sqrt{(1+\rho)/2}\big) \in [1/2, 1]$. *At equal vols* the sticky-CVC vega is never larger than the sticky-ρ vega, and near $\rho \approx 1$ it is almost exactly half. With mismatched vols only the signed inequality (??) survives; in magnitude sticky-CVC can amplify, as Remark ??'s $\mathcal{R} = 1.89$ counterexample shows.

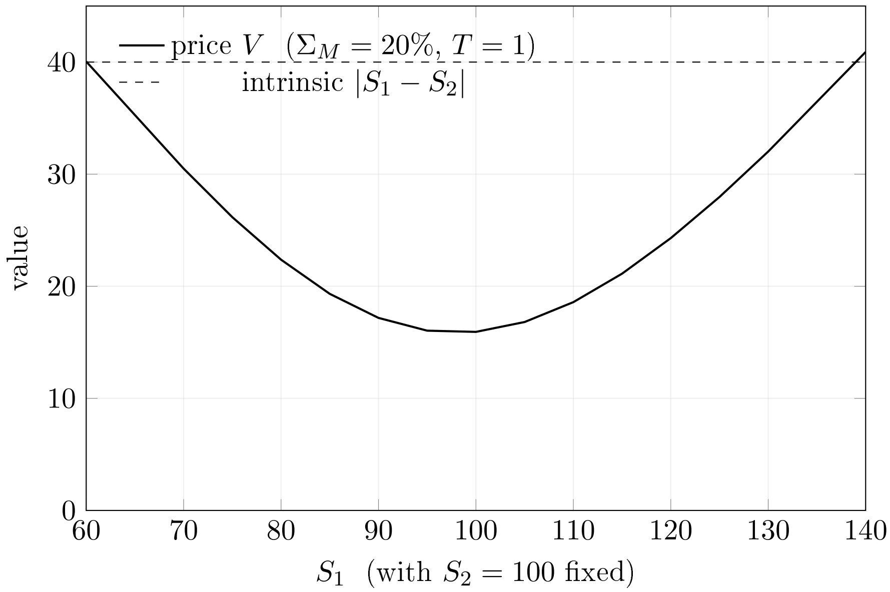

Figure 3.2: The absolute spread at the anchor ($\Sigma_M = 20\%$, $T = 1$, $S_2 = 100$): a straddle on the spread. The minimum-value point sits at $S_1 = S_2$, where the deltas (**??**) are both small and positive and the rank-one gamma (**??**) peaks; as $T \to 0$ the price collapses onto the intrinsic kink and the position pins.

(4) **The correlation thread closes.** At the volume's anchor ($\sigma = 20\%$, $\rho = 0.50$, $\xi = 0.02679$) the spread option's shadow vega, -0.368 per vol point, equals the gap between its sticky-CVC and sticky-ρ parallel vegas to the digit, and its shadow delta is $+9.2$ at vol.slope $= -0.25$ — equal channel, opposite sign to the basket book. The shadow Greeks are exactly the difference between the two conventions, they vanish under sticky-ρ, and they net across a dispersion pair. What must never happen is a book marked half in one convention and half in the other.

Chapter 4

Correlation Admissibility for n Names: Schur Complements and Blockwise Modification

4.1 The admissibility ladder

Every rung of this corpus's no-arbitrage theory has the same shape: a quoted object is admissible if and only if it lies in a convex feasible set, and the interesting economics happens at the boundary of that set.

- **Rung 1 (one name).** Volume I, Chapter 2: a single-name surface $\sigma(m, \ell_F, T) = \sigma_{\text{ATM}}(\ell_F, T) + \sigma_{\text{smile}}(m, T)$ is admissible only if the call prices $C(K, T)$ it induces are convex in K (no butterfly arbitrage — equivalently, the risk-neutral density is nonnegative), the wings of $\sigma_{\text{smile}}(m, T)$ grow slowly enough in $|m|$ that the density integrates, and the total variance $\sigma^2 T$ is non-decreasing in T (no calendar-spread arbitrage) — conditions (A2), the Lee wing control, and (A3) of the admissible set of that chapter. The feasible set is a convex set of smile functions; its boundary is a zero of the density.
- **Rung 2 (two names).** Chapter 1 of the present volume: the basket volatility σ_B, the mean volatility $\bar{\sigma} = \frac{1}{2}(\sigma_1 + \sigma_2)$, and the implied correlation ρ are tied by $\sigma_B^2 = \frac{1}{4}\sigma_1^2 + \frac{1}{4}\sigma_2^2 + \frac{1}{2}\rho\sigma_1\sigma_2$, and the pair $(\sigma_1, \sigma_2, \rho)$ is admissible if and only if $\rho \in [-1, 1]$ — equivalently, if and only if the 2×2 matrix $\left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ is positive semidefinite. The upper endpoint is the vanishing of the basket vol spread: $\xi = \bar{\sigma} - \sigma_B \geq 0$ with equality exactly at $\rho = 1$. The feasible set is an interval; its boundary is a singular 2×2 correlation matrix.
- **Rung 3 (n names).** This chapter. With n names the quoted object is the full matrix Ω of pairwise implied correlations, and admissibility is positive semidefiniteness of Ω. The feasible set is a compact convex body (the set of correlation matrices); its boundary is the locus of singular Ω. The tool that makes this rung computable — that reduces "is the whole matrix admissible?" to conditions on the piece being changed — is the Schur complement.

Throughout, the per-name volatilities that accompany Ω are the single-name ATM volatilities of the mandated decomposition, $\sigma_i \equiv \sigma_{\text{ATM},i}(\ell_{F,i}, T)$ evaluated at each name's current forward (constraint C1). No forward appears in any derivation of this chapter, so no zero-rate collapse is invoked anywhere in it: correlation admissibility is a statement of pure linear algebra and Gaussian conditioning, sitting on top of — and independent of — the surface dynamics of the earlier rungs.

Rung 3 — n names (this chapter). $\Omega \succeq 0$, certified blockwise: $\Omega \succeq 0 \iff \mathrm{Schur}_B(\Omega) \succeq 0$ given $B \succ 0$.

↑ condition on the rest

Rung 2 — two names (Ch. 1). $\xi = \bar{\sigma} - \sigma_B \geq 0 \iff \rho \leq 1$; admissibility is $\left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right) \succeq 0$, i.e. $|\rho| \leq 1$.

↑ add a name

Rung 1 — one name (Vol. I, Ch. 2). Convexity of $C(K, T)$ in K, wing growth of $\sigma_{\mathrm{smile}}(m, T)$, and calendar monotonicity of total variance: the risk-neutral density is nonnegative at every expiry.

Figure 4.1: The admissibility ladder. Each rung is the boundary of a convex feasible set; rung 2 is recovered from rung 3 by taking an empty conditioning block (Remark **??**).

4.1.1 Local notation

The following symbols are scoped to this chapter. Block letters A, B, C are those of the ratified Schur entry $\mathrm{Schur}_B(M) = A - C^{\mathsf{T}} B^{-1} C$; one of them collides with the ratified table, and the collision is resolved by declaration rather than denied. The glyph C is owned in the ratified table by the European call price $C(K, T)$ (owner I.1). In this chapter, *bare* C always denotes the cross-block of the partition (**??**) — the contract's own gloss for $\mathrm{Schur}_B(M)$ already uses the block letter — while the call price is always written with its arguments, $C(K, T)$, and appears only in the rung-1 ladder prose of Sections **??** and **??**, never in a derivation of this chapter. No other local symbol collides with the ratified table. Index sets are calligraphic, $\mathcal{I}$ and $\mathcal{J}$, with sizes $n_{\mathcal{I}} = |\mathcal{I}|$ and $n_{\mathcal{J}} = |\mathcal{J}|$ (the glyphs $\mathcal{J}$ and $n_{\mathcal{J}}$ are distinct from the observation count J of the calibration chapters). Id_p is the $p \times p$ identity, $\mathbf{1}$ the all-ones column vector, $\mathbf{1}\mathbf{1}^{\mathsf{T}}$ the all-ones matrix. $\Theta := \Omega^{-1}$ is the precision matrix. $\Pi := C^{\mathsf{T}} B^{-1} C$ is the explained block, $\Sigma_{\mathcal{I}|\mathcal{J}}$ the conditional covariance, R the conditional correlation matrix, D the diagonal matrix of conditional variances, $G := C^{\mathsf{T}} B^{-1}$ the regression matrix, $\lambda \in [0, 1]$ the blending weight, and $\mu_{\min}(\cdot)$, $\mu_{\max}(\cdot)$ the extreme eigenvalues of a symmetric matrix. For the single-entry bound: $q_u := u^{\mathsf{T}} B^{-1} u$, $q_v := v^{\mathsf{T}} B^{-1} v$, centre $\bar{\rho}_{ij} := u^{\mathsf{T}} B^{-1} v$, half-width $\eta_{ij} := \sqrt{(1 - q_u)(1 - q_v)}$, and $t \in [-1, 1]$ the partial-correlation coordinate. Inner products are written $x^{\mathsf{T}} y$ throughout; the bra-ket calculus is reserved for the calibration block of Volume I and does not appear in this chapter. Two near-misses are declared, though neither is a contract collision: the conditional covariance is written $\Sigma_{\mathcal{I}|\mathcal{J}}$ (bare Σ appears only as the local abbreviation $\Sigma := \mathrm{Schur}_B(M)$ inside the statements and proofs of Section **??**) and is unrelated to the Margrabe volatility Σ_M of Chapter 3; the extreme eigenvalues $\mu_{\min}(\cdot)$, $\mu_{\max}(\cdot)$ are unrelated to the kernel centres μ_i of Volume I.

Definition 4.1 (Admissible correlation matrix)**.** A matrix $\Omega \in \mathbb{R}^{n \times n}$ is an *admissible correlation matrix* if it is symmetric, has unit diagonal ($\Omega_{ii} = 1$ for all i), and is positive semidefinite, $\Omega \succeq 0$; $\Omega \succ 0$ denotes the positive-definite case. The set of admissible $n \times n$ correlation matrices is denoted $\mathcal{E}_n$ and called the *elliptope*; it is compact and convex. Convexity and closedness hold because $\mathcal{E}_n$ is an intersection of the PSD cone (closed, convex) with the affine constraints $\Omega_{ii} = 1$; boundedness needs one more line: PSD-ness of every 2×2 principal minor together with the unit diagonal forces $1 - \Omega_{ij}^2 \geq 0$, i.e. $|\Omega_{ij}| \leq 1$ for all i, j, so $\mathcal{E}_n$ is bounded, and closed plus bounded in $\mathbb{R}^{n \times n}$ is compact.

Admissibility is not decoration. If $\Omega \not\succeq 0$ there is a vector x with $x^{\mathsf{T}} \Omega x < 0$: the portfolio with weights proportional to x (in vol-normalised units) has *negative variance.* Every Monte Carlo engine, every basket vol, every Margrabe volatility Σ_M built on such a matrix is pricing an object that cannot exist.

4.2 The Schur complement and the positivity criterion

Definition 4.2 (Schur complement)**.** Let $M \in \mathbb{R}^{n\times n}$ be symmetric, partitioned as

$$M \;=\; \begin{pmatrix} A & C^{\mathsf{T}} \\ C & B \end{pmatrix}, \qquad A \in \mathbb{R}^{n_{\mathcal{I}}\times n_{\mathcal{I}}}, \quad B \in \mathbb{R}^{n_{\mathcal{J}}\times n_{\mathcal{J}}}, \quad C \in \mathbb{R}^{n_{\mathcal{J}}\times n_{\mathcal{I}}}, \tag{4.1}$$

with $n_{\mathcal{I}} + n_{\mathcal{J}} = n$. If $B \succ 0$, the *Schur complement of B in M* is

$$\mathrm{Schur}_B(M) \;:=\; A - C^{\mathsf{T}} B^{-1} C. \tag{4.2}$$

The entire chapter rests on one algebraic identity: the block partition (**??**) can be *diagonalised by congruence*, with the Schur complement appearing as the residual block.

Lemma 4.3 (Block congruence)**.** *Let M be as in* (**??**) *with $B \succ 0$, and set $\Sigma := \mathrm{Schur}_B(M)$. Then*

$$M \;=\; L^{\mathsf{T}} \begin{pmatrix} \Sigma & 0 \\ 0 & B \end{pmatrix} L, \qquad L \;=\; \begin{pmatrix} \mathrm{Id}_{n_{\mathcal{I}}} & 0 \\ B^{-1}C & \mathrm{Id}_{n_{\mathcal{J}}} \end{pmatrix}, \tag{4.3}$$

and L is invertible.

Proof. Multiply out the right-hand side:

$$L^{\mathsf{T}} \begin{pmatrix} \Sigma & 0 \\ 0 & B \end{pmatrix} L = \begin{pmatrix} \mathrm{Id} & C^{\mathsf{T}}B^{-1} \\ 0 & \mathrm{Id} \end{pmatrix} \begin{pmatrix} \Sigma & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} \mathrm{Id} & 0 \\ B^{-1}C & \mathrm{Id} \end{pmatrix} = \begin{pmatrix} \Sigma + C^{\mathsf{T}}B^{-1}C & C^{\mathsf{T}} \\ C & B \end{pmatrix} = M,$$

using $C^{\mathsf{T}}B^{-1}B = C^{\mathsf{T}}$ and $\Sigma + C^{\mathsf{T}}B^{-1}C = A$. The matrix L is block lower-triangular with identity diagonal blocks, hence has determinant 1 and is invertible. □

Theorem 4.4 (Schur positivity criterion)**.** *Let M be as in* (**??**) *with $B \succ 0$. Then*

$$M \succeq 0 \quad\Longleftrightarrow\quad \mathrm{Schur}_B(M) \succeq 0,$$

and likewise $M \succ 0 \iff \mathrm{Schur}_B(M) \succ 0$.

Proof. By Lemma **??**, for any $x \in \mathbb{R}^n$, writing $y = Lx$,

$$x^{\mathsf{T}} M x = y^{\mathsf{T}} \begin{pmatrix} \Sigma & 0 \\ 0 & B \end{pmatrix} y = y_{\mathcal{I}}^{\mathsf{T}} \Sigma\, y_{\mathcal{I}} + y_{\mathcal{J}}^{\mathsf{T}} B\, y_{\mathcal{J}}.$$

Since L is invertible, $x \mapsto y$ is a bijection of $\mathbb{R}^n$. If $\Sigma \succeq 0$, both terms on the right are nonnegative ($B \succ 0$ by hypothesis), so $M \succeq 0$. Conversely, if $\Sigma \not\succeq 0$, pick z with $z^{\mathsf{T}}\Sigma z < 0$ and take $y = (z, 0)$: the corresponding $x = L^{-1}y$ gives $x^{\mathsf{T}}Mx = z^{\mathsf{T}}\Sigma z < 0$, so $M \not\succeq 0$. The positive-definite statement is identical with strict inequalities, noting that $y \neq 0 \iff x \neq 0$. □

4.2.1 The Schur complement is a conditional covariance

The criterion of Theorem **??** is pure linear algebra, but its content is probabilistic, and the probabilistic reading is what makes every construction in this chapter interpretable. It is the same Gaussian-conditioning identity that drives the Bayesian update of the calibration filter in Volume I, Chapter 4: there, the posterior covariance of the parameter ket given an observation is the Schur complement of the observation block in the joint covariance; here, the roles are played by two blocks of names.

Proposition 4.5 (Conditional covariance identity)**.** *Let $(X_{\mathcal{I}}, X_{\mathcal{J}})$ be a zero-mean jointly Gaussian vector with covariance matrix M partitioned as in* (**??**)*, $B \succ 0$. Then*

$$\operatorname{Cov}\big(X_{\mathcal{I}} \mid X_{\mathcal{J}}\big) \;=\; \operatorname{Schur}_B(M) \;=\; A - C^{\mathsf{T}} B^{-1} C, \tag{4.4}$$

and the conditional mean is $\mathbb{E}[X_{\mathcal{I}} \mid X_{\mathcal{J}}] = G X_{\mathcal{J}}$ with $G = C^{\mathsf{T}} B^{-1}$.

Proof (regression / best linear predictor). Because the vector is jointly Gaussian, the conditional expectation is linear: $\mathbb{E}[X_{\mathcal{I}} \mid X_{\mathcal{J}}] = G X_{\mathcal{J}}$ for some $G \in \mathbb{R}^{n_{\mathcal{I}} \times n_{\mathcal{J}}}$, and the conditional covariance is the covariance of the residual $X_{\mathcal{I}} - G X_{\mathcal{J}}$. The matrix G is pinned down by the orthogonality of the residual to the conditioning variables:

$$0 = \operatorname{Cov}\big(X_{\mathcal{I}} - G X_{\mathcal{J}},\, X_{\mathcal{J}}\big) = \operatorname{Cov}(X_{\mathcal{I}}, X_{\mathcal{J}}) - G \operatorname{Cov}(X_{\mathcal{J}}, X_{\mathcal{J}}) = C^{\mathsf{T}} - GB,$$

so $G = C^{\mathsf{T}} B^{-1}$. The residual covariance is then

$$\begin{aligned} \operatorname{Cov}\big(X_{\mathcal{I}} - G X_{\mathcal{J}}\big) &= A - GC - C^{\mathsf{T}} G^{\mathsf{T}} + G B G^{\mathsf{T}} \\ &= A - C^{\mathsf{T}} B^{-1} C - C^{\mathsf{T}} B^{-1} C + C^{\mathsf{T}} B^{-1} B B^{-1} C \;=\; A - C^{\mathsf{T}} B^{-1} C, \end{aligned}$$

the middle two terms cancelling against the last. For a Gaussian vector the residual is independent of $X_{\mathcal{J}}$, so its covariance is exactly the conditional covariance. □

A second, purely algebraic route to the same identity runs through the precision matrix; it is recorded because the precision-matrix block is needed for the partial-correlation coordinate of Section **??**.

Proposition 4.6 (Block inverse)**.** *Let $M \succ 0$ be partitioned as in* (**??**) *and set $\Sigma := \operatorname{Schur}_B(M)$. Then the $\mathcal{I} \times \mathcal{I}$ block of M^{-1} is Σ^{-1}:*

$$M^{-1} \;=\; \begin{pmatrix} \Sigma^{-1} & * \\ * & * \end{pmatrix}.$$

Proof. Write $M^{-1} = \left(\begin{smallmatrix} X & Y^{\mathsf{T}} \\ Y & Z \end{smallmatrix}\right)$ with $X \in \mathbb{R}^{n_{\mathcal{I}} \times n_{\mathcal{I}}}$. The identity $M M^{-1} = \mathrm{Id}_n$ gives, in its first block column,

$$AX + C^{\mathsf{T}} Y = \mathrm{Id}_{n_{\mathcal{I}}}, \qquad CX + BY = 0.$$

From the second equation $Y = -B^{-1} C X$; substituting into the first, $\big(A - C^{\mathsf{T}} B^{-1} C\big) X = \mathrm{Id}_{n_{\mathcal{I}}}$, i.e. $\Sigma X = \mathrm{Id}$, so $X = \Sigma^{-1}$. □

Combining Proposition **??** with the standard Gaussian conditioning formula $\operatorname{Cov}(X_{\mathcal{I}} \mid X_{\mathcal{J}}) = \big([M^{-1}]_{\mathcal{I}\mathcal{I}}\big)^{-1}$ recovers (**??**) a second time: the conditional covariance is the inverse of the $\mathcal{I} \times \mathcal{I}$ precision block, which is $(\Sigma^{-1})^{-1} = \Sigma$.

Remark 4.7 (Explained plus residual). Proposition **??** splits the block A of the generic matrix M of (**??**) into two PSD pieces,

$$A \;=\; \Pi + \Sigma_{\mathcal{I}|\mathcal{J}}, \qquad \Pi := C^{\mathsf{T}} B^{-1} C = \operatorname{Cov}\big(\mathbb{E}[X_{\mathcal{I}} \mid X_{\mathcal{J}}]\big), \qquad \Sigma_{\mathcal{I}|\mathcal{J}} := \operatorname{Schur}_B(M) = \operatorname{Cov}\big(X_{\mathcal{I}} \mid X_{\mathcal{J}}\big): \tag{4.5}$$

the covariance inside $\mathcal{I}$ that is *explained* through $\mathcal{J}$, plus the covariance of what remains after the best linear prediction. Both pieces are PSD: for any x, $x^{\mathsf{T}}\Pi x = (Cx)^{\mathsf{T}} B^{-1}(Cx) \geq 0$ since $B \succ 0$, and $\Sigma_{\mathcal{I}|\mathcal{J}} \succeq 0$ by Theorem **??** whenever $M \succeq 0$. The Ω-specialised version of (**??**) appears as (**??**) below. This is the analysis-of-variance decomposition of multivariate regression, and it is the load-bearing structure of Sections **??**–**??**: once B and C are frozen, Π is frozen, and *all* remaining freedom lives in the residual term.

4.3 One entry at a time: the Schur bound

The smallest possible modification changes a single correlation entry and keeps everything else. This is precisely the bump that defines Cega hedging in Chapters 1–2 — a one-correlation-point move of one pair — and the theorem below says exactly how far such a bump may go before the model leaves the elliptope.

Let $\Omega \in \mathcal{E}_n$ and fix two distinct indices i, j. Reorder the names so that i and j come first, and let $\mathcal{R} = \{1, \dots, n\} \setminus \{i, j\}$ be the conditioning set. In this ordering,

$$\Omega \;=\; \begin{pmatrix} 1 & \rho & u^{\mathsf{T}} \\ \rho & 1 & v^{\mathsf{T}} \\ u & v & B \end{pmatrix}, \qquad u = \Omega_{\mathcal{R}i}, \quad v = \Omega_{\mathcal{R}j}, \quad B = \Omega_{\mathcal{R}\mathcal{R}}, \tag{4.6}$$

where $\rho := \Omega_{ij}$ is the entry allowed to move, and u, v, B are held fixed.

Assumption 4.8 (Nondegenerate conditioning block)**.** Throughout Sections **??**–**??**, the fixed block B is positive definite, $B \succ 0$. (If B is merely PSD and singular, some name in the conditioning set is a deterministic linear combination of the others; remove the redundant names first.)

Define the scalars

$$q_u := u^{\mathsf{T}} B^{-1} u, \qquad q_v := v^{\mathsf{T}} B^{-1} v, \qquad \bar{\rho}_{ij} := u^{\mathsf{T}} B^{-1} v, \qquad \eta_{ij} := \sqrt{(1-q_u)(1-q_v)}. \tag{4.7}$$

Theorem 4.9 (Schur bound for a single entry)**.** *Let Ω be as in* (**??**) *with $B \succ 0$, and suppose $q_u \leq 1$ and $q_v \leq 1$ (automatic when the unmodified matrix is admissible). Then the matrix Ω with entry $\Omega_{ij} = \rho$ is an admissible correlation matrix if and only if*

$$\bar{\rho}_{ij} - \eta_{ij} \;\leq\; \rho \;\leq\; \bar{\rho}_{ij} + \eta_{ij}. \tag{4.8}$$

Proof. First, the parenthetical in the hypothesis. If the unmodified matrix is admissible, its principal submatrix on the index set $\{i\} \cup \mathcal{R}$ is PSD; applying Theorem **??** to that submatrix with conditioning block $B \succ 0$ shows that its Schur complement, the scalar $1 - u^{\mathsf{T}} B^{-1} u = 1 - q_u$, is nonnegative, i.e. $q_u \leq 1$; likewise $q_v \leq 1$ from the submatrix on $\{j\} \cup \mathcal{R}$.

For the main statement, apply Theorem **??** with $\mathcal{I} = \{i, j\}$ and $\mathcal{J} = \mathcal{R}$. The Schur complement of B is the 2×2 matrix

$$\Sigma_{ij|\mathcal{R}} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} - \begin{pmatrix} u^{\mathsf{T}} \\ v^{\mathsf{T}} \end{pmatrix} B^{-1} \begin{pmatrix} u & v \end{pmatrix} = \begin{pmatrix} 1 - q_u & \rho - \bar{\rho}_{ij} \\ \rho - \bar{\rho}_{ij} & 1 - q_v \end{pmatrix}.$$

A symmetric 2×2 matrix is PSD if and only if both diagonal entries and the determinant are nonnegative. The diagonal conditions are the hypotheses $q_u \le 1$, $q_v \le 1$; the determinant condition is

$$(1 - q_u)(1 - q_v) - (\rho - \bar{\rho}_{ij})^2 \ \ge\ 0 \quad \iff \quad |\rho - \bar{\rho}_{ij}| \ \le\ \eta_{ij},$$

which is (??). Unit diagonal and symmetry are untouched by the modification. □

Corollary 4.10 (Partial-correlation coordinate)**.** *Assume additionally $q_u < 1$, $q_v < 1$. Then every admissible value of the entry can be written*

$$\rho \ =\ \bar{\rho}_{ij} + \eta_{ij}\, t, \qquad t \in [-1, 1]. \tag{4.9}$$

On the open *interval $t \in (-1, 1)$ the matrix is positive definite — $\Sigma_{ij|\mathcal{R}}$ has positive diagonal ($q_u, q_v < 1$) and determinant $\eta_{ij}^2(1 - t^2) > 0$, hence $\Sigma_{ij|\mathcal{R}} \succ 0$ and $\Omega \succ 0$ by Theorem **??** — so $\Theta := \Omega^{-1}$ exists and the coordinate is the partial correlation:*

$$t \ =\ \rho_{ij\cdot\mathcal{R}} \ :=\ -\frac{\Theta_{ij}}{\sqrt{\Theta_{ii}\Theta_{jj}}} \ =\ \frac{\rho - \bar{\rho}_{ij}}{\eta_{ij}}, \qquad t \in (-1, 1). \tag{4.10}$$

The endpoints $t = \pm 1$ are exactly the singular boundary of the elliptope: there $\Sigma_{ij|\mathcal{R}}$, and hence Ω, drops rank, Θ is undefined, and the identity (??) holds only in the limit $t \to \pm 1$ (both sides being continuous in ρ on the open interval).

Proof. The parametrisation (??) over the closed interval is a restatement of Theorem **??**. Now fix $|t| < 1$, so that $\Omega \succ 0$ as computed in the statement and Θ exists. By Proposition **??** (with i, j ordered first), the top-left 2×2 block of Θ is $\Sigma_{ij|\mathcal{R}}^{-1}$. Inverting the 2×2 matrix from the previous proof,

$$\Sigma_{ij|\mathcal{R}}^{-1} = \frac{1}{\det \Sigma_{ij|\mathcal{R}}} \begin{pmatrix} 1 - q_v & -(\rho - \bar{\rho}_{ij}) \\ -(\rho - \bar{\rho}_{ij}) & 1 - q_u \end{pmatrix},$$

so

$$-\frac{\Theta_{ij}}{\sqrt{\Theta_{ii}\Theta_{jj}}} = \frac{\rho - \bar{\rho}_{ij}}{\sqrt{(1 - q_u)(1 - q_v)}} = \frac{\rho - \bar{\rho}_{ij}}{\eta_{ij}},$$

the determinant prefactors cancelling in the ratio. Theorem **??** says admissibility is $|t| \le 1$, with equality forcing $\det \Sigma_{ij|\mathcal{R}} = 0$. □

The reading of (??) is the Bayesian one: $\bar{\rho}_{ij}$ is the correlation between i and j *already implied by their common exposure to the rest of the market* (the explained part, through Π), and η_{ij} is the room left over — the geometric mean of the two conditional standard deviations. The market may quote any residual (partial) correlation t in $[-1, 1]$ on top of the explained centre; it may not quote more than perfect conditional correlation.

Remark 4.11 (Rung 2 recovered). For $n = 2$ the conditioning set $\mathcal{R}$ is empty: $q_u = q_v = 0$, $\bar{\rho}_{ij} = 0$, $\eta_{ij} = 1$, and (??) collapses to $-1 \le \rho \le 1$ — the two-name admissibility interval of Chapter 1, whose upper endpoint is $\xi = 0$. The ladder is consistent: rung 2 is rung 3 with nothing to condition on.

Example 4.12 (A three-name Schur bound, worked in full)**.** Take $n = 3$ and let name 3 be the conditioning set for the pair $(1, 2)$, with

$$\Omega_{13} = 0.6, \qquad \Omega_{23} = 0.5, \qquad B = (1),$$

so $u = (0.6)$, $v = (0.5)$. The scalars (**??**) are

$$q_u = 0.36, \qquad q_v = 0.25, \qquad \bar{\rho}_{12} = 0.6 \times 0.5 = 0.30, \qquad \eta_{12} = \sqrt{0.64 \times 0.75} = \sqrt{0.48} \approx 0.6928.$$

Theorem **??** gives the feasible interval

$$\rho_{12} \in [\,0.30 - 0.6928,\ 0.30 + 0.6928\,] = [-0.3928,\ 0.9928].$$

Cross-check against the determinant of the full 3×3 matrix, $\det \Omega = 1 + 2\rho_{12}\Omega_{13}\Omega_{23} - \rho_{12}^2 - \Omega_{13}^2 - \Omega_{23}^2 = -\rho_{12}^2 + 0.6\,\rho_{12} + 0.39$: its roots are $\rho_{12} = \big(0.6 \pm \sqrt{0.36 + 1.56}\big)/2 = (0.6 \pm 1.3856)/2$, i.e. -0.3928 and 0.9928 — the interval endpoints, as they must be. A quoted $\rho_{12} = 0.80$ is admissible, with partial-correlation coordinate $t = (0.80 - 0.30)/0.6928 \approx 0.72$; a quoted $\rho_{12} = -0.50$ is *not* admissible: no joint distribution of the three names exists with these pairwise correlations, however plausible each pair looks in isolation. The interval is asymmetric: the common exposure to name 3 shifts it upward, squeezing out anticorrelation while leaving near-perfect positive correlation available.

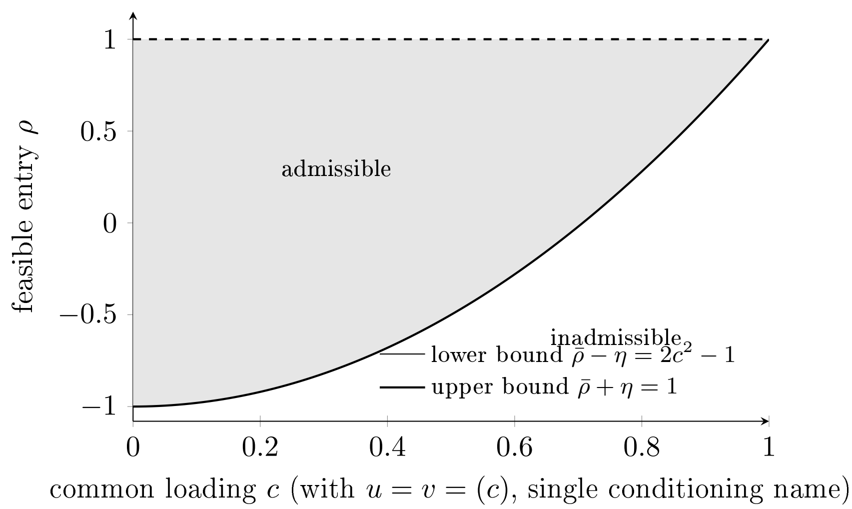

Figure 4.2: The Schur interval (**??**) for a pair with a symmetric loading c on a single common conditioning name: $\bar{\rho} = c^2$, $\eta = 1 - c^2$, feasible range $[2c^2 - 1,\ 1]$. A strong common driver never forbids $\rho = 1$, but it progressively squeezes out anticorrelation: at $c = 0.9$ the pair cannot be quoted below 0.62.

Remark 4.13 (Consequences for Cega bumps). Cega is quoted per correlation point: a bump of 0.01 in one entry of Ω. Theorem **??** is the statement that this bump is only well-defined while $\rho \pm 0.01$ stays inside $[\bar{\rho}_{ij} - \eta_{ij},\ \bar{\rho}_{ij} + \eta_{ij}]$. Near the boundary — high $|t|$ in (**??**) — a symmetric bump exits the elliptope on one side, and the sensitivity must be computed one-sided, or better, in the coordinate t, which is unconstrained on $(-1, 1)$ and geometry-respecting by construction.

Remark 4.14 (Conditioning of the fixed block B). Assumption **??** excludes *exact* singularity of B only. Near-singularity is not excluded, and it is the operationally dangerous regime: whenever the rest-of-book correlations crowd upward — the ambient market state in exactly the crisis scenarios for which the stress machinery of Section **??** is designed — $\mu_{\min}(B)$ collapses and every quantity built on B^{-1} inflates. A single rest-of-book pair quoted at 0.99 already puts $\mu_{\min}(B)$ at order 10^{-2} and entries of B^{-1} at order 10^2; the scalars q_u, q_v, $\bar{\rho}_{ij}$ are then differences of $O(10^2)$ quantities, and the interval endpoints $\bar{\rho}_{ij} \pm \eta_{ij}$ lose precision to catastrophic cancellation — even though the bound itself remains exact and admissibility of Ω still guarantees $q_u \le 1$, $q_v \le 1$. Three implementation rules follow. First, never form B^{-1} explicitly: compute q_u, q_v, $\bar{\rho}_{ij}$ through a stable factorisation of B (a Cholesky solve). Second, report $\mu_{\min}(B)$, or the condition number $\kappa(B)$, alongside every Schur interval, and flag the interval as numerically suspect when $\kappa(B)$ exceeds the implementation's threshold. Third, the eigenvalue floor of Remark **??** below conditions the *modified* block only and never touches B; it is no defence here. The fragility is worst precisely where the tool is most wanted; it must be surfaced, not assumed away.

4.4 Blockwise modification and conditional correlation

The construction now scales from one entry to a whole sub-block. Partition the names into a block $\mathcal{I}$ to be modified (say, the equities in a hybrid basket) and its complement $\mathcal{J}$ to be kept exactly as quoted (say, the FX names, plus every equity–FX cross correlation). After reordering,

$$\Omega \;=\; \begin{pmatrix} A & C^{\mathsf T} \\ C & B \end{pmatrix}, \qquad A = \Omega_{\mathcal{I}\mathcal{I}}, \quad B = \Omega_{\mathcal{J}\mathcal{J}}, \quad C = \Omega_{\mathcal{J}\mathcal{I}}, \tag{4.11}$$

with $B \succ 0$ (Assumption **??**). The explained/residual split (**??**) reads

$$A \;=\; \Pi + \Sigma_{\mathcal{I}|\mathcal{J}}, \qquad \Pi = C^{\mathsf T} B^{-1} C, \qquad \Sigma_{\mathcal{I}|\mathcal{J}} = \mathrm{Schur}_B(\Omega)\,. \tag{4.12}$$

Theorem 4.15 (Blockwise modification criterion)**.** *Fix $B \succ 0$ and C, and let A_{new} be symmetric. The modified matrix*

$$\Omega_{\text{new}} = \begin{pmatrix} A_{\text{new}} & C^{\mathsf T} \\ C & B \end{pmatrix}$$

is an admissible correlation matrix if and only if

$$\Sigma_{\text{new}} := A_{\text{new}} - \Pi \;\succeq\; 0 \qquad \textit{and} \qquad \mathrm{diag}(\Sigma_{\text{new}}) = \mathbf{1} - \mathrm{diag}(\Pi). \tag{4.13}$$

Proof. Since B and C are unchanged, Π is unchanged, and $\mathrm{Schur}_B(\Omega_{\text{new}}) = A_{\text{new}} - \Pi = \Sigma_{\text{new}}$. By Theorem **??**, $\Omega_{\text{new}} \succeq 0 \iff \Sigma_{\text{new}} \succeq 0$. The block B keeps its unit diagonal; the unit-diagonal requirement on A_{new} is $\mathrm{diag}(A_{\text{new}}) = \mathrm{diag}(\Pi) + \mathrm{diag}(\Sigma_{\text{new}}) = \mathbf{1}$, which is the second condition of (**??**). Symmetry of Ω_{new} is symmetry of A_{new}. □

The freedom left inside the block is therefore exactly: a PSD matrix with a *prescribed* diagonal. The natural coordinates for that object are a correlation matrix and a fixed diagonal scaling — and this is where the ladder's recursive structure appears: the admissible modifications of a block of an n-name correlation matrix are themselves parametrised by a smaller correlation matrix.

Definition 4.16 (Conditional correlation matrix)**.** Let $\Sigma_{\mathcal{I}|\mathcal{J}}$ be as in (**??**) and suppose its diagonal entries — the conditional variances $D_{ii} = 1 - \Pi_{ii}$ — are strictly positive. Write $D := \mathrm{diag}\big(\Sigma_{\mathcal{I}|\mathcal{J}}\big)$ for the diagonal matrix of conditional variances. The *conditional correlation matrix* of the block $\mathcal{I}$ given $\mathcal{J}$ is

$$R \;:=\; D^{-1/2}\, \Sigma_{\mathcal{I}|\mathcal{J}}\, D^{-1/2} \;\in\; \mathcal{E}_{n_{\mathcal{I}}}. \tag{4.14}$$

It is the correlation matrix of the regression residuals $X_{\mathcal{I}} - G X_{\mathcal{J}}$ of Proposition **??**.

Proposition 4.17 (The block's only degree of freedom). *Fix $B \succ 0$ and C with $\Pi_{ii} < 1$ for all $i \in \mathcal{I}$, and let $D = \operatorname{diag}(\mathbf{1} - \operatorname{diag}(\Pi))$. The map*

$$R \longmapsto A_{\text{new}}(R) \;=\; \Pi + D^{1/2} R\, D^{1/2}$$

is a bijection between the elliptope $\mathcal{E}_{n_\mathcal{I}}$ and the set of admissible replacement blocks A_{new} in Theorem **??**.

Proof. If $R \in \mathcal{E}_{n_\mathcal{I}}$ then $\Sigma_{\text{new}} = D^{1/2}RD^{1/2}$ is PSD (congruence by the invertible $D^{1/2}$ preserves positivity) and $(D^{1/2}RD^{1/2})_{ii} = D_{ii}R_{ii} = D_{ii} = 1-\Pi_{ii}$, so both conditions of (**??**) hold. Conversely, any admissible A_{new} has $\Sigma_{\text{new}} = A_{\text{new}} - \Pi \succeq 0$ with $\operatorname{diag}(\Sigma_{\text{new}}) = \operatorname{diag}(D)$, and $R = D^{-1/2}\Sigma_{\text{new}}D^{-1/2}$ is PSD with unit diagonal. The two maps are mutually inverse. □

Everything else — B, C, hence Π, G, and D — is pinned by the part of the market left untouched. The modeller who wants to restructure dependence inside a block, without disturbing a single quoted correlation outside it, has exactly one dial: the conditional correlation matrix R.

4.5 Convex blending: stressing a block toward comonotonicity

The canonical application of Proposition **??** is the correlation stress: in a crisis scenario the names inside $\mathcal{I}$ should become more strongly dependent, while the quoted structure of the rest of the book — block B and all cross-correlations C — must be held exactly at market. The construction is a convex blend of the conditional correlation matrix toward the comonotone extreme.

Theorem 4.18 (Convex blending). *Let $R_0 \in \mathcal{E}_{n_\mathcal{I}}$ be the initial conditional correlation matrix* (**??**)*, and for $\lambda \in [0,1]$ define*

$$R(\lambda) \;=\; (1-\lambda)\, R_0 + \lambda\, \mathbf{1}\mathbf{1}^\mathsf{T}, \qquad A(\lambda) \;=\; \Pi + D^{1/2}R(\lambda)\, D^{1/2}, \qquad \Omega(\lambda) \;=\; \begin{pmatrix} A(\lambda) & C^\mathsf{T} \\ C & B \end{pmatrix}. \tag{4.15}$$

Then for every $\lambda \in [0,1]$, $\Omega(\lambda)$ is an admissible correlation matrix, it agrees with Ω outside the block $\mathcal{I} \times \mathcal{I}$, and $\Omega(0) = \Omega$.

Proof. The all-ones matrix $\mathbf{1}\mathbf{1}^\mathsf{T}$ is symmetric, rank one, with eigenvalues $\{n_\mathcal{I}, 0, \dots, 0\}$, hence PSD, and has unit diagonal; so it lies in $\mathcal{E}_{n_\mathcal{I}}$. The elliptope is convex (Definition **??**), so $R(\lambda) \in \mathcal{E}_{n_\mathcal{I}}$ for all $\lambda \in [0,1]$: explicitly, $x^\mathsf{T}R(\lambda)x = (1-\lambda)\, x^\mathsf{T}R_0x + \lambda\,(\mathbf{1}^\mathsf{T}x)^2 \geq 0$ and $\operatorname{diag}(R(\lambda)) = (1-\lambda)\mathbf{1} + \lambda\mathbf{1} = \mathbf{1}$. Proposition **??** then delivers admissibility of $\Omega(\lambda)$; blocks B and C are untouched by construction, and $\lambda = 0$ recovers $A(0) = \Pi + D^{1/2}R_0D^{1/2} = \Pi + \Sigma_{\mathcal{I}|\mathcal{J}} = A$. □

As $\lambda \uparrow 1$ the block's residuals become perfectly conditionally correlated: given the rest of the market, the names in $\mathcal{I}$ move as one. The stress is *maximal* in a precise sense — $\lambda = 1$ puts R at the comonotone vertex $\mathbf{1}\mathbf{1}^\mathsf{T}$, a rank-one point of the elliptope's boundary, and in fact an extreme point: a rank-r correlation matrix is an extreme point of $\mathcal{E}_n$ exactly when $r(r+1)/2 \leq n$ (Grone, Pierce & Watkins, *Extremal correlation matrices*, Linear Algebra Appl. 134 (1990), 63–70), and rank one always qualifies — and *surgical* in an equally precise sense: no quoted correlation outside the block moves at all.

Example 4.19 (Blending traverses the Schur interval). Return to Example **??** with $\mathcal{I} = \{1,2\}$, $\mathcal{J} = \{3\}$, and take the initially quoted pair correlation to be exactly the explained centre, $\rho_{12} = \bar{\rho}_{12} = 0.30$. Then

$$\Pi = \begin{pmatrix} 0.36 & 0.30 \\ 0.30 & 0.25 \end{pmatrix}, \qquad \Sigma_{\mathcal{I}|\mathcal{J}} = A - \Pi = \begin{pmatrix} 0.64 & 0 \\ 0 & 0.75 \end{pmatrix}, \qquad D = \begin{pmatrix} 0.64 & 0 \\ 0 & 0.75 \end{pmatrix}, \qquad R_0 = \mathrm{Id}_2 :$$

the two names are conditionally *uncorrelated* given name 3; their entire quoted correlation of 0.30 is explained through the common name. Blending per (**??**) gives $R(\lambda) = \left(\begin{smallmatrix} 1 & \lambda \\ \lambda & 1 \end{smallmatrix}\right)$, so

$$\left(D^{1/2}R(\lambda)D^{1/2}\right)_{12} = \lambda\sqrt{0.64 \times 0.75} = 0.6928\,\lambda, \qquad A(\lambda)_{12} = 0.30 + 0.6928\,\lambda.$$

The blend traverses exactly the upper half of the Schur interval of Example **??**, with λ coinciding with the partial-correlation coordinate t of Corollary **??**; at $\lambda = 1$ the quoted pair correlation reaches the boundary value 0.9928 and $\Omega(1)$ is singular. The single-entry bound and the block blend are one construction seen at two scales.

4.5.1 Approach to singularity and the choice of λ

The comonotone extreme $\mathbf{1}\mathbf{1}^\top$ has rank one, so if $R_0 \succ 0$ and λ is close to 1, the blend $R(\lambda)$ is close to rank one and the conditional covariance $D^{1/2}R(\lambda)D^{1/2}$ is nearly singular. $\Omega(\lambda)$ remains admissible — Theorem **??** is uniform on $[0,1]$ — but it may cease to be *positive definite*, and with it go the precision matrix, the Gaussian density, and the Cholesky factor a Monte Carlo engine needs.

Example 4.20 (Equicorrelated block: exact eigenvalues along the blend)**.** Let R_0 be equicorrelated on $n_{\mathcal{I}} = 3$ names with common conditional correlation $r = 0.4$. An equicorrelated matrix with parameter r has eigenvalues $1 + (n_{\mathcal{I}} - 1)r$ (eigenvector $\mathbf{1}$) and $1 - r$ (multiplicity $n_{\mathcal{I}} - 1$). Blending preserves equicorrelation with parameter $r(\lambda) = (1-\lambda)r + \lambda = 0.4 + 0.6\lambda$, so

$$\mu_{\max}\big(R(\lambda)\big) = 1 + 2\,r(\lambda) = 1.8 + 1.2\lambda, \qquad \mu_{\min}\big(R(\lambda)\big) = 1 - r(\lambda) = 0.6\,(1-\lambda),$$

the latter with multiplicity two. The smallest eigenvalue decays *linearly* to zero: at $\lambda = 1$ the spectrum is $\{3, 0, 0\}$, the rank-one comonotone limit. The linear decay to zero is generic, not an artefact of equicorrelation: for any $R_0 \succ 0$, $R(\lambda) \succeq (1-\lambda)R_0$ gives $\mu_{\min}(R(\lambda)) \geq (1-\lambda)\,\mu_{\min}(R_0)$. For the matching upper bound, write $R(\lambda) = M_0 + ww^\top$ with $M_0 = (1-\lambda)R_0$ and $w = \sqrt{\lambda}\,\mathbf{1}$: the interlacing theorem for a rank-one positive-semidefinite update states $\mu_{\min}(M_0 + ww^\top) \leq \mu_2(M_0)$, where μ_2 denotes the second-smallest eigenvalue (Horn & Johnson, *Matrix Analysis*, 2nd ed., Corollary 4.3.9), and $\mu_2(M_0) = (1-\lambda)\,\mu_2(R_0)$, so $\mu_{\min}(R(\lambda)) \leq (1-\lambda)\,\mu_2(R_0)$. The smallest eigenvalue is thus pinched between two linear functions of $(1-\lambda)$ and vanishes at $\lambda = 1$ exactly to first order; the closed-form floor $(1-\lambda)\,\mu_{\min}(R_0)$ is available for any block. As a consistency check, when $\mu_{\min}(R_0)$ has multiplicity at least two, $\mu_2(R_0) = \mu_{\min}(R_0)$, the two bounds coincide, and $\mu_{\min}(R(\lambda)) = (1-\lambda)\,\mu_{\min}(R_0)$ exactly — which is precisely the equicorrelated computation above (multiplicity $n_{\mathcal{I}} - 1 = 2$).

Remark 4.21 (Practical selection of the blend weight). When strict positive definiteness of $\Omega(\lambda)$ is required (an invertible covariance, a Cholesky factorisation, a well-conditioned precision matrix), impose an eigenvalue floor $\epsilon > 0$. Two floors must be distinguished, because they are not equivalent.

Conditional-covariance floor: require $\mu_{\min}\big(D^{1/2}R(\lambda)D^{1/2}\big) \geq \epsilon$. Here a conservative closed-form cap is available, from two one-line inequalities. First, $R(\lambda) \succeq (1-\lambda)R_0$ (the omitted term $\lambda\,\mathbf{1}\mathbf{1}^\top$ is PSD), so $\mu_{\min}(R(\lambda)) \geq (1-\lambda)\,\mu_{\min}(R_0)$. Second, for any unit vector x, $x^\top D^{1/2}RD^{1/2}x \geq \mu_{\min}(R)\,\|D^{1/2}x\|^2 \geq \mu_{\min}(R)\,\mu_{\min}(D)$, so $\mu_{\min}\big(D^{1/2}RD^{1/2}\big) \geq \mu_{\min}(D)\,\mu_{\min}(R)$. Chaining the two, the floor holds whenever $\lambda \leq 1 - \epsilon/\big(\mu_{\min}(D)\,\mu_{\min}(R_0)\big)$. This cap is scoped to the conditional-covariance floor and to it alone.

Ω-level floor: require $\mu_{\min}(\Omega(\lambda)) \geq \epsilon$ — this is the floor the Monte Carlo engine actually needs, since $\Omega(\lambda)$ is the matrix it Cholesky-factorises. The cap above does *not* deliver it: the congruence of Lemma **??** preserves inertia, not eigenvalues, so $\mu_{\min}(\Omega(\lambda))$ can lie far below $\mu_{\min}\big(\Sigma(\lambda) \oplus B\big)$

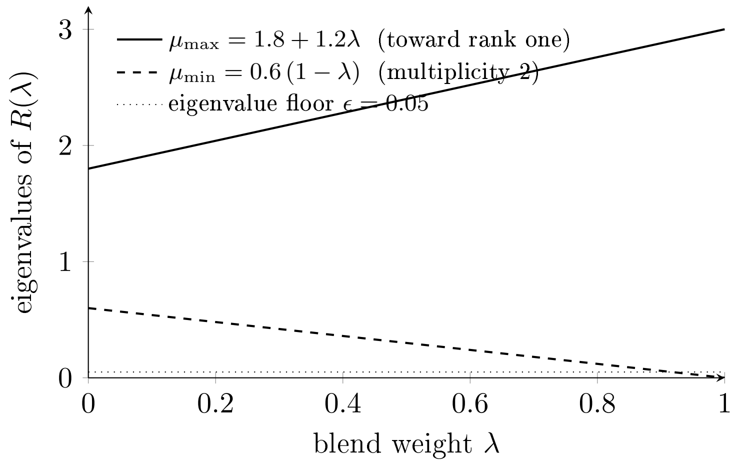

Figure 4.3: Spectrum of the blended conditional correlation matrix of Example **??** ($n_{\mathcal{I}} = 3$, $r = 0.4$). The smallest eigenvalue decays linearly to zero; a floor ϵ caps the usable stress at $\lambda^* = 1-\epsilon/(1-r)$, here $\lambda^* \approx 0.917$. This λ^* floors the *bare* conditional correlation $R(\lambda)$ only and is illustrative; the object a Monte Carlo engine Cholesky-factorises is $\Omega(\lambda)$, whose operational floor is the $\mu_{\min}(D)$-weighted cap of Remark **??**.

with $\Sigma(\lambda) = D^{1/2}R(\lambda)D^{1/2}$. What the congruence does give (Ostrowski's theorem on congruence transformations) is the corrected bound

$$\mu_{\min}\big(\Omega(\lambda)\big) \;\geq\; \sigma_{\min}(L)^2 \cdot \min\big(\mu_{\min}(\Sigma(\lambda)),\ \mu_{\min}(B)\big),$$

with L the unit-triangular factor of (**??**) (fixed along the blend, since B and C are). Since $\sigma_{\min}(L)$ must itself be computed, an implementation either uses this corrected bound or — simpler, and recommended — checks $\mu_{\min}(\Omega(\lambda))$ (or the Cholesky factorisation of $\Omega(\lambda)$) directly at the target λ, backtracking λ toward zero until the floor is restored.

The floor ϵ is a numerical-conditioning parameter of the implementation, not of the theory: every $\lambda \in [0,1]$ is admissible in exact arithmetic.

4.6 Why not global partial correlations

Corollary **??** suggests a tempting alternative: since the partial correlations $\rho_{ij\cdot\mathcal{R}}$ are each free in $[-1,1]$, stress dependence by editing the global partial-correlation matrix

$$P_{ij} \;=\; -\frac{\Theta_{ij}}{\sqrt{\Theta_{ii}\,\Theta_{jj}}}, \qquad \Theta = \Omega^{-1},$$

and mapping back to Ω.

The obstruction is locality. The map from P back to Ω runs through a matrix inversion, and inversion is a *global* operation: perturbing even a single entry P_{ij} with $i,j \in \mathcal{I}$ changes, in general, *every* entry of the reconstructed Ω — including the block B and the cross-block C that the mandate requires held at market. (This is visible already in Proposition **??**: the $\mathcal{I}\times\mathcal{I}$ block of Θ is $\Sigma^{-1}_{\mathcal{I}|\mathcal{J}}$, which entangles A, B, and C; inverting a modified Θ redistributes the change across all three.) Partial-correlation coordinates are the right chart for checking admissibility of a *given* matrix and for one-entry moves; they are the wrong chart for constrained blockwise modification. The Schur-based construction of Sections **??**–**??** exists precisely because it localises: it fixes B and C *exactly*, by construction, and exposes the residual freedom in coordinates (R) where admissibility is manifest.

4.7 Closing the ladder

The three rungs now stand in one pattern. At rungs 2 and 3, admissibility is positive semidefiniteness of a conditional object and the boundary of the feasible set is its loss of rank; at rung 1 the feasible set is a different convex cone, and the correspondence is one of shape, not an identity:

- *One name.* The density constraint of Volume I, Chapter 2 has the same shape one level down: the feasible set is the cone of *nonnegative measures* (risk-neutral densities), and the no-butterfly condition — convexity of the call price $C(K,T)$ in K — is nonnegativity of a *linear* functional of the density (the price of the butterfly portfolio), not of a quadratic form. The cone of nonnegative measures is not the PSD cone, and no identity between the rungs is claimed; the shared shape is a convex cone constraint whose boundary is degeneracy — loss of mass of the density at rung 1, loss of rank of the matrix at rungs 2 and 3.

- *Two names.* $|\rho| \leq 1$ is Theorem **??** with an empty conditioning set (Remark **??**); its upper boundary $\rho = 1$ is $\xi = 0$, the collapse of the basket vol spread $\xi = \bar{\sigma} - \sigma_B$ of Chapter 1. Admissibility is also what keeps the Margrabe volatility of Chapter 3 real: for $\rho \in [-1,1]$, $\Sigma_M^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 = (\sigma_1 - \sigma_2)^2 + 2\sigma_1\sigma_2(1-\rho) \geq 0$, with equality only at $\rho = 1$, $\sigma_1 = \sigma_2$.

- *n names.* $\Omega \in \mathcal{E}_n$, certified blockwise by Theorem **??**, moved one entry at a time within the Schur interval (**??**), and stressed one block at a time by the blend (**??**) — without ever leaving the elliptope, and without touching what was quoted outside the block.

Two forward couplings deserve to be stated explicitly, because they connect this static chapter to the dynamics of Chapter 2.

Remark 4.22 (Admissibility under the quoting conventions)*.* The entries of Ω do not sit still: the constituent volatilities are the per-name ATM vols $\sigma_{\mathrm{ATM},i}(\ell_{F,i}, T)$ of constraint C1, and as the forwards move, the vols move. Under sticky-ρ the correlation is held fixed, $\mathrm{d}\rho = 0$, and admissibility of Ω is a static check. Under sticky-CVC the correlation is the determined function $\rho(\sigma_1, \sigma_2, \xi)$ — the spread ξ is held invariant and ρ moves with the vols, which is exactly what generates the shadow Greeks of Chapter 2. Every convention-induced correlation path $\rho(\cdot)$ must remain inside the Schur interval (**??**) *given the contemporaneous rest of the matrix*: the interval is not static — under a book-wide convention the bounds $\bar{\rho}_{ij}$ and η_{ij} are themselves functions of the instantaneous u, v, B, which co-move with the pair being tracked as the vols move — so the check is pointwise in time, with every quantity evaluated on the same instant's matrix. For a pair embedded in an n-name book, the binding constraint on a sticky-CVC-driven correlation move is $\rho \leq \bar{\rho}_{ij} + \eta_{ij}$, and the ceiling obeys

$$\bar{\rho}_{ij} + \eta_{ij} \;\leq\; 1 \qquad \text{always,}$$

by two applications of Cauchy–Schwarz: $\bar{\rho}_{ij} = u^{\mathsf{T}} B^{-1} v \leq \sqrt{q_u q_v}$ (Cauchy–Schwarz in the B^{-1} inner product), and $\sqrt{q_u q_v} + \sqrt{(1-q_u)(1-q_v)} \leq \sqrt{(q_u + 1 - q_u)(q_v + 1 - q_v)} = 1$ (Cauchy–Schwarz in $\mathbb{R}^2$ applied to $(\sqrt{q_u}, \sqrt{1-q_u})$ and $(\sqrt{q_v}, \sqrt{1-q_v})$). Equality throughout holds if and only if $u = v$: the first inequality is tight iff u and v are positively parallel in the B^{-1} geometry, the second iff $q_u = q_v$, and together these force $u = v$. So the Schur ceiling is *weakly* tighter than the two-name bound $\rho \leq 1$, and strictly tighter exactly when the two names load *asymmetrically* on the rest of the book ($u \neq v$). $\eta_{ij} < 1$ is *not* the trigger: for symmetric loadings $u = v$ the ceiling is $q_u + (1 - q_u) = 1$ exactly, however strong the common loading — consistent with Figure **??**, whose upper bound is identically 1. Finally, this pointwise check does not settle admissibility of the full book-wide sticky-CVC *path*, with all entries of Ω moving simultaneously as the vols move: that is a coupled dynamic question, open at the level of this static chapter and deferred to the dynamics of Chapter 2. The two-name bound of Chapter 1 is the correct constraint only for a two-name book.

Remark 4.23 (The same identity, one chapter earlier). Proposition **??** is the workhorse of the Bayesian calibration engine of Volume I, Chapter 4, in different clothing: there the joint Gaussian vector is (parameters, observations), the regression matrix G is the gain applied to the innovation e_t, and the posterior covariance of the parameter ket is the Schur complement of the observation block in the joint covariance. Conditioning shrinks covariance by the explained part Π — whether what is being conditioned is a block of names in a correlation matrix or a vector of smile parameters given a day's quotes. The ladder of this chapter and the filter of that one are the same piece of Gaussian linear algebra.

The admissibility theory of the corpus is now closed: densities nonnegative and total variance calendar-monotone on every single-name surface, $\xi \geq 0$ on every pair, and $\Omega \succeq 0$ — blockwise certified, boundary understood, and stressable without leaving the feasible set — on the full book.

Chapter 5

Numerical Methods: Regression Monte Carlo, Control Variates, and Surrogate Pricing

Every price in this corpus so far has been either a closed form or the solution of a well-posed differential equation. This chapter treats the remaining case: a payoff whose price is available only as an expectation, estimated by simulation. The engineering question is not *how to simulate* — the Euler scheme with Itô correction carries the two-asset dynamics — but *how to spend a finite simulation budget so that the estimator is accurate.* The answer developed here is compositional: a complex payoff is never priced from scratch. It is priced *relative to a library of simpler instruments whose prices are already known*, and only the residual is simulated — a random variable engineered to be small, smooth, and cheap to learn.

The mathematics is elementary and is proved in full. The variance-reduction step is an orthogonal projection in L^2; the regularisation of that projection is the same conjugate Gaussian update that calibrates the single-name volatility surface in Volume I; and the surrogate that interpolates the residual across states is Gaussian-process regression, the nonparametric limit of that same Bayesian regression. Three ideas, one spine: *represent, regress, interpolate.*

Remark 5.1 (Local notation for this chapter). This chapter introduces symbols scoped to it. Several reuse glyphs owned by the contract table; each such reuse is *declared here explicitly* (Convention 3(a)) and is local to this chapter. The number of simulated paths is N_{p}; the number of state points on a training grid is N_x; the number of hedge instruments is H. The Monte Carlo path index is $p = 1, \ldots, N_{\mathrm{p}}$ — *not* the retired k, which is not used as an index (ruling R3); the hedge index is $i = 1, \ldots, H$; the state-grid index is $j = 1, \ldots, N_x$. A bare N is never written for a count: $N(\cdot)$ is the standard normal CDF (ruling R5), and it alone owns that glyph.

The hedge payoffs are written $g_1, \ldots, g_H$ — *not* h_i, reserved by the contract for the Epanechnikov kernel bandwidth (owner I.2); the stencil bump of Section **??** is b_x, for the same reason. The hedge-ratio vector is $\lambda \in \mathbb{R}^H$, accompanied by a regression intercept λ_0. The Gaussian-process covariance function is $\mathcal{K}(\cdot, \cdot)$ and its query cross-covariance vector is $\boldsymbol{\kappa}_*$ (not k_*, so that k genuinely never reappears); its signal amplitude is a_{gp} — distinct from the Margrabe argument a (owner II.3, used unaltered in Remark **??**) — and its per-coordinate length scales are $\ell_{\mathrm{gp},d}$, distinct from the surface coordinate $\ell_F = \ln(F/F_{\mathrm{ref}})$. The GP posterior variance is v_{GP}, written with a v because σ is reserved for implied volatility; the Gaussian-regression noise scale is ς (not s, reserved for the smile skew $s(m)$), and the Monte Carlo observation noise on the training grid is ζ_j. Network weights are written w, not θ (reserved for the Black–Scholes theta, owner I.1, which does not appear in this chapter). The diagonal training anchors of Example **??** are S_q, never S_0 (a retired, banned pivot glyph, Convention 7 / ruling R1).

The probability measure is $\mathbb{Q}$ and $L^2(\mathbb{Q})$ denotes the space of square-integrable payoffs; the sample space is *not* named, since its usual glyph Ω is reserved by the contract for the n-name correlation matrix (owner II.4). Inner products of random variables are written as expectations, $\mathbb{E}[UV]$, never in bra-ket: the bra-ket $|\cdot\rangle$, $\langle\cdot|$, Φ is reserved for the feature/parameter calibration block of Volume I (Convention 6).

5.1 The pricing problem as an estimation problem

Fix the risk-neutral filtered probability space carrying the measure $\mathbb{Q}$ and filtration $\{\mathcal{F}_t\}$ (its sample space is left unnamed, per Remark **??**), under the standing assumption of zero rates and zero dividends, so the discount factor is unity and every forward equals its spot at the horizon it is read: $F = S$ (flagged here at its first use; it recurs throughout and is flagged again at each point where it does analytic work, including in Section **??**). Let x_0 denote the current market state — spots, and the volatility-surface and correlation parameters that the dynamics of Volumes I and II consume.

Definition 5.2 (Target and hedge payoffs)**.** Let $P \in L^2(\mathbb{Q})$ be the discounted terminal payoff of the instrument to be priced, and let $g_1, \dots, g_H \in L^2(\mathbb{Q})$ be the discounted terminal payoffs of a *hedge library*: instruments whose arbitrage-free prices

$$\mathrm{Price}(g_i; x_0) \;=\; \mathbb{E}[g_i \mid X_0 = x_0], \qquad i = 1, \dots, H,$$

are known to high accuracy from a closed form, a PDE solve, or an independent converged simulation. The object required is $\mathrm{Price}(P; x_0) = \mathbb{E}[P \mid X_0 = x_0]$. When a library price is itself only an approximation — as the basket leg of Section **??** will be — unbiasedness holds only *relative to the plugged-in price*; the resulting bias is isolated in Remark **??**.

Square-integrability is the only regularity needed: it makes P and the g_i vectors in the Hilbert space $L^2(\mathbb{Q})$ with inner product $\mathbb{E}[UV]$ and it guarantees the projection of Section **??** exists and is unique.

Definition 5.3 (Crude Monte Carlo estimator)**.** Simulate N_p independent paths from x_0 and record the pathwise payoffs $P^{(p)}$, $p = 1, \dots, N_\mathrm{p}$. The crude estimator is the sample mean $\bar{P} = N_\mathrm{p}^{-1} \sum_p P^{(p)}$. It is unbiased with

$$\mathrm{Var}(\bar{P}) \;=\; \frac{\mathrm{Var}(P)}{N_\mathrm{p}}, \qquad \text{standard error} \;=\; \frac{\sqrt{\mathrm{Var}(P)}}{\sqrt{N_\mathrm{p}}}.$$

The $N_\mathrm{p}^{-1/2}$ law is the entire economic problem. Halving the standard error costs four times the paths. For a payoff with large $\mathrm{Var}(P)$ — a spread or worst-of option, whose value is a thin sliver of a large gross exposure — the constant $\sqrt{\mathrm{Var}(P)}$ is punishing. Every method in this chapter attacks that constant, never the exponent.

5.2 Control variates as orthogonal projection

5.2.1 The estimator and its unbiasedness

Definition 5.4 (Control-variate estimator)**.** Given a hedge-ratio vector $\lambda \in \mathbb{R}^H$, define the *residual* random variable

$$R \;=\; R(\lambda) \;:=\; P - \sum_{i=1}^{H} \lambda_i\, g_i \;=\; P - \lambda^\mathsf{T} g, \tag{5.1}$$

and the control-variate estimator

$$\widehat{\mathrm{Price}}(P;x_0) \;:=\; \bar{R} \;+\; \sum_{i=1}^{H} \lambda_i\,\mathrm{Price}(g_i;x_0), \qquad \bar{R} = \frac{1}{N_\mathrm{p}}\sum_{p=1}^{N_\mathrm{p}} R^{(p)}, \tag{5.2}$$

where $R^{(p)} = P^{(p)} - \lambda^\mathsf{T} g^{(p)}$ is the pathwise residual.

Proposition 5.5 (Unbiasedness for every λ)**.** *For any fixed $\lambda \in \mathbb{R}^H$, the estimator (**??**) is unbiased:* $\mathbb{E}[\widehat{\mathrm{Price}}(P;x_0)] = \mathrm{Price}(P;x_0)$.

Proof. Taking risk-neutral expectations of (**??**), $\mathbb{E}[R] = \mathbb{E}[P] - \sum_i \lambda_i \mathbb{E}[g_i] = \mathrm{Price}(P;x_0) - \sum_i \lambda_i \mathrm{Price}(g_i;x_0)$. The sample mean $\bar{R}$ is unbiased for $\mathbb{E}[R]$, and the second term of (**??**) is the deterministic constant $\sum_i \lambda_i \mathrm{Price}(g_i;x_0)$. Adding the two recovers $\mathrm{Price}(P;x_0)$. □

Proposition **??** is the load-bearing fact, with one caveat it makes explicit: it charges the second term of (**??**) with the *exact* hedge price $\mathbb{E}[g_i \mid X_0 = x_0]$. Because the hedge prices are *known*, subtracting the hedges in simulation and adding them back analytically changes the variance of the estimator but never its mean; λ may therefore be chosen purely to minimise variance. If instead an approximate price $\widetilde{\mathrm{Price}}(g_i;x_0)$ is plugged in, the estimator acquires a deterministic bias $\sum_i \lambda_i \big(\widetilde{\mathrm{Price}}(g_i;x_0) - \mathbb{E}[g_i \mid X_0 = x_0]\big)$ that does *not* shrink with N_p; this is the subject of Remark **??**.

5.2.2 The variance-optimal hedge ratio

Write $\tilde{U} := U - \mathbb{E}[U]$ for the centring of a random variable, and let

$$\Sigma_{gg} := \mathrm{Cov}(g) \in \mathbb{R}^{H\times H}, \qquad \Sigma_{gP} := \mathrm{Cov}(g,P) \in \mathbb{R}^H,$$

with entries $(\Sigma_{gg})_{i\ell} = \mathrm{Cov}(g_i, g_\ell)$ and $(\Sigma_{gP})_i = \mathrm{Cov}(g_i, P)$.

Theorem 5.6 (Optimal control variate as L^2 projection)**.** *Suppose Σ_{gg} is nonsingular. Then*

$$\lambda^\star \;=\; \arg\min_{\lambda\in\mathbb{R}^H} \mathrm{Var}\big(P - \lambda^\mathsf{T} g\big)$$

exists, is unique, and is the solution of the normal equations

$$\Sigma_{gg}\,\lambda^\star \;=\; \Sigma_{gP}, \qquad \textit{i.e.} \qquad \lambda^\star = \Sigma_{gg}^{-1}\Sigma_{gP}. \tag{5.3}$$

The optimal residual $R^\star = P - \lambda^{\star\mathsf{T}} g$ is uncorrelated with every hedge*: $\mathrm{Cov}(g_i, R^\star) = 0$ for all i. Its variance is*

$$\mathrm{Var}(R^\star) \;=\; \mathrm{Var}(P) - \Sigma_{gP}^\mathsf{T}\Sigma_{gg}^{-1}\Sigma_{gP} \;=\; \mathrm{Var}(P)\,\big(1-\varrho^2\big), \tag{5.4}$$

where $\varrho^2 := \Sigma_{gP}^\mathsf{T}\Sigma_{gg}^{-1}\Sigma_{gP}/\mathrm{Var}(P) \in [0,1]$ is the squared multiple correlation between P and the hedge span. (The symbol is ϱ, not ρ: ρ is reserved for pairwise asset correlation.) Both $\lambda^\star$ and ϱ^2 depend on P and g only through their centred *second moments, so the optimum is invariant to any additive constant — the reason the empirical fit below must carry an intercept.*

Proof. The map $\lambda \mapsto \mathrm{Var}(\tilde{P} - \lambda^\mathsf{T}\tilde{g}) = \mathrm{Var}(P) - 2\lambda^\mathsf{T}\Sigma_{gP} + \lambda^\mathsf{T}\Sigma_{gg}\lambda$ is a quadratic form with Hessian $2\Sigma_{gg}$, positive definite by hypothesis, hence strictly convex with a unique stationary point. Setting the gradient $-2\Sigma_{gP} + 2\Sigma_{gg}\lambda$ to zero gives (**??**). Orthogonality is the stationarity condition read componentwise: $\mathrm{Cov}(g_i, R^\star) = (\Sigma_{gP} - \Sigma_{gg}\lambda^\star)_i = 0$. Substituting $\lambda^\star$ into the quadratic form yields (**??**). Geometrically, $\sum_i \lambda_i^\star \tilde{g}_i$ is the orthogonal projection of $\tilde{P}$ onto the closed subspace $\mathrm{span}\{\tilde{g}_1,\dots,\tilde{g}_H\} \subset L^2(\mathbb{Q})$, and $\tilde{R}^\star = \tilde{P} - \mathrm{proj}\,\tilde{P}$ is the projection residual, orthogonal to the subspace by construction — the Hilbert-space Pythagorean identity (**??**). □

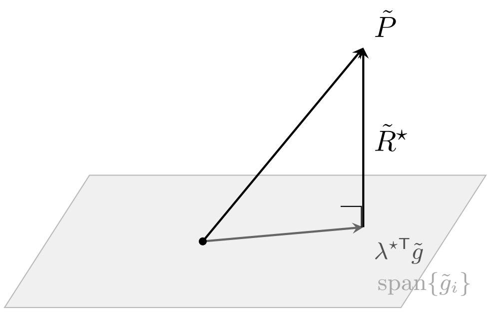

Figure 5.1: Control variate as orthogonal projection in $L^2(\mathbb{Q})$. The residual $\tilde{R}^\star$ is orthogonal to the hedge span; by (**??**) its length squared is $\mathrm{Var}(P)(1-\varrho^2)$. Simulation is spent only on $\tilde{R}^\star$, the component of P that the known hedge prices cannot already supply.

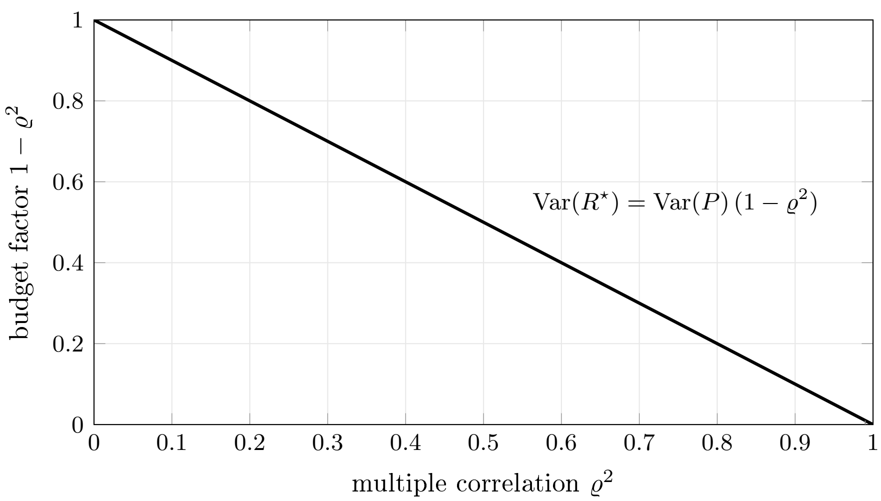

Figure 5.2: The fraction of the crude-Monte-Carlo path budget retained after control-variate correction, as a function of how much of $\mathrm{Var}(P)$ the hedge library explains. The regime that makes surrogate modelling worthwhile is the right edge, $\varrho^2 \to 1$.

Figure **??** is the whole theorem in one picture: P is a vector, the hedges span a plane, and the crude estimator wastes its budget on the part of P that already lies in the plane — the part whose expectation is known exactly and for free.

Corollary 5.7 (Variance-reduction budget)**.** *Under* (**??**)*, the paths required to reach a fixed standard error scale by the factor* $1-\varrho^2$ *relative to crude Monte Carlo. A hedge span capturing* $\varrho^2 = 0.99$ *of the variance of* P *cuts the path count by a factor of* 100 *at equal accuracy.*

This is the "core efficiency gain": the residual $R^\star$ converges rapidly in N_p precisely because its variance has been contracted by $1-\varrho^2$ before a single path is averaged. Figure **??** plots the budget factor.

5.2.3 The empirical regression

In practice Σ_{gg} and Σ_{gP} are unknown and are replaced by their sample analogues on the simulated paths. Because the optimum of Theorem **??** is centring-invariant, the empirical fit must include an *intercept* λ_0; profiling it out is exactly what centres the regressors. Stacking the centred pathwise hedge payoffs into $\tilde{\mathbf{g}} \in \mathbb{R}^{N_\mathrm{p} \times H}$ (each column with its sample mean removed) and the centred targets into $\tilde{\mathbf{P}} \in \mathbb{R}^{N_\mathrm{p}}$, the intercept-augmented ordinary least-squares fit is

$$(\hat{\lambda}_0, \hat{\lambda}) \;=\; \arg\min_{\lambda_0,\lambda} \frac{1}{N_\mathrm{p}} \sum_{p=1}^{N_\mathrm{p}} \big(P^{(p)} - \lambda_0 - \lambda^\mathsf{T} g^{(p)}\big)^2, \qquad \hat{\lambda} \;=\; \big(\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{g}}\big)^{-1}\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{P}}, \quad \hat{\lambda}_0 = \bar{P} - \hat{\lambda}^\mathsf{T}\bar{g}, \tag{5.5}$$

the slope solving the *centred* normal equations (the finite-sample analogue of (**??**)) and converging to $\lambda^\star$ as $N_\mathrm{p} \to \infty$. It is only this centred fit — not an intercept-free one — that recovers the projection ratio of Theorem **??**. This is the regression identified in the source note as "the standard control variate construction, with $\{g_i\}$ serving as control variates for P."

Remark 5.8 (Two caveats). First, estimating $\hat{\lambda}$ on the *same* paths used to average $\bar{R}$ introduces a bias of order $O(1/N_\mathrm{p})$, lower order than the $O(N_\mathrm{p}^{-1/2})$ standard error, and removable by estimating $\hat{\lambda}$ on an independent path batch. Second, when hedges are collinear — a spread option and its two constituent legs, say — $\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{g}}$ is near-singular and $\hat{\lambda}$ explodes, fitting Monte Carlo noise rather than structure. The Moore–Penrose pseudoinverse (a truncated-spectrum solve of (**??**)) is the minimal remedy; the principled one is the prior of the next section.

5.3 Bayesian regularisation of the hedge ratios

The collinearity pathology of Remark **??** is exactly the ill-conditioning that the surface calibration of Volume I controls with a prior. The construction is identical, and it is stated in full so that the control-variate regression and the surface regression are visibly one machine wearing two hats.

Definition 5.9 (Gaussian prior on hedge ratios)**.** Model the pathwise target as an intercept-plus-linear fit with Gaussian noise, $P^{(p)} = \lambda_0 + \lambda^\mathsf{T} g^{(p)} + \varepsilon^{(p)}$ with $\varepsilon^{(p)} \sim \mathcal{N}(0, \varsigma^2)$ i.i.d., give the intercept λ_0 a flat prior, and place a conjugate prior $\lambda \sim \mathcal{N}(\mu_0, \Sigma_0)$ on the slope. Here μ_0 encodes prior beliefs (small or structured hedge positions) and Σ_0 the prior confidence.

Proposition 5.10 (Posterior hedge ratio)**.** *Under Definition* **??***, profiling out the flat-prior intercept centres the data, and the posterior for the slope λ given the simulated data is Gaussian, $\lambda \mid \text{data} \sim \mathcal{N}(\mu_\star, \Sigma_\star)$, with*

$$\Sigma_\star^{-1} = \Sigma_0^{-1} + \frac{1}{\varsigma^2}\,\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{g}}, \qquad \mu_\star = \Sigma_\star\Big(\Sigma_0^{-1}\mu_0 + \frac{1}{\varsigma^2}\,\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{P}}\Big), \tag{5.6}$$

the design matrix and target being the centred $\tilde{\mathbf{g}}$, $\tilde{\mathbf{P}}$ *of* (**??**)*. The posterior mean $\mu_\star$ is the regularised hedge vector used in* (**??**)*.*

Proof. Conjugate Gaussian update: with a flat prior on λ_0, integrating it out (equivalently, maximising over it) replaces $\mathbf{g}, \mathbf{P}$ by their column-centred versions $\tilde{\mathbf{g}}, \tilde{\mathbf{P}}$. The resulting log-posterior in the slope is, up to an additive constant, $-\frac{1}{2\varsigma^2}\|\tilde{\mathbf{P}} - \tilde{\mathbf{g}}\lambda\|^2 - \frac{1}{2}(\lambda - \mu_0)^\mathsf{T}\Sigma_0^{-1}(\lambda - \mu_0)$, a quadratic in λ. Completing the square gives a Gaussian with precision and mean (**??**). □

In the diffuse-prior limit $\Sigma_0^{-1} \to 0$ the posterior mean collapses to $\mu_\star \to (\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{g}})^{-1}\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{P}} = \hat{\lambda}$, the centred OLS fit (**??**), which converges to $\lambda^\star = \Sigma_{gg}^{-1}\Sigma_{gP}$ as $N_\mathrm{p} \to \infty$. The Bayesian layer therefore regularises *the* projection hedge ratio of Theorem **??**, not a different uncentred target — this is what the intercept buys.

Remark 5.11 (Same estimator, three readings). Equation (**??**) is simultaneously (i) the ridge-regularised least-squares solution — take $\mu_0 = 0$, $\Sigma_0 = \tau^2 I$ (with $\tau > 0$ a prior scale, local to this remark, *not* time to expiry, per ruling R6), giving $\mu_\star = (\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{g}} + (\varsigma^2/\tau^2)I)^{-1}\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{P}}$, the penalty ς^2/τ^2 curing the collinearity of Remark **??**; (ii) the matrix-weighted (Tikhonov) shrinkage of the OLS estimator (**??**) toward μ_0, the two combined with weights Σ_0^{-1} and $\varsigma^{-2}\tilde{\mathbf{g}}^\mathsf{T}\tilde{\mathbf{g}}$; and (iii) the identical conjugate update that fits the parameter ket $|\beta\rangle$ to the market volatility quotes through the design matrix Φ in the calibration block of Volume I. The volatility surface and the hedge library are regularised by the same Bayesian arithmetic; only the feature matrix differs.

5.3.1 Scenario enrichment

The source note recommends feeding the regression additional scenarios — perturbed spots, perturbed volatilities, stressed states — beyond samples from the baseline dynamics. In the projection language of Theorem **??** this widens the empirical measure under which the covariances Σ_{gg}, Σ_{gP} are computed, so the estimated $\hat{\lambda}$ is the projection under a *mixture* of market states rather than a single one. The result is a mixture-optimal hedge ratio: variance-optimal for no single state, but a good compromise across the states in the mixture, trading per-state optimality for reusability across the state grid of the next section. Unbiasedness is untouched — Proposition **??** holds for *any* λ — so treating this mixture ratio as slowly varying (or constant) in x costs only variance efficiency, and only where the true $\lambda^\star(x)$ genuinely varies.

5.4 The residual field and its Gaussian-process surrogate

So far λ and the estimate $\bar{R}$ live at a single state x_0. The efficiency gain becomes a *pricing surface* when the state varies.

Definition 5.12 (Residual field)**.** Let x denote the state vector (spots, and any surface/correlation parameters carried by the dynamics). The *residual field* is the conditional expectation

$$R(x) \;:=\; \mathbb{E}\Big[P - \sum_{i=1}^{H} \lambda_i(x)\, g_i \;\Big|\; X_0 = x\Big]. \tag{5.7}$$

Its value at x is what (**??**) estimates when the simulation is launched from $X_0 = x$.

The glyph R is deliberately overloaded across three roles, reconciled here once: $R(\lambda)$ of (**??**) is the pathwise *random* residual at a fixed state; $R(x)$ of (**??**) is its conditional-expectation *field*, a deterministic function of the launch state; and Definition **??** places a prior on this field, making R a *random field* (a Gaussian process) in the modeller's epistemic sense. It is the deterministic field $R(x)$ that the process models.

Two structural facts make R a good regression target. By construction it is *small* — its scale is $\sqrt{\mathrm{Var}(P)(1-\varrho^2)}$, not $\sqrt{\mathrm{Var}(P)}$. And it is *smooth*: the non-smooth features of P (kinks at exercise boundaries, the corner of a spread payoff) are largely inherited by the hedge library, which was chosen for exactly that resemblance, and are subtracted off in (**??**). A small, smooth field on a low-dimensional state space is the textbook setting for Gaussian-process regression.

5.4.1 Gaussian-process regression as nonparametric Bayes

Definition 5.13 (GP prior on the residual field)**.** Model R as a Gaussian process,

$$R(\cdot) \;\sim\; \mathcal{GP}\big(g_0(\cdot), \mathcal{K}(\cdot,\cdot)\big),$$

with mean function g_0 and covariance function $\mathcal{K}$. The canonical choice is the squared-exponential (radial basis) kernel with a signal amplitude a_{gp} and per-coordinate length scales $\ell_{\text{gp},d}$ (both carrying the gp tag to separate them from the Margrabe argument a and the surface coordinate ℓ_F),

$$\mathcal{K}(x,x') \;=\; a_{\text{gp}}^2 \exp\Big(-\tfrac{1}{2}\sum_d \frac{(x_d - x'_d)^2}{\ell_{\text{gp},d}^2}\Big), \tag{5.8}$$

whose hyperparameters $(a_{\text{gp}}, \ell_{\text{gp},d})$ are fitted by marginal likelihood.

Run the control-variate procedure at a design $\{x_j\}_{j=1}^{N_x}$ of states, with a modest path count at each, obtaining noisy observations $\hat{R}_j = R(x_j) + \eta_j$ of the field, where the observation noise $\eta_j \sim \mathcal{N}(0, \zeta_j^2)$ is the Monte Carlo error at x_j, of size $\zeta_j^2 = \mathrm{Var}(R^\star \mid X_0 = x_j)/N_{\text{p}}$. This is the second place the control variate pays off: it lowers not only the estimator variance but the *noise floor of the surrogate's training data.*

Theorem 5.14 (GP posterior)**.** *Write* $\mathbf{K} \in \mathbb{R}^{N_x \times N_x}$ *with* $\mathbf{K}_{j\ell} = \mathcal{K}(x_j, x_\ell)$*, the noise matrix* $\mathbf{E} = \mathrm{diag}(\zeta_1^2, \ldots, \zeta_{N_x}^2)$*, and for a query state* x_* *the vector* $\boldsymbol{\kappa}_* = (\mathcal{K}(x_*, x_j))_{j=1}^{N_x}$*. Then the posterior* $R(x_*) \mid \{\hat{R}_j\}$ *is Gaussian with mean and variance*

$$\hat{R}_{\text{GP}}(x_*) = g_0(x_*) + \boldsymbol{\kappa}_*^\mathsf{T}(\mathbf{K}+\mathbf{E})^{-1}\big(\hat{\mathbf{R}} - \mathbf{g}_0\big), \tag{5.9}$$

$$v_{\text{GP}}(x_*) = \mathcal{K}(x_*, x_*) - \boldsymbol{\kappa}_*^\mathsf{T}(\mathbf{K}+\mathbf{E})^{-1}\boldsymbol{\kappa}_*. \tag{5.10}$$

Proof. The joint law of the training observations $\hat{\mathbf{R}}$ and the field value $R(x_*)$ is Gaussian: $\hat{\mathbf{R}}$ has mean $\mathbf{g}_0$ and covariance $\mathbf{K}+\mathbf{E}$; $R(x_*)$ has mean $g_0(x_*)$ and variance $\mathcal{K}(x_*, x_*)$; their cross-covariance is $\boldsymbol{\kappa}_*$. Conditioning a joint Gaussian on $\hat{\mathbf{R}}$ gives the standard conditional mean and variance, which are (**??**)–(**??**). □

Remark 5.15 (The surrogate is the parametric posterior, taken to the limit). Equation (**??**) is a kernel-weighted interpolant of the training residuals, and (**??**) a closed-form uncertainty band that shrinks near design points and widens in the gaps — the honesty the crude estimator lacks. Structurally it is Proposition **??** with the finite feature vector replaced by the infinite feature map implicit in $\mathcal{K}$: Gaussian-process regression is Bayesian linear regression in an infinite-dimensional feature space, the nonparametric limit of the surface calibration of Volume I.

5.4.2 The composed price and its cost

The full price is reassembled by adding the known hedge prices back to the surrogate residual:

$$\mathrm{Price}(P; x) \;\approx\; \hat{R}_{\text{GP}}(x) \;+\; \sum_{i=1}^{H} \lambda_i(x)\,\mathrm{Price}(g_i; x), \tag{5.11}$$

with $\lambda_i(x)$ constant (fitted at a reference state) or state-dependent (fitted across the grid). Once the factorisation of $\mathbf{K}+\mathbf{E}$ is cached, each evaluation of (**??**) is a dot product plus H closed-form hedge prices — orders of magnitude cheaper than a fresh simulation, which is what makes scenario grids, sensitivity ladders, and calibration loops tractable.

Remark 5.16 (Sensitivities: the full derivative of the composed price). A sensitivity ladder differentiates (**??**) in x, and the total derivative carries *three* terms, not one:

$$\frac{\mathrm{d}\,\mathrm{Price}(P;x)}{\mathrm{d}x} = \underbrace{\frac{\mathrm{d}\hat{R}_{\text{GP}}(x)}{\mathrm{d}x}}_{\text{surrogate slope}} + \sum_{i=1}^{H}\Big[\underbrace{\frac{\mathrm{d}\lambda_i(x)}{\mathrm{d}x}\,\mathrm{Price}(g_i;x)}_{\text{ratio drift}} + \underbrace{\lambda_i(x)\,\frac{\mathrm{d}\,\mathrm{Price}(g_i;x)}{\mathrm{d}x}}_{\text{hedge sensitivity}}\Big],$$

where $\mathrm{d}\hat{R}_{\mathrm{GP}}/\mathrm{d}x = (\partial_x \boldsymbol{\kappa}_*)^\mathsf{T}(\mathbf{K}+\mathbf{E})^{-1}(\hat{\mathbf{R}}-\mathbf{g}_0)+\mathrm{d}g_0/\mathrm{d}x$ from (**??**). Under the "λ constant" convention the *ratio-drift* term is dropped; this is exact only where $\lambda^\star(x)$ is genuinely flat, and elsewhere introduces an *unquantified* error of order $\sum_i \|\mathrm{d}\lambda_i/\mathrm{d}x\|\,\mathrm{Price}(g_i;x)$ — the same carried-term hazard as freezing a state-dependent quantity in any total derivative. Under the "$\lambda(x)$ fitted across the grid" convention the term is retained, but then a differentiable construction of $\lambda(x)$ (e.g. a second GP on the fitted ratios) must be supplied; (**??**) alone does not provide one.

Remark 5.17 (Admissibility of the surrogate price). Nothing in (**??**) *enforces* the static no-arbitrage bounds of Volume II, Chapter 4 — an intrinsic-value floor, monotonicity, positivity of the residual-corrected price. A GP posterior mean can dip below intrinsic in the band's tails. For the GP the band (**??**) at least *certifies* this: the surrogate violates a given static bound by more than $z\sqrt{v_{\mathrm{GP}}(x)}$ with probability at most the Gaussian tail $N(-z)$, so clamping the evaluated price to the known bound costs at most that tail. The neural surrogate of Section **??** carries *no* such certificate and can violate a bound without warning. Feeding an inadmissible surrogate into a calibration loop is the dangerous case; clamp to the admissibility machinery of Volume II, Chapter 4 at evaluation.

Remark 5.18 (Where the cost lives). Training the GP requires factorising $\mathbf{K}+\mathbf{E}$, an $O(N_x^3)$ operation, and storing it, $O(N_x^2)$. This is the reason the design $\{x_j\}$ is kept small and placed intelligently — along the states the book actually visits — rather than on a dense grid. It is also the reason the next section's neural surrogate exists: it trades the exact $O(N_x^3)$ posterior for an amortised parametric fit that scales to large designs at the cost of the closed-form uncertainty band (**??**).

5.5 Recursive decomposition and neural surrogates

5.5.1 The hierarchy

The decomposition $P = R + \lambda^\mathsf{T} g$ presumes the hedge prices $\mathrm{Price}(g_i;x)$ are known. When a hedge is itself too complex to price in closed form, the same construction applies to it: choose a *secondary* library of still-simpler instruments, regress, and surrogate the secondary residual. Each node in the resulting hierarchy is variance-reduced relative to *its own* target. It is tempting to conclude that these factors multiply down to the root; they do not, and the correct budget is *additive*, as the following makes precise.

Proposition 5.19 (Additive error budget of the hierarchy). *Let P be priced against library $\{g_i\}$ with root residual $R^\star$ of variance* $\mathrm{Var}(R^\star) = \mathrm{Var}(P)(1-\varrho^2)$, *and suppose each hedge price* $\mathrm{Price}(g_i;x)$ *is not known exactly but is itself* estimated *by an independent sub-estimator* $\widehat{\mathrm{Price}}(g_i)$ *of variance* $\mathrm{Var}(g_i)(1-\varrho_i^2)/N_i$ *(its own control-variate contraction, at path budget N_i). If the root residual is averaged over N_0 independent paths and the $H+1$ estimators use mutually independent path batches, then the composed root estimator* (**??**) *has variance*

$$\mathrm{Var}\big(\widehat{\mathrm{Price}}(P)\big) \;=\; \frac{\mathrm{Var}(R^\star)}{N_0} \;+\; \sum_{i=1}^{H} \lambda_i^2\,\frac{\mathrm{Var}(g_i)(1-\varrho_i^2)}{N_i}. \tag{5.12}$$

The sub-level contractions therefore enter additively, *each weighted by λ_i^2, not as a product. Without the independence of batches the identity carries the usual covariance cross-terms; the product form $\prod_\ell(1-\varrho_\ell^2)$ holds only in the different construction where a* single *residual is recursively re-decomposed against nested libraries (each level's target being the previous level's residual), which is not the "hedge-the-hedges" recursion described here.*

Proof. By Proposition **??** the composed estimator is $\bar{R}+\sum_i \lambda_i \widehat{\text{Price}}(g_i)$. The root average $\bar{R}$ has variance $\text{Var}(R^\star)/N_0$; each $\widehat{\text{Price}}(g_i)$ enters linearly with coefficient λ_i and contributes $\lambda_i^2\,\text{Var}(\widehat{\text{Price}}(g_i))$; independence of the batches kills all cross-terms, giving (**??**). Substituting each sub-estimator's contracted variance completes the identity. The claimed product form would require the target of level i to *be* the root residual $R^\star$; here it is the hedge g_i, whose variance is unrelated to $\text{Var}(R^\star)$, so no telescoping occurs. □

The design lesson is that budget flows to the *worst-weighted* node: a large λ_i or a poorly hedged sub-instrument (ϱ_i^2 small) can dominate the root error however small $\text{Var}(R^\star)$ is, so paths are allocated across nodes to equalise the terms of (**??**), not lavished on the root. The picture remains compositional pricing — a complex payoff expressed relative to simpler blocks, themselves expressed relative to still-simpler blocks — with simulation confined at every node to a small, smooth residual.

5.5.2 Neural surrogates for ubiquitous hedges

For a hedge g_i that is priced repeatedly across the library — "ubiquitous" — amortising its residual model pays off. Decompose it once,

$$g_i(x) \;=\; r_i(x) \;+\; \sum_\ell \alpha_{i\ell}(x)\, g_\ell^{\text{simpler}}(x), \tag{5.13}$$

and replace the GP surrogate of the residual r_i by a neural network $\hat{r}_i(x;w)$, with weight vector w (not θ; see Remark **??**), trained on Monte Carlo samples of r_i over a rich state design. The composed price is then

$$\text{Price}(g_i;x) \;\approx\; \hat{r}_i(x;w) \;+\; \sum_\ell \alpha_{i\ell}(x)\,\text{Price}(g_\ell^{\text{simpler}};x). \tag{5.14}$$

Remark 5.20 (GP versus neural surrogate: the trade-off). The two surrogates sit at opposite ends of one axis. The GP (Theorem **??**) gives an exact posterior with a calibrated uncertainty band but costs $O(N_x^3)$ and does not amortise across products. The neural network gives no closed-form uncertainty and requires a training loop, but its cost is paid once and its evaluation is $O(1)$ in the design size, so it scales to the large, shared training sets that a ubiquitous hedge justifies. A sufficiently wide network approximates the smooth residual r_i to arbitrary accuracy (universal approximation), but — the leaky abstraction — only where the training design has covered; outside it, unlike the GP, the network gives no warning that it is extrapolating. Use the GP where the uncertainty band is the product; use the network where throughput is.

5.6 Worked example: a two-asset generic product

The whole pipeline is instantiated on a concrete two-asset book, the product used by the reference engine. Every number in this section is recomputed from the closed forms of Volume II, Chapter 3 (Margrabe) and the basket approximation of Volume II, Chapter 1.

Flag (constraint C4): constant-volatility regime.

The reference engine prices the constituents with *flat, constant* single-name volatilities $\sigma_1 = 0.10$, $\sigma_2 = 0.35$ and constant correlation $\rho = 0.30$. In the contract's decomposition this is the corner $\sigma_{\text{ATM}}(\ell_F, T) \equiv \text{const}$ (so vol.slope $= 0$) and $\sigma_{\text{smile}}(m,T) \equiv 0$: *the entire surface dynamics of Volume I are switched off in this illustration.* Every price quoted below is therefore flagged "computed under σ_{ATM} = const" and must be re-derived through the live surface before it is trusted in a book that quotes a skew. The example teaches the *method*; the surface-consistent version is Section **??**.

Example 5.21 (The generic product and its hedge library)**.** Fix expiry $T = 0.10$ years and basket strike $K = 100$. The target payoff is a weighted sum of an absolute spread option and a basket call,

$$P \;=\; w_M\,\big|S_1 - S_2\big| \;+\; w_B\left(\tfrac{1}{2}(S_1+S_2) - K\right)^+, \qquad w_M = w_B = 1,$$

priced under zero rates and zero dividends, so $F = S$ (flagged) and every constituent is a spot-quoted closed form. The hedge library is the set of instruments whose prices are already known:

(i) the two exchange legs of the spread, $\max(S_1 - S_2, 0)$ and $\max(S_2 - S_1, 0)$, priced by the Margrabe formula of Volume II, Chapter 3 with total volatility $\omega_M = \Sigma_M\sqrt{T}$ and $\Sigma_M^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$;

(ii) the basket call, priced *approximately* as a Black–Scholes call on $B = \frac{1}{2}(S_1 + S_2)$ with the basket volatility σ_B of Volume II, Chapter 1, $\sigma_B^2 = \frac{1}{4}\sigma_1^2 + \frac{1}{4}\sigma_2^2 + \frac{1}{2}\rho\sigma_1\sigma_2$, whose constituent vols are the single-name ATM vols $\sigma_{\text{ATM},1}, \sigma_{\text{ATM},2}$ by constraint C1 (here flat). A sum of lognormals is not lognormal, so this is the moment-matching approximation of Volume II, Chapter 1, *not* a closed form.

Remark 5.22 (Where the basket approximation enters the error). Item (ii) plugs an *approximate* basket price into the second term of (**??**), so by Proposition **??** the estimator error decomposes into two pieces that do not mix:

$$\widehat{\text{Price}}(P;x_0) - \text{Price}(P;x_0) \;=\; \underbrace{\big(\bar{R} - \mathbb{E}[R]\big)}_{\text{Monte Carlo, } O(N_{\text{p}}^{-1/2})} \;+\; \underbrace{\lambda_{\text{basket}}\big(\sigma_B\text{-approx price} - \mathbb{E}[\text{basket} \mid X_0 = x_0]\big)}_{\text{deterministic, does not shrink with } N_{\text{p}}}.$$

Unbiasedness holds only *relative to the plugged-in basket price*. Two routes close the gap: bound the moment-matching error at these parameters (small here — the two vols are close to lognormal over $T = 0.10$ — but it must be stated, not assumed), or price the basket leg by an independent converged simulation, which Definition **??** already permits and which makes the second term genuinely negligible. The latter is done in any book that quotes against this product; the table below uses the approximation and is labelled accordingly.

At the given parameters the two derived volatilities are

$$\Sigma_M \;=\; \sqrt{0.10^2 + 0.35^2 - 2(0.30)(0.10)(0.35)} \;=\; 0.33392, \qquad \omega_M \;=\; \Sigma_M\sqrt{0.10} \;=\; 0.10559, \tag{5.15}$$

and, on the diagonal $S_1 = S_2$ where the two basket weights are each $\frac{1}{2}$,

$$\sigma_B \;=\; \sqrt{\tfrac{1}{4}(0.10^2) + \tfrac{1}{4}(0.35^2) + \tfrac{1}{2}(0.30)(0.10)(0.35)} \;=\; 0.19590.$$

Table **??** lists the recomputed reference prices along the training diagonal.

Remark 5.23 (Reading the diagonal). The spread value is strictly positive at $S_1 = S_2$ — 8.4212 at $S = 100$ — because $|S_1 - S_2|$ is an option on the vol-driven dispersion of the two legs over $[0, T]$, worth the symmetric Margrabe value even when the legs start equal. At $S_1 = S_2 = S$ the two Margrabe arguments coincide at $a = b = \omega_M/2$, so the closed form of Volume II, Chapter 3 collapses to the symmetric at-the-money value

$$2S\big[N(\omega_M/2) - N(-\omega_M/2)\big] \;=\; 200\big[N(0.05280) - N(-0.05280)\big] \;=\; 8.421,$$

Table 5.1: Reference prices of the generic product along the diagonal $S_1 = S_2 = S$, recomputed from the Margrabe closed form and the basket *moment-matching approximation* (not a closed form; see Remark **??**) at $T = 0.10$, $K = 100$, $\sigma_1 = 0.10$, $\sigma_2 = 0.35$, $\rho = 0.30$. *Flag (C4): computed under* $\sigma_{\mathrm{ATM}} =$ *const,* $\sigma_{\mathrm{smile}} \equiv 0$*; and* $F = S$ *(zero rates). The spread column is nonzero on the diagonal because the exchange option is long the Margrabe volatility* ω_M *over the horizon.*

S	spread $w_M\lvert S_1 - S_2\rvert$	basket call	generic P
80	6.7370	0.0002	6.7372
90	7.5791	0.1072	7.6863
95	8.0002	0.6905	8.6906
100	8.4212	2.4710	10.8922
102	8.5897	3.6221	12.2117
105	8.8423	5.7796	14.6219
110	9.2634	10.1742	19.4376
120	10.1055	20.0032	30.1086

matching the table, with the symmetry constant $\varkappa = S\varphi(\omega_M/2)$ governing the local behaviour off the diagonal. The basket call, by contrast, is a plain in-/out-of-the-money vanilla and vanishes as $S \downarrow$. The generic price is dominated by the spread deep out-of-the-money and by the basket near and above the strike — exactly the regime split a single hedge instrument cannot span, which is why the library needs both.

Example 5.24 (Surrogate design)**.** The residual field (**??**) is trained on a design of $N_x = 40$ states: eight diagonal anchors $S_q \in \{80, 90, 95, 100, 102, 105, 110, 120\}$ ($q = 1, \ldots, 8$; the anchor glyph is S_q, never the banned S_0), each carrying a five-point stencil (S_q, S_q), $(S_q \pm b_x, S_q)$, $(S_q, S_q \pm b_x)$ with relative bump $b_x = 0.025\, S_q$ (so $b_x = 2.5$ points at $S_q = 100$). The GP uses the squared-exponential kernel (**??**) with two length scales (one per spot), inputs standardised, mean function fitted as a constant. Two evaluation clouds probe generalisation: an independent uniform cloud on $[75, 125]^2$ (a stress of states the book need not visit), and a *realistic* correlated cloud — a bivariate lognormal on $(\log S_1, \log S_2)$ around the central spot 100 (i.e. F_{ref} under $F = S$) propagated over a short spot horizon $t_{\mathrm{h}} = 3/252$ years, distinct from the option expiry $T = 0.10$ and carrying the same $\rho = 0.30$.

The off-diagonal spread of this cloud is computed inline. The standard deviation of $\log(S_1/S_2)$ over t_{h} is $\Sigma_M\sqrt{t_{\mathrm{h}}} = 0.33392\sqrt{3/252} = 0.03643$, so the median absolute spread is

$$\mathrm{med}\,\lvert S_1 - S_2\rvert \;\approx\; 100 \times N^{-1}(0.75) \times 0.03643 \;=\; 100 \times 0.6745 \times 0.03643 \;\approx\; 2.46 \text{ points}$$

(Monte Carlo check: 2.44). This is *not* a fraction of a point — it is comparable to the stencil half-width $b_x = 2.5$ points, so a large part of the realistic cloud sits at or just beyond the off-diagonal edge of the training stencil. The cloud therefore tests *mild extrapolation* off the diagonal, not pure interpolation. Two things keep this safe: the residual is smooth (Section **??**), so short-range extrapolation is benign, and — decisively — the posterior band (**??**) *widens* exactly in the off-diagonal gaps and flags any state where the surrogate is reaching beyond its support. If tighter interpolation is wanted, widen the off-diagonal stencil bump or shorten t_{h}; either brings the cloud back inside the design.

Remark 5.25 (Deferred: empirical error metrics). The reference engine reports mean-absolute, root-mean-square, and maximum surrogate errors on each cloud, together with the posterior band (**??**) used for a two-sigma containment check. Those figures are outputs of the fitted Gaussian process (a full kernel optimisation), not of a closed form, and are *not* recomputed here; they are deferred rather than transcribed, since (i) reproducing them requires running the surrogate, and (ii) the prices they compare against are the constant-volatility prices of Table **??**, already flagged under C4. The qualitative claim that survives independent of the fit is structural: because the residual inherits the smoothness argument of Section **??** and the realistic cloud sits at the off-diagonal edge of the design (mild extrapolation, per Example **??**), the surrogate error on the realistic cloud is the relevant one, and the posterior band (**??**) is what certifies where it can be trusted.

5.6.1 A surface-consistent control variate: the worst-of book

The constant-vol illustration isolates the numerics; the production use joins them to the live surface. Consider a worst-of book simulated under the two-asset local-volatility, local-correlation dynamics of Volumes I and II — the Euler scheme with Itô correction, driving each leg through its own $\sigma_{\mathrm{ATM}}(\ell_F, T) + \sigma_{\mathrm{smile}}(m, T)$ surface. Here σ_{ATM} is *not* constant: the simulation carries the skew, so no C4 flag is needed.

An *exact* algebraic identity disciplines the choice of target. The terminal worst-of satisfies

$$\min(S_{1,T}, S_{2,T}) \;=\; \tfrac{1}{2}(S_{1,T} + S_{2,T}) \;-\; \tfrac{1}{2}\big|S_{1,T} - S_{2,T}\big|. \tag{5.16}$$

Both legs on the right — the basket and the absolute spread — are in the control-variate library below, so with $\lambda_{\mathrm{basket}} = 1$ and $\lambda_{\mathrm{spread}} = -\frac{1}{2}$ the residual $R^\star$ of the terminal worst-of is *identically zero*: $\varrho^2 = 1$, and there is no residual field for a surrogate to learn. The linear worst-of is a *perfect* control variate but a *vacuous* surrogate demonstration. The genuine target is a *nonlinear* member of the book — a put on the worst-of, $(K - \min(S_{1,T}, S_{2,T}))^+$ — whose payoff is *not* spanned by the library: the outer $(\cdot)^+$ breaks the exact identity (**??**), leaving a small, smooth, genuinely nonzero residual field, which is the object the GP of Section **??** interpolates. That put is the target.

The control-variate library is

$$\{\,\text{basket } \tfrac{1}{2}(S_1 + S_2),\;\; \text{absolute spread } |S_1 - S_2|,\;\; \text{CVC},\;\; \text{put}_1,\;\; \text{put}_2\,\},$$

where the *CVC* instrument is the call-vs-call basket spread

$$\mathrm{CVC}(K) \;=\; \tfrac{1}{2}\big[(S_{1,T} - K)^+ + (S_{2,T} - K)^+\big] \;-\; \big(\tfrac{1}{2}(S_{1,T} + S_{2,T}) - K\big)^+, \tag{5.17}$$

an at-the-money proxy for the basket vol spread $\xi = \bar\sigma - \sigma_B \geq 0$: by convexity it is nonnegative and it grows with the dispersion that ξ measures. The basket and spread legs span the *linear* part of the payoff by (**??**); the two vanilla puts and the CVC leg absorb the strike-and-convexity part that the $(\cdot)^+$ introduces, leaving the small residual for the surrogate.

Its presence in the library is what ties the numerical scheme to the correlation conventions of Volume II. Pricing the worst-of under sticky-ρ ($\mathrm{d}\rho = 0$) versus sticky-CVC ($\mathrm{d}\xi = 0$) is a change of the simulated *law*, so the same drawn paths do not carry the same probability under both: differencing prices requires either a common-random-numbers coupling (both convention-parameterisations driven by the *same* Brownian increments, with a convention-dependent parameter map) or a likelihood-ratio reweighting. The CVC leg carries the ξ-direction that separates the conventions; the coupled-path difference estimator is then the control-variate expression of the shadow Greeks, whose construction is deferred to the shadow-Greek differencing scheme of Volume II, Chapter 2 rather than asserted here. The hedge ratios are the regularised $\mu_\star$ of Proposition **??**, and constraint C1 threads the surface dynamics of Volume I into every basket and CVC leg through $\sigma_i \equiv \sigma_{\mathrm{ATM},i}(\ell_{F,i}, T)$, where the surface is read at $\ell_F = \ln(F/F_{\mathrm{ref}})$ off the Euler-simulated *spot* paths — *under zero rates, $F = S$ along each path* (the F=S collapse, flagged here at this analytic use).

Remark 5.26 (Engine outputs versus theoretical guarantees). The worst-of prices, hedge ratios, and residual standard deviations are functions of the simulated surface and correlation convention; they are engine outputs at a chosen seed and path count, not closed forms, and are not transcribed here (running the full stochastic engine is out of scope, and any transcribed figure would carry the engine's own conventions rather than this contract's). What the theory guarantees, and what needs no run, is the structure: the nonlinear put-on-worst-of residual variance contracts by $1-\varrho^2$ (Theorem **??**) — while the *linear* worst-of residual is exactly zero by (**??**) — the CVC leg carries the ξ-direction that separates sticky-ρ from sticky-CVC (differenced on coupled paths, per the construction deferred to Volume II, Chapter 2), and the same Bayesian regression that calibrates the Volume I surface regularises the hedge vector. The numbers are the engine's; the guarantees are the chapter's.

Chapter summary

The method is three moves. *Represent*: write the target price as a known linear combination of hedge prices plus a residual, unbiased for every hedge ratio (Proposition **??**). *Regress*: choose the hedge ratio as the L^2 projection of the target onto the hedge span, contracting the simulation variance by $1-\varrho^2$ (Theorem **??**), and stabilise it with the same Gaussian prior that regularises the Volume I surface calibration (Proposition **??**). *Interpolate*: treat the residual as a small, smooth field over the state space and learn it with a Gaussian process — Bayesian regression in an infinite feature space, with a closed-form uncertainty band (Theorem **??**) — or, for a ubiquitous hedge, with an amortised neural surrogate at the cost of that band (Remark **??**). The construction recurses: complex hedges are themselves priced relative to simpler ones, and the sub-level variances enter the root budget *additively*, λ_i^2-weighted — not as a product (Proposition **??**). The worked example prices a Margrabe-plus-basket product; its constant-volatility numbers are flagged under C4, and the surface-consistent worst-of book of Section **??** shows where the live Volume I surface and the Volume II correlation conventions re-enter through constraint C1 and the CVC control variate.